

		L.J Institute of Engineering and Technology, Ahmedabad. <u>Introduction to Probability Theory and Stochastic Processes</u> Practice Book (Sem- III)							
		Note: This practice book is only for reference purpose. LJU Test question paper may not be completely set from this practice book.							
Sr. No.	unit_number	Question text	Answer	Marks	Previous Year	Option 1	Option 2	Option 3	Option 4
1	1	What are the chances that no two boys are sitting together for a photograph if there are 5 girls and 2 boys?	5/7	1		1/21	4/7	2/7	5/7
2	1	An event in the probability that will never be happened is called as -	Impossible event	1		Unsure event	Sure event	Possible event	Impossible event
3	1	A coin is tossed up 4 times. The probability that tails turn up in 3 cases is _____	1/4	11	LJU-2024	1/2	1/3	1/4	1/5
4	1	What will be the probability of getting odd numbers if a die is thrown?	1/2	1		1/2	2	4/2	5/2
5	1	What is the probability of getting a sum as 3 if two dice are thrown?	1/18	1		2/18	1/18	4	1/36
6	1	12 numbered cards are there from 1 to 12, and two cards a chosen at random. What is the probability that the sum of chosen cards is odd?	6/11	1	LJU-2024	5/22	6/11	5/11	NONE
7	1	What is the probability of getting the sum as a prime number if two dice are thrown?	5/12	1		5/24	5/12	5/30	1/4
8	1	If two dice are thrown together, what is the probability of getting an even number on one dice and an odd number on the other dice?	1/2	1		1/4	3/5	3/4	1/2
9	1	When two coins are tossed simultaneously, what are the chances of getting at least one tail?	3/4	1		3/4	1/5	4/5	1/4
10	1	A coin and a dice are tossed consecutively. What is the probability to get a 6 on a dice given that a head on a coin?	1/6	1	LJU - 2023	1/12	1/6	1/3	NONE
11	1	What is the probability of having 53 Thursdays in a non-leap year?	1/7	1	LJU - 2025	6/7	1/7	1/365	53/365
12	1	Three unbiased coins are tossed. What is the probability of getting at least 2 tails?	0.5	1		0.75	0.5	0.25	0.2
13	1	Probability of getting a sum 5 on the top of the dice is 1/36. How many times a dice is rolled?	3	1	LJU - 2023	2	4	3	NONE
14	1	Two friends A and B apply for a job in the same company. The chances of A getting selected is 2/5 and that of B is 4/7. What is the probability that both of them get selected?	8/35	1		8/35	34/35	27/35	None of these
15	1	A dice is rolled twice. What is the probability of getting sum 9?	1/9	1		2/3	1/3	1/9	4/9
16	1	For two equally likely, exhaustive and independent events A and B, $P(A \cap B)$ is:	1	1	LJU-2024	0	1	0.5	0.25
17	1	In how many different ways can the letters of the word ‘GEOGRAPHY’ be arranged such that the vowels must always come together?	15120	1	LJU-2024	15130	2530	15120	2540
18	1	Students likes either mathematics or physics. 65% likes physics and 25% likes both the subjects. What is the probability that who likes mathematics also likes physics.	5/12	1	LJU-2023	5/12	5/13	12/13	NONE
19	1	If a number is selected at random from the first 50 natural numbers, what will be the probability that the selected number is a multiple of 3 and 4?	2/25	1	LJU-2025	7/50	4/25	2/25	3/25
20	1	An urn contains 9 balls, two of which are red, three blue and four black. Three balls are drawn at random. The chance that they are of the same color is	5/84	1	LJU - 2025	5/84	7/84	3/9	4/9
21	1	A and B are two independent events such that $P(\bar{A}) = 0.7, P(\bar{B}) = k$ and $P(A \cup B) = 0.8$, then k is	2/7	1	LJU-2025	2/7	5/7	6/7	1
22	1	The probability that Jay will solve a problem is 2/3 and the probability that Vijay will solve it is 3/4. What is the probability that the problem will be solved?	11/12	1		11/12	11/24	1/12	1/6

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23	1	3 people A, B and C are sitting in a circular fashion. Find the probability that A and B do not sit together.	0	1		1	0	0.5	0.33
24	1	If four dice are thrown together, the probability that the sum of the numbers appearing on them is 13 is,	$\frac{35}{324}$	1		$\frac{7}{108}$	$\frac{23}{324}$	$\frac{35}{1296}$	$\frac{35}{324}$
25	1	What will be the probability of an impossible event?	0	1		0	1	Infinity	None of the above
26	1	What will be the number of events if 10 coins are tossed simultaneously?	1024	1		512	90	1000	1024
27	1	48 ice-creams are placed on a table. Some are chocolate flavored and others are vanilla flavored. If a person takes an ice -cream at random from the table, the probability that it is chocolate flavored is $\frac{2}{3}$.Find the number of vanilla flavored ice-cream on the table.	16	1	LJU-2024	16	8	32	24
28	1	A box contains 5 red and 10 green balls. Eight (8) of them are placed with another box. The chances that the letter box contains 2 red and 6 green balls are ____.	140/429	1	LJU-2024	240/429	140/367	140/429	240/367
29	1	A box Contain 20 defective and 80 non-defective items. If two items are selected at random without replacement. What will be the probability that both items are defective?	19/495	1		1/5	1/25	20/99	19/495
30	1	A 1 box contains 2 white balls, 3 black balls, and 4 red balls. What is the number of ways of drawing 3 balls from the box with at least one black ball?	64	1	LJU-2024	84	72	48	64
31	1	As per De Morgan's Law, $(A \cap B)'$ is equal to	$A' \cup B'$	1		$A' \cap B'$	$A' \cup B'$	$(A \cap B)'$	$A' + B'$
32	1	As per De Morgan's Law, $(A \cup B)'$ is equal to	$A' \cap B'$	1		$A' \cap B'$	$A' \cup B'$	$(A \cap B)'$	$A' + B'$
33	1	If E and F are two events associated with the same sample space of a random experiment then $P(E F)$ is given by _____	$\frac{P(E \cap F)}{P(F)}$, provided $P(F) \neq 0$	1		$\frac{P(E \cap F)}{P(F)}$, provided $P(F) \neq 0$	$\frac{P(E \cap F)}{P(F)}$, provided $P(F) = 0$	$\frac{P(E \cap F)}{P(F)}$	$\frac{P(E \cap F)}{P(E)}$
34	1	Let E and F be events of a sample space S of an experiment, if $P(S F) = P(F F)$ then value of $P(S F)$ is _____	1	1		0	-1	1	2
35	1	Given that E and F are events such that $P(E) = 0.6$, $P(F) = 0.3$ and $P(E \cap F) = 0.2$, then $P(E F)$?	2/3	1	LJU-2025	2/3	1/3	3/4	1/4
36	1	If $P(A) = \frac{7}{11}$, $P(B) = \frac{6}{11}$ and $P(A \cup B) = \frac{8}{11}$, then $P(A B) =$ _____	5/6	1		3/5	2/3	1/2	5/6
37	1	If $P(A) = \frac{1}{5}$, $P(B) = 0$, then what will be the value of $P(A B)$?	Not defined	1		0	1	Not defined	1/5
38	1	Which of this represents the multiplication theorem of probability?	$P(A \cap B) = P(A) P(B/A)$	1		$P(A \cap B) = P(B) P(B/A)$	$P(A \cap B) = P(A) P(B/A)$	$P(A \cap B) = P(A) P(B/B)$	$P(A \cap B) = P(A) P(A/A)$
39	1	What is the probability that a 5 digits number using 0, 2, 4, 6 and 8 without repeating which is a multiple of 4?	5/8	1	LJU-2023 LJU-2025	5/8	7/12	11/24	NONE
40	1	A person is known to hit the target in 3 out of 4 shots, whereas another person is known to hit the target in 2 out of 3 shots. What is the Probability of Target being hit at all when they both try?	0.9167	1		0.0833	0.9167	0.25	0.3333
41	1	A person is known to hit the target in 3 out of 4 shots, whereas another person is known to miss the target in 1 out of 4 shots. What is the Probability of Target is not being hit at all when they both try independently?	1/16	1	LJU-2024 LJU-2025	1/16	1/4	1/8	NONE
42	1	There are 4 red pencils,5 blue pencils, 3 yellow pencils, and 10 white pencils in a bag. What is the probability of getting a pencil that is red or blue?	$\frac{9}{22}$	1	LJU-2022 LJU-2025	$\frac{9}{22}$	$\frac{13}{22}$	$\frac{8}{22}$	$\frac{7}{22}$

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43	1	What is the probability of getting at least one green ball from bag contains 2 red, 3 green and 5 brown balls when three ball are selected at random?	17/24	1	LJU-2024	3/10	17/24	7/12	NONE
44	1	In a group of 100 computer buyers, 40 bought CPU, 30 purchased monitor, and 20 purchased CPU and monitors. If a computer buyer chose at random and bought a CPU, what is the probability they also bought a Monitor?	0.5	1		0	1	0.5	0.3
45	1	You are making a sub. You have 4 choices of meat, 3 choices of cheese,6 choices of vegetables and 4 different dressings. How many different subs are possible for you to make, if you are allowed one from each category?	288	1		288	388	480	399
46	1	You are making a password. The password must consist of 3 letters followed by 3 digits. All letters and digits are permitted and repetition of the letter and digits are allowed. How many different passwords are possible?	17576000	1		178500	1788200	1875570	17576000
47	1	A bag contains 10 red marbles,5 blue marbles and 8 yellow marbles. What is the probability of choosing a marble which is not yellow?	$\frac{15}{23}$	1	LJU-2025	$\frac{18}{23}$	$\frac{15}{23}$	$\frac{13}{23}$	$\frac{8}{23}$
48	1	Two dice are rolled. What is the probability of getting an even number on the first die and a 2 on second die?	$\frac{1}{12}$	1		$\frac{1}{2}$	$\frac{1}{6}$	$\frac{1}{12}$	$\frac{3}{12}$
49	1	Each of ten tickets is marked with a different number from 0 to 9 and put into a box. If you draw a ticket from the box, what is the probability that you will draw a 2 or 8 or 4?	$\frac{3}{10}$	1		$\frac{3}{10}$	$\frac{1}{10}$	$\frac{2}{10}$	$\frac{8}{10}$
50	1	Bag A contains 12 blue marbles and 5 red marbles. Bag B contain 7 orange marbles and 5 grey marbles. Find the probability of selecting a red marble from bag A and an orange marble from bag B in one draw from each bag.	$\frac{35}{204}$	1		$\frac{204}{35}$	$\frac{7}{35}$	$\frac{35}{204}$	$\frac{12}{204}$
51	1	The probability that it will be sunny on Friday is 4/5. The probability that an ice cream shop will sell ice creams on a sunny Friday is 2/3 and the probability that the ice cream shop sells ice creams on a non-sunny Friday is 1/3. Then what is the probability that it will be sunny and the ice cream shop sells the ice creams on Friday?	$\frac{8}{15}$	1		$\frac{8}{15}$	$\frac{15}{8}$	$\frac{11}{15}$	$\frac{9}{15}$
52	1	If 12 persons are seated at a round table, what is the probability that two particular persons sit together?	$\frac{2}{11}$	1		$\frac{2}{11}$	$\frac{1}{11}$	$\frac{1}{12}$	$\frac{5}{12}$
53	1	If a fair die is rolled twice, observe the numbers that face up. Find the conditional probability that the sum of the numbers is 7, given that the first number is 2.	$\frac{1}{6}$	1		$\frac{6}{7}$	$\frac{8}{7}$	$\frac{7}{6}$	$\frac{1}{6}$
54	1	70% of your friends like Chocolate, and 35% like Chocolate AND like Strawberry. What percent of those who like Chocolate also like Strawberry?	50%	1	LJU-2022 LJU-2024	20%	40%	50%	45%
55	1	Company A produces 10% defective products, Company B produces 20% defective products and Company C produces 5% defective Products. If we choosing a company is an equally like event, then what is the probability that product chosen is defective?	0.12	1	LJU-2022 LJU-2025	0.22	0.12	0.11	0.21
56	1	The probability that a person stopping at a gas station will ask to have his tires checked 0.12, probability that he will ask to have his oil checked is 0.29and probability that he will ask to them both checked is 0.07. The probability that a person who has his tries checked will also have oil checked is	0.58	1	LJU-2025	0.34	0.58	0.24	0.41

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Sr. No.	unit_number	Question text	Answer	Marks	Previous Year	Option 1	Option 2	Option 3	Option 4
57	1	In given day in the rainy season, it may rain 70 % of the time. If it rains, chances that a village fair will make a loss on that day is 80%. However, if it does not rain, chance that the fair will make a loss on that day is only 10%. If the fair has not made a loss on a given day in the rainy season, what is the probability that it has not rained on that day?	$\frac{27}{41}$	1	LJU-2025	$\frac{27}{41}$	$\frac{3}{10}$	$\frac{9}{11}$	$\frac{14}{17}$
58	1	Let A and B be two events such that $P(\overline{A \cup B}) = \frac{1}{6}$, $P(A \cap B) = \frac{1}{4}$, $P(\bar{A}) = \frac{1}{4}$, where $\bar{A}$ stands for complement of event Then, events A and B are:	Independ ent but not equally likely	1		Equally likely but not independent	Mutually exclusive and independent	Independent but not equally likely	Equally likely and mutually exclusive
59	1	If A and B be two arbitrary events, then	$P(A \cup B) \leq P(A) + P(B)$	1		$P(A \cap B) = P(A) \cdot P(B)$	$P(A \cup B) = P(A) + P(B)$	$P(A/B) = P(A \cap B)P(B)$	$P(A \cup B) \leq P(A) + P(B)$
60	1	If two fair coins are flipped and at least one of the outcomes is known to be head, what is the probability that both outcomes are heads?	$\frac{1}{3}$	1	LJU-2022	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{2}{3}$	$\frac{1}{3}$
61	1	15 numbered cards are there from 1 to 15, and two cards a chosen at random such that the sum of the numbers on both the cards is even. Find the probability that the chosen cards are odd-numbered.	NONE	1	LJU - 2023	2/27	1/24	3/17	NONE
62	1	In a certain residential suburb, 60% of all household s get internet service from the local cable company, 80% get television service from that company, and 50% get both services from that company. If a household is randomly selected, what is the probability that it gets at least one of these services from the company?	0.9	1		0.5	0.6	0.9	None of these
63	1	The probability that a randomly selected person has high blood pressure (the event H) is $P(H) = 0.2$ and the probability that a randomly selected person is a runner (the event R) is $P(R) = 0.3$.The probability that a randomly selected person has high blood pressure and is a runner is 0.1. What is the probability that a randomly selected person has high blood pressure and is not a runner?	0.1	1		0.2	0.3	0.5	0.1
64	1	Using the digits 1, 2, 3, 4, and 5, a number having five digits is formed without any repetition. What is the probability that the number is divisible by 4?	0.2	1	LJU-2025	0.34	0.2	0.58	0.44
65	1	If A and B are two independent events, then the probability of occurrence of at least of A and B is given by	$1 - P(\bar{A}) \cdot P(\bar{B})$	1		$P(A) \cdot P(B)$	$1 - P(\bar{A}) \cdot P(\bar{B})$	0	$1 - P(A) \cdot P(\bar{B})$
66	1	A man is known to speak truth 2 out of 3 times. He throws a die and report that number obtained is a four. Find the Probability that the number obtained is actually a four.	2/7	1	LJU-2022	1/14	2/7	3/7	5/7
67	1	Find P(E F), where E: no tail appears, F: no head appears, when two coins are tossed in the air.	0	1		0	1	2	3
68	1	The chances of a defective screw in the three boxes A, B, C are 1/5, 1/6 and 1/7 respectively. A box is selected at random and a screw drawn from it at random is found to be defective. Find the probability that it came from box A.	42/107	1	LJU-2022	40/107	41/107	42/107	43/107
69	1	The Probability that India wins a cricket test match against England is $\frac{1}{3}$. If India and England play 3 matches, the probability that India will win at least one match is	$\frac{19}{27}$	1	LJU-2024 LJU-2025	$\frac{8}{27}$	$\frac{19}{27}$	$\frac{9}{27}$	$\frac{5}{27}$
70	1	If six dice are rolled, then the probability that all show different face is	$\frac{6!}{6^6}$	1	LJU-2025	$\frac{1}{6^6}$	$\frac{6!}{6^6}$	$\frac{1}{2^6}$	$\frac{6}{6^6}$

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71	1	The letters of the word ‘article’ are arranged at random then the probability that the vowels may occupy the even places is _____.	$\frac{1}{35}$	1		$\frac{1}{35}$	$\frac{3}{35}$	$\frac{4}{35}$	$\frac{12}{35}$
72	1	Four dice are thrown simultaneously. What will be the probability that all of them have the same face?	1/216	1		1/36	1/216	1/316	1/6
73	1	Four dice are rolled then what is the probability to get all have different numbers on top face?	5/18	1	LJU-2024 LJU-2025	5/12	5/18	1/36	NONE
74	1	A single letter is selected at random from the word ‘probability’. The probability that it is a vowel is	4/11	1		0	4/11	7/11	1
75	1	In a bucket there are 5 purple, 15 grey and 25 green balls. If the ball is picked up randomly, find the probability that it is neither grey nor purple?	5/9	1		5/9	12/13	51/43	2/7
76	1	The probabilities that a student will solve Question A and Question B are 0.4 and 0.5 respectively. What is the probability that he solves at least one of the two questions?	0.7	1		0.6	0.7	0.8	0.9
77	1	At a party, there are 45 balloons. Some are red, and the others are blue. If the probability of picking a red balloon at random is $\frac{4}{9}$, how many blue balloons are there?	30	1	LJU-2025	15	20	30	25
78	1	The probability of getting heads when tossing a fair coin is 1/2. If the coin is tossed 5 times, what is the probability of getting at least one head?	31/32	1	LJU-2025	31/32	25/32	15/32	20/27
79	1	Let A and B be two events with $P(A) = \frac{3}{8}$, $P(B) = \frac{5}{8}$, $P(A \cup B) = \frac{3}{4}$. Find P(A/B) and P(B/A)		3					
80	1	An urn contains 10 white and 3 black balls, while another urn contains 3 white and 5 black balls. Two balls are drawn from the first urn and put into the second urn and then a ball is drawn from the latter. What is the probability that it is a white ball?		4					
81	1	Find the probability of drawing a queen and a king from a pack of cards in two consecutive draws, the cards drawn not being replaced.		3					
82	1	In a certain assembly plant, three machines, B1, B2, and B3, make 30%, 45%, and 25%, respectively, of the products. It is known from past experience that 2%, 3%, and 2% of the products made by each machine, respectively, are defective. Now, suppose that a finished product is randomly selected. What is the probability that it is defective?		4					
83	1	Three boxes contain 10%, 20% and 30% of defective finger joints. A finger joint is selected at random which is defective. Determine the probability that it comes from (i) 1 st box (ii) 2 nd box (iii) 3 rd box		4					
84	1	What is the chance that a leap year selected at random will contain 53 Sundays?		3					
85	1	In tossing 3 balanced coins, what is the probability of getting 2 heads?		2					
86	1	Compute $P(A/B)$, If $P(A) = 0.6$, $P(B) = 0.7$ and $P(A \cap B) = 0.3$		2					
87	1	In a box, 100 bulbs are supplied out of which 10 bulbs have defects of type A, 5 bulbs have defects of type B and 2 have defects of both types. Find the probabilities that a bulb to be drawn at random has a B type defect under the condition that it has an A type defect.		3					
88	1	One shot is fired from each of the three guns. E_1, E_2, E_3 denotes the events that the target is hit by		3					

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		the first, second and third guns respectively. If $P(E_1)=0.5, P(E_2)=0.6$ and $P(E_3)=0.8$ and E_1, E_2, E_3 are independent event. Find the probability that exactly one hit is registered.																															
89	1	Two students x and y work independently on a problem. The probability that x will solve it is 3/4 and the probability that y will solve it is 2/3. What is the probability that problem will be solved?		2																													
90	1	There are two bags. The first contains 2 red and 1 white ball, whereas the second bag has only 1 red and 2 white balls. One ball is taken out at random from the first bag and put in second. Then a ball is chosen at random from the second bag. What is the probability that this last ball is red?		4	LJU-2025																												
91	1	A bag contains 2 black, 3 red, and 5 blue balls. Three balls are drawn at random. Find the probability that the three balls drawn: (1) are blue (2) consist of 2 blue and 1 red ball, and (3) consist of exactly one black ball.		3																													
92	1	Probability of solving specific independently by A and B are $\frac{1}{2}$ and $\frac{1}{3}$ respectively. If both try to solve the problem independently, find the probability that (i) the problem is solved (ii) exactly one of them solves the problem.		3																													
93	1	From the employees of a company 5 persons are selected to represent them in the managing committee of the company. Particulars of five persons are as follows: <table border="1"><thead><tr><th>Sr.No</th><th>Name</th><th>Sex</th><th>Age in years</th></tr></thead><tbody><tr><td>1</td><td>Harish</td><td>M</td><td>30</td></tr><tr><td>2</td><td>Rohan</td><td>M</td><td>33</td></tr><tr><td>3</td><td>Sheetal</td><td>F</td><td>46</td></tr><tr><td>4</td><td>Alis</td><td>F</td><td>28</td></tr><tr><td>5</td><td>Salim</td><td>M</td><td>41</td></tr></tbody></table> A person is selected at random from this group to act as a spokesperson. What is the probability that the spokesperson will be either male or over 35 years?	Sr.No	Name	Sex	Age in years	1	Harish	M	30	2	Rohan	M	33	3	Sheetal	F	46	4	Alis	F	28	5	Salim	M	41		3					
Sr.No	Name	Sex	Age in years																														
1	Harish	M	30																														
2	Rohan	M	33																														
3	Sheetal	F	46																														
4	Alis	F	28																														
5	Salim	M	41																														
94	1	In a bolt factory, three machines A, B and C manufacture 25%, 35% and 40% of the total product respectively. Of these outputs 5%, 4% and 2% respectively, are defective bolts. A bolt is picked up at random and found to be defective. What are the probabilities that it was manufactured by machines A, B and C?		4	LJU-2024 LJU-2025																												
95	1	A factory produces a certain type of outputs by three types of machines. The respective daily production figures are: Machine I: 3000 Units; Machine II: 2500 Units; Machine III: 4500 Units Past experience shows that 1 Per cent of the output produced by Machine I is defective. The corresponding fraction of defectives for the other		3																													

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		two machines are 1.2 Per cent and 2 Per cent respectively. An item is drawn at random from the day’s production run and is found to be defective. What is probability that it comes from the output of Machine II?							
96	1	It is observed that 50% of mails are spam. There is a software that filters spam mail before reaching the inbox. Its accuracy for detecting a spam mail is 99% and chances of tagging a non-spam mail as spam mail is 5%. If a certain mail is tagging as spam find the probability that it is not a spam mail.		4	LJU-2023				
97	1	Three boxes contained 10%, 20% and 30% red colors pens. A pen is selected at random whose color is red. Determine the probability that it came from 3rd box, 2nd box, 1st box.		4					
98	1	An aircraft emergency locator transmitter (ELT) is a device designed to transmit a signal in the case of crash. The Altigauge Manufacturing company makes 80% of the ELTs, the Bryant company makes 15% of them, and the Chartair company makes the order 5%. The ELTs made by Altigauge have a 4% rate of defects, the Bryant ELTs have a 6% rate of defects, and Chartair have 9% defects. (1) If a randomly selected ELT is then tested and found to be defective, find the probability that it was made by the Chartair manufacturing company. (2) If a randomly selected ELT is then tested and is found to be defective, find the probability that it was made by the Altigauge Manufacturing company.		4					
99	1	A card is lost from a pack of 52 cards. From the remaining cards two are drawn randomly and found to be both clubs. Find the probability that the lost card is also a club.		3	LJU-2024				
100	1	A card is lost from a standard deck of 52 cards. From the remaining cards, three cards are drawn at random and all are found to be hearts. What is the probability that the lost card is also a heart?		4	LJU-2025				
101	1	There are 3 boxes, the first one containing 1 white, 2 red and 3 black balls; the second one containing 2 white, 3 red and 1 black ball and the third one containing 3 white, 1 red and 2 black balls. A box is chosen at random and from it two balls are drawn at random. One ball is red and the other, white. What is the probability that they come from the second box?		4					
102	1	A container has 4 red, 4 blue, and 4 black balls. Three balls are drawn at random with replacement. Find the probability that the three drawn, (1) are red (2) consist of exactly two blue, and (3) contain no black ball.		3	LJU-2025				
103	1	A class consists of 6 girls and 10 boys. If a committee of three is chosen at random from the class, find the probability that, (i) three boys are selected; (ii) exactly two girls are selected.		3					
104	1	In producing screws, let A mean “screw too slim” and B “screw too short”. Let $P(A) = 0.1$ and let the conditional probability that a slim screw is also too small be $P(B/A) = 0.2$. What is the probability that the screw that we pick randomly from the lot produced will be both too slim and too short?		3					

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		Note: This practice book is only for reference purpose. LJU Test question paper may not be completely set from this practice book.							
Sr. No.	unit_number	Question text	Answer	Marks	Previous Year	Option 1	Option 2	Option 3	Option 4
105	1	Define conditional probability. A bag contains 19 tickets numbered from 1 to 19. Two tickets are drawn successively without replacement. Find the probability that both tickets will show even number?		3					
106	1	If A and B are independent events with $P(A) = 0.26$, and $P(B) = 0.45$, find (a) $P(A \cap B)$; (b) $P(A \cap \bar{B})$; (c) $P(\bar{A} \cap \bar{B})$.		3					
107	1	A room has three lamp sockets. From a collection of 10 light bulbs of which only 6 are good. A person selects 3 at random and puts them in the socket. What is the probability that the room will have light?		3					
108	1	Data on readership of a certain magazine show that the proportion of male readers under 35 is 0.40 and that over 35 is 0.20. if the proportion of readers under 35 is 0.70. find the probability of subscribers that are females over 35 years. Also, calculate the probability that a randomly selected male subscriber is under 35 years of age.		3	LJU-2024				
109	1	The completion of a construction job may be delayed due to strike. The probability of strike is 0.6. the probability that the construction job gets completed on time if there is no strike is 0.85 and the probability that the construction job gets completed on time if there is a strike is 0.35. what is the probability that the construction job will not be completed on time?		3	LJU-2024				
110	1	If 3 balls are “randomly drawn” from a bowl containing 6 white and 5 black balls. What is the probability that one of the balls is white and the other two black?		3					
111	1	A microchip company has two machines that produce the chips. Machine I produce 65% of the chips, but 5% of its chips are defective. Machine II produces 35% of the chips and 15% of its chips are defective. A chip is selected at random and found to be defective. What is the probability that it came from Machine I?		4					
112	1	A factory has two machines, A and B. Past records show that the machine A produce 30% of the total output and the machine B, the remaining 70%. Machine A produces 5% of defective items and Machine B produce 1% defective items. An item is drawn at random and found to be defective. What is the probability that it was produced by (i) the machine A, and (ii) the machine B?		3	LJU-2022				
113	1	A businessman goes to hotels X, Y, Z for 20%, 50%, 30% of the time respectively. It is known that 5%, 4%, 8% of the rooms in X, Y, Z hotels have faulty plumbing. What is the probability that the businessman’s room having faulty plumbing is assigned to (1) Hotel X (2) Hotel Y (3) Hotel Z		3					
114	1	Four cards are drawn from a pack of cards. Find the probability that (i) all are diamonds, (ii) there is one card of each suit, (iii) there are two spades and two hearts.		3					
115	1	If 6 of 18 new buildings in a city violate the building code, what is the probability that a building		4					

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Sr. No.	unit_number	Question text	Answer	Marks	Previous Year	Option 1	Option 2	Option 3	Option 4
		inspector, who randomly selects 4 of the new buildings for inspection, will catch (i) None, (ii) One, (iii) at least 3, of the new buildings that violate the building code?							
116	1	There are 5 yellow, 2 red and 5 white balls in the box. Three balls are randomly selected from the box. Find the probability of the following events. (1) All are of different colors (2) 2 yellow and 1 red color (3) All are of same color		4					
117	1	A problem in statistics is given to two students <i>A</i> and <i>B</i> . The odds in favour of <i>A</i> solving the problem are 6 to 9 and against <i>B</i> solving the problem are 12 to 10. If both <i>A</i> and <i>B</i> attempt, find the probability of the problem being solved.		3					
118	1	In a class of 75 students, 15 were considered to be very intelligent, 45 as medium and the rest below average. The probability that a very intelligent student fails in a viva-voce examination is 0.005; the medium student failing has a probability 0.05; and the corresponding probability for a below average student is 0.15. If a student is known to have passed the viva-voce examination, what is the probability that he is below average?		3	LJU-2023				
119	1	A problem in statistics is given to three students A, B and C, whose chances of solving it independently are 1/2, 1/3 and 1/4 respectively. Find the probability that (i) the problem is solved (ii) at least two of them are able to solve the problem (iii) exactly two of them are able to solve the problem		3	LJU-2022 LJU-2024				
120	1	Suppose we have 3 cards identical in form except that both sides of the first card are colored red, both sides of the second card are colored black, and one side of the third card is colored red and the other side is colored black. The 3 cards are mixed up in a hat, and 1 card is randomly selected and put down on the ground. If the upper side of the chosen card is colored red, what is the probability that the other side is colored black?		3					
121	1	In a group of 1000 persons, there are 650 who can speak Hindi, 400 can speak English, and 150 can speak both Hindi and English. If a person selected at random, what is the probability that he speaks (1) Hindi only, (2) English only, (3) at least one of the two languages?		3					
122	1	(i) An unbiased coin is tossed 3 times. What is the probability of obtaining two heads? (ii) A 4-sided fair die is thrown twice. What is the probability that the sum of the two outcomes is equal to 6?		2					
123	1	A problem in statistics is given to three students A, B and C. Whose chances of solving are $\frac{1}{2}$, $\frac{3}{4}$ and $\frac{1}{4}$ respectively. What is the probability that the problem will be solved if all of them try independently?		3					
124	2	A random variable is also called	Chance Variable	1		Constant	Variable	Attribute	Chance Variable

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Sr. No.	unit_ number	Question text	Answer	Marks	Previous Year	Option 1	Option 2	Option 3	Option 4														
125	2	Suppose, four coins are tossed, the value of a random variable H (No. of heads) is:	0,1,2,3,4	1		1,2,3,4	0,1,2,3,4	0,1,2,3	0,1														
126	2	The sum of probabilities of a discrete random variable is	One	1		Zero	Four	Three	One														
127	2	A random variable assuming only a finite number of values is called:	Discrete random v ariable	1		Discrete random varia ble	Continuous random varia ble	Random var iable	None of these														
128	2	A random variable assuming an infinite number of values is called	Continuo us random v ariable	1		Absolute variable	Discrete random varia ble	Continuous random vari able	None of these														
129	2	If X and Y are random variables then E(X+Y) is equal to	E(X)+ E(Y)	1		E(X)+E(Y)	E(X)-E(Y)	E(X)+Y	None of these														
130	2	A variable which can assume each and every value within a given range is called	Continuo us variabl e	1		Discrete vari able	Random vari able	Qualitative variable	Continuous va riable														
131	2	A random variable X has the Probability density function given by $f(x) = kx^2 + x, 0 \leq x \leq 1$. The value of k is	3/2	1		3/2	2	2/3	1														
132	2	A continuous random variable X has the pdf $f(x) = \begin{cases} mx^2, & 0 < x < 3 \\ 0, & otherwise \end{cases}$ then find m.	1/9	1	LJU-2024	-3/9	1/9	2/9	-2/9														
133	2	A fair die is tossed thrice. If the probabilities of zero, one, two and three successes are 8/27, 4/9, 2/9 and 1/27 respectively. Find the mean of the number of successes	1	1	LJU-2022	1/2	1	2/3	3/4														
134	2	The probability distribution of a random variable X is given by <table border="1"><tr><td>X</td><td>-2</td><td>-1</td><td>0</td><td>1</td><td>2</td><td>3</td></tr><tr><td>P(X =x)</td><td>0.1</td><td>0.1</td><td>0.2</td><td>0.2</td><td>0.3</td><td>0.1</td></tr></table> Find mean.	X	-2	-1	0	1	2	3	P(X =x)	0.1	0.1	0.2	0.2	0.3	0.1	0.8	1		0	0.2	0.8	1
X	-2	-1	0	1	2	3																	
P(X =x)	0.1	0.1	0.2	0.2	0.3	0.1																	
135	2	For which value of k the function $P(X = x) = k(x^2 + 1), x = 0, 1, 2, 3$ can be considered as probability mass function	1/18	1	LJU-2025	1/9	1/18	2/9	-1/18														
136	2	Find the value of the λ such that the function f(x) is a valid probability density function $f(x) = \lambda(x - 1)(2 - x), \quad 1 \leq x \leq 2$ $= 0, \text{ Otherwise}$	6	1	LJU-2022	6	5	1	4														
137	2	A random variable X has the Probability density function given by $f(x) = \begin{cases} k(x^2 - 2), & -1 \leq x \leq 3 \\ 0, & otherwise \end{cases}$. Then find the value of k.	None of these	1	LJU-2025	3/28	3/27	3/16	5/28														
138	2	A continuous random variable X has the pdf $f(x) = \frac{k}{1 + x^2}, \quad -\infty < x < \infty$ then find k.	$1/\pi$	1		$1/\pi$	$2/\pi$	$3/\pi$	$\pi/2$														
139	2	Let X be a continuous random variable denoting the temperature measured. The range of temperature is [0, 100] degree Celsius and let the probability density function of x be $f(x) = 0.01$ for $0 \leq x \leq 100$. The mean of X is.	50.0	1		5.0	2.5	25.0	50.0														
140	2	A continuous random variable X has the pdf $f(x) = \begin{cases} m(x^2 - 3x + 2), & 1 < x < 2 \\ 0, & otherwise \end{cases}$ then find m	-6	1	LJU-2025	6	3	-6	-3														
141	2	Find the value of P(X=3). If x is the discrete random variable, Taking values x_1, x_2, x_3 where $P(X = 0) = 0, P(X = 1) = 1/4, P(X = 2) = 1/4$	1/2	1	LJU-2024	1	1/2	1/3	1/4														

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Sr. No.	unit_ number	Question text	Answer	Marks	Previous Year	Option 1	Option 2	Option 3	Option 4												
142	2	What is mean value of pmf $f(x) = \frac{x}{10}$, $x = 0, 1, 2, 3, 4$?	3	1	LJU-2024	1	3	10	25												
143	2	A fair six-sided die is rolled, with X being the number on the uppermost face. The variance of X is...	35/12	1		35/6	35/12	25/12	25/6												
144	2	What will be the value of P (not E) if P(E) = 0.07?	0.93	1		0.90	0.007	0.93	0.093												
145	2	A probability density function f(x) for the continuous random variable X is denoted as _____	$\int f(x)dx = 1,$ $-\infty \leq x \leq \infty$	1		$\int f(x)dx = \infty,$ $-1 \leq x \leq 1$	$\int f(x)dx = 1,$ $-\infty \leq x \leq \infty$	$\int f(x)dx = 0,$ $-\infty \leq x \leq \infty$	$\int f(x + 2)dx = 0.5$ $-\infty \leq x \leq \infty$												
146	2	In a card game Reena wins 3 Rs. if she draws a king or a spade and 7 Rs. if a heart or a queen from a pack of 52 playing cards. If she pays a certain amount of money each time she will lose the game. What will be the amount so that the game will come out a fair game?	8	1		15	6	8	2												
147	2	A Random Variable X can take only two values, 4 and 5 such that P(4) = 0.32 and P(5) = 0.68. Determine the Variance of X.	0.2176	1	LJU-2025	0.8271	0.1236	17.44	0.2176												
148	2	A, B, and C are three mutually and Exhaustive Event P(A)= 2P(B)= 6P(C). Find P(B).	0.3	1	LJU-2022 LJU - 2023	0.1	0.3	0.25	0.4												
149	2	A fair cubical die is thrown twice and their scores summed up. If the sum of the scores of upper sides faces by throwing two times a die is an event. Find the Expected Value of that event.	7	1		48	76	7	132												
150	2	A random variable X can take only two values, 2 and 4 i.e., P(2) = 0.45 and P(4) = 0.55. What is the Expected value of X?	3.1	1	LJU-2024	3.8	3.1	4.78	5.32												
151	2	Mean of a constant ‘a’ is _____	a	1		0	a	a/2	1												
152	2	Variance of a constant ‘a’ is _____	0	1		0	a	a/2	1												
153	2	The probability that it rains tomorrow is 0.72. Find the probability that it does not rain tomorrow?	28%	1		65%	43%	28%	32%												
154	2	If E(x) = 2 and E(z) = 4, then E(z – x) =?	2	1		2	6	0	Insufficient data												
155	2	A table with all possible value of a random variable and its corresponding probabilities is called _____	Probability Distribution	1		Probability Mass Function	Probability Density Function	Cumulative distribution function	Probability Distribution												
156	2	What is the value of $E(2X - 3)$ for given Probability distribution of a Random Variable X is given Below: <table border="1"><tr><td>X</td><td>-2</td><td>-1</td><td>0</td><td>1</td><td>2</td></tr><tr><td>$P(X = x)$</td><td>0.2</td><td>0.1</td><td>0.3</td><td>0.3</td><td>0.1</td></tr></table>	X	-2	-1	0	1	2	$P(X = x)$	0.2	0.1	0.3	0.3	0.1	-3	1	LJU-2025	0	1	0.5	-3
X	-2	-1	0	1	2																
$P(X = x)$	0.2	0.1	0.3	0.3	0.1																
157	2	A coin is tossed up 4 times. The probability that tails turn up in 3 cases is _____	1/4	1		1/2	1/3	1/4	1/6												
158	2	A player tosses two fair coins. He wins 100Rs if a head appears and 200 Rs. If two heads appear. On the other hand, he loses 500 Rs. If no head appears. What is the expected value of the game?	-25 Rs.	1	LJU-2025	25 Rs.	-25 Rs	100 Rs.	200 Rs.												
159	2	Amit plays a game of tossing a dice. If a number less than 3 appears, he gets “a” Rs, otherwise he has to pay 10 Rs.if the game is fair, what is the value of “a”?	20	1		25	20	21	None of these												
160	2	If a random variable X assumes the values 0 and 1 with $P(X = 0) = 3P(X = 1)$, then $V(X)$ is _____	$\frac{3}{16}$	1	LJU-2024	$\frac{1}{16}$	$\frac{3}{16}$	$\frac{5}{16}$	$\frac{4}{16}$												

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Sr. No.	unit_number	Question text	Answer	Marks	Previous Year	Option 1	Option 2	Option 3	Option 4
161	2	The random variables X and Y have variances 0.2 and 0.5 respectively. Let $Z = 5X - 2Y$. The variance of Z is?	7	1	LJU-2023 LJU-2024 LJU-2025	2	5	7	NONE
162	2	A continuous random variable X has pdf $f(x) = 2x, 0 \leq x \leq 1$ (pdf has value 0 for other values of x) then $F(4)$ will be	1	1	LJU-2024	1	4	0	16
163	2	A Joint pmf of x & y is $f(x,y) = \frac{2x+y}{30}, x = 1,2, y = 1,2,3$. What is $P(y \leq 2)$?	3/5	1	LJU-2024	4/15	3/5	3/10	8/15
164	2	A discrete random variable X has the following probability distribution $X = 0,1,2,3,4,5$ $P(X = x) = 0, k, 0.2, 2k, 0.3, 2k$ Find (i) k , (ii) $P(X < 3)$, (iii) $P(X \geq 3)$		4					
165	2	If is the function $f(x)$ defined by $f(x) = \begin{cases} e^{-x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$ Is $f(x)$ probability density function? If so, find the probability that the variate having this density falls in the interval (1,2).		4					
166	2	Find the constant k such that the function $f(x) = \begin{cases} kx^2, & 0 < x < 3 \\ 0, & otherwise \end{cases}$ is a probability density function and compute (i) $P(X < 2)$, (ii) $P(X \geq 2)$		4					
167	2	Is $f(x) = \frac{x}{6}, x = 0,1,2,3,4$ define probability distribution? Justify your answer.		3					
168	2	A machine produces on average of 500 items during the first week of the month and on average of 400 items during the last week of the month. The probability for these being 0.68 and 0.32. Determine the expected value of the production.		3					
169	2	The following table gives the probabilities that a certain computer will malfunction 0, 1, 2, 3, 4, 5 or 6 times on any one day: Number of malfunctions x : 0, 1, 2, 3, 4, 5, 6 Probability $f(x)$: 0.17, 0.29, 0.27, 0.16, 0.07, 0.03, 0.01 Find the mean and variance of this probability distribution.		4					
170	2	The life in hours of a certain kind of radio tube has the probability density $f(x) = \frac{100}{x^2}, \quad for\ x \geq 100$ $= 0, \quad elsewhere,$ find the distribution function and use it to determine the probability that the life of tube is more than 150 hrs.		5	LJU-2023				
171	2	Three coins are tossed to gather and let random variable X be the number of heads in each outcome. Then find (a) Probability distribution, (b) Mean and (c) standard deviation.		3					
172	2	Given that $f(x) = \frac{k}{2^x}$ is probability distribution for a random variable that can take on the values $x = 0, 1, 2, 3, 4$. Find k .		4					
173	2	For the following probability distribution $X = 1,2,3,4,5$ and $f(x) = 0.1, 0.1, 0.2, 0.3, 0.3$ (i) Find the mean and variance. (ii) Find the distribution function.		5					

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Sr. No.	unit_ number	Question text	Answer	Marks	Previous Year	Option 1	Option 2	Option 3	Option 4																		
174	2	The probability distribution of a commodity is given below. <table><tr><td>Dema nd</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td></tr><tr><td>Proba bility</td><td>0.05</td><td>0.10</td><td>0.30</td><td>0.40</td><td>0.10</td><td>0.05</td></tr></table> Find expected demand.	Dema nd	5	6	7	8	9	10	Proba bility	0.05	0.10	0.30	0.40	0.10	0.05		4									
Dema nd	5	6	7	8	9	10																					
Proba bility	0.05	0.10	0.30	0.40	0.10	0.05																					
175	2	The Probability distribution of a random variable x is given below <table><tr><td>x</td><td>-2</td><td>-1</td><td>0</td><td>1</td><td>2</td></tr><tr><td>$p(x)$</td><td>$\frac{1}{12}$</td><td>$\frac{1}{3}$</td><td>p</td><td>$\frac{1}{4}$</td><td>$\frac{1}{6}$</td></tr></table> Find (i) $E(x)$ (ii) $E(2x + 3)$ (iii) $E(5x - 2)$.	x	-2	-1	0	1	2	$p(x)$	$\frac{1}{12}$	$\frac{1}{3}$	p	$\frac{1}{4}$	$\frac{1}{6}$		5											
x	-2	-1	0	1	2																						
$p(x)$	$\frac{1}{12}$	$\frac{1}{3}$	p	$\frac{1}{4}$	$\frac{1}{6}$																						
176	2	Find the expectation for the following discrete probability distribution: <table><tr><td>x</td><td>10</td><td>14</td><td>18</td><td>25</td><td>35</td></tr><tr><td>$p(x)$</td><td>0.125</td><td>0.225</td><td>0.325</td><td>0.200</td><td>0.125</td></tr></table>	x	10	14	18	25	35	$p(x)$	0.125	0.225	0.325	0.200	0.125		3											
x	10	14	18	25	35																						
$p(x)$	0.125	0.225	0.325	0.200	0.125																						
177	2	Find the probability distribution of the number of heads when three coins are tossed.		3																							
178	2	A random variable X has a probability mass function given by <table><tr><td>X</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>$P(X= x)$</td><td>0.1</td><td>0.2</td><td>0.5</td><td>0.2</td></tr></table> Find (i) $P(2 \leq x \leq 4)$, (ii) $P(x \leq 2)$	X	1	2	3	4	$P(X= x)$	0.1	0.2	0.5	0.2		4													
X	1	2	3	4																							
$P(X= x)$	0.1	0.2	0.5	0.2																							
179	2	A random variable X has the probability distribution $X = 0,1,2,3,4,5,6,7$ $P(X = x) = a, 4a, 3a, 7a, 8a, 10a, 6a, 9a$ Find (i) the value of a , (ii) $P(X < 3)$		4																							
180	2	The probability mass function of a random variable x is zero except at the points $X= 0, 1, 2$. At these points it has the values $P(X = 0) = 3c^3$, $P(X = 1) = 4c - 10c^2, P(X = 2) = 5c - 1$ Find (i) c , (ii) $P(X < 1)$, (ii) $P(1 < X \leq 2)$		4																							
181	2	If $x = \{0,1\}$ and $y = \{0,1\}$ be two independent binary random variables. If $P(X = 0) = p$ and $P(Y = 0) = q$. Probability of (i) $P(X + Y \geq 1)$ (ii) $P(X = 1)$ (iii) $P(Y = 1)$		3																							
182	2	A random variable X has the following probability function: <table><tr><td>X</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr><tr><td>$P(X= x)$</td><td>0</td><td>k</td><td>2k</td><td>2k</td><td>3k</td><td>k^2</td><td>$2k^2$</td><td>$7k^2 + k$</td></tr></table> (i) Determine k , (ii) Evaluate $P(X < 6)$, (iii) Find the mean and variance	X	0	1	2	3	4	5	6	7	$P(X= x)$	0	k	2k	2k	3k	k^2	$2k^2$	$7k^2 + k$		3	LJU-2022				
X	0	1	2	3	4	5	6	7																			
$P(X= x)$	0	k	2k	2k	3k	k^2	$2k^2$	$7k^2 + k$																			
183	2	A random variable X has the probability distribution $X = -2, -1, 0, 1, 2$ $P(X = x) = 0.2, 0.1, 0.3, 0.3, 0.1$ Find (i) Mean, (ii) Variance, (iii) $E(2X + 3)$, (iv) $E(2X - 3)$		4																							
184	2	Show that the function $f(x)$ defined by $f(x) = \begin{cases} \frac{1}{7}, & 1 < x < 8 \\ 0, & otherwise \end{cases}$ is a probability density function for a random variable. Hence find $P(3 < x < 10)$		3																							

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		Note: This practice book is only for reference purpose. LJU Test question paper may not be completely set from this practice book.																																			
Sr. No.	unit_number	Question text	Answer	Marks	Previous Year	Option 1	Option 2	Option 3	Option 4																												
185	2	Find the value of k such that f(x) is a probability density function. Also find $P(X < 1.5)$. $f(x) = \begin{cases} kx & ; 0 \leq x \leq 1 \\ k & ; 1 < x \leq 2 \\ k(3 - x) & ; 2 < x \leq 3 \end{cases}$		4																																	
186	2	Let X be a continuous random variable with pdf $f(x) = kx(1 - x), 0 \leq x \leq 1$ Find k and determine a number b such that $P(X \leq b) = P(X > b)$.		3																																	
187	2	Check whether the function $f(x) = \begin{cases} x & , -1 < x < 1 \\ 0 & , otherwise \end{cases}$ Is a probability density function? If yes, find mean and variance.		3																																	
188	2	For the continuous random variable having pdf $f(x) = \begin{cases} k x & , -2 \leq x \leq 2 \\ 0 & , otherwise \end{cases}$ Find mean and variance.		4	LJU-2024																																
189	2	For the continuous random variable having pdf $f(x) = \begin{cases} kx, & 0 < x \leq 1 \\ k(2 - x)^2, & 1 < x \leq 2 \\ 0, & otherwise \end{cases}$ Find k, mean and variance of X.		4	LJU-2025																																
190	2	If the random variable X takes the value 1, 2, 3 and 4 such that $2P(X = 1) = 3P(X = 2) = P(X = 3) = 5P(X = 4)$. Find the probability distribution.		3																																	
191	2	The joint distribution of X and Y is given by $f(x, y) = \frac{x+y}{21}, x = 1,2,3 \text{ and } y = 1,2$ Find the marginal distribution.		4																																	
192	2	Give is the joint distribution of X and Y <table border="1"><tr><td>X \ Y</td><td>0</td><td>1</td><td>2</td></tr><tr><td>0</td><td>0.02</td><td>0.08</td><td>0.10</td></tr><tr><td>1</td><td>0.05</td><td>0.20</td><td>0.25</td></tr><tr><td>2</td><td>0.03</td><td>0.12</td><td>0.15</td></tr></table> Find (i) Marginal distributions (ii) the conditional distributions of X given Y= 0	X \ Y	0	1	2	0	0.02	0.08	0.10	1	0.05	0.20	0.25	2	0.03	0.12	0.15		3	LJU-2022																
X \ Y	0	1	2																																		
0	0.02	0.08	0.10																																		
1	0.05	0.20	0.25																																		
2	0.03	0.12	0.15																																		
193	2	From the following table for the bivariate distribution of (x, y). Find $P(X \leq 1), P(Y \leq 3), P(X \leq 1, Y \leq 3),$ $P(X \leq 1/Y \leq 3), P(Y \leq 3/X \leq 1), P(X + Y \leq 4)$ <table border="1"><tr><td>X \ Y</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>0</td><td>0</td><td>0</td><td>$\frac{1}{32}$</td><td>$\frac{2}{32}$</td><td>$\frac{2}{32}$</td><td>$\frac{3}{32}$</td></tr><tr><td>1</td><td>$\frac{1}{16}$</td><td>$\frac{1}{16}$</td><td>$\frac{1}{8}$</td><td>$\frac{1}{8}$</td><td>$\frac{1}{8}$</td><td>$\frac{1}{8}$</td></tr><tr><td>2</td><td>$\frac{1}{32}$</td><td>$\frac{1}{32}$</td><td>$\frac{1}{64}$</td><td>$\frac{1}{64}$</td><td>0</td><td>$\frac{1}{32}$</td></tr></table>	X \ Y	1	2	3	4	5	6	0	0	0	$\frac{1}{32}$	$\frac{2}{32}$	$\frac{2}{32}$	$\frac{3}{32}$	1	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	2	$\frac{1}{32}$	$\frac{1}{32}$	$\frac{1}{64}$	$\frac{1}{64}$	0	$\frac{1}{32}$		4	LJU-2024 LJU-2025				
X \ Y	1	2	3	4	5	6																															
0	0	0	$\frac{1}{32}$	$\frac{2}{32}$	$\frac{2}{32}$	$\frac{3}{32}$																															
1	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$																															
2	$\frac{1}{32}$	$\frac{1}{32}$	$\frac{1}{64}$	$\frac{1}{64}$	0	$\frac{1}{32}$																															
194	2	A random variable X has the following distribution $X = 1, 2, 3, 4, 5, 6$ $P(X = x) = \frac{1}{36}, \frac{3}{36}, \frac{5}{36}, \frac{7}{36}, \frac{9}{36}, \frac{11}{36}$ Find (i) mean, (ii) variance, (iii) $P(1 < X < 6)$		3																																	
195	2	A random variable X takes the values $-3, -2, -1, 0, 1, 2, 3$ such that $P(X = 0) = P(X > 0) = P(X < 0), P(X = -3) = P(X = -2) = P(X = -1) = P(X = 1) = P(X = 2) =$		3																																	

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		Note: This practice book is only for reference purpose. LJU Test question paper may not be completely set from this practice book.																																
Sr. No.	unit_ number	Question text	Answer	Marks	Previous Year	Option 1	Option 2	Option 3	Option 4																									
		$P(X = 3)$. Obtain the probability distribution and distribution function of X.																																
196	2	The probability density function of a continuous random variable X is given by $f(x) = \begin{cases} ax & 0 \leq x \leq 1 \\ a & 1 < x \leq 2 \\ 3a - ax & 2 < x \leq 3 \\ 0 & \text{otherwise} \end{cases}$ (i) Find the value of a, and (ii) Find the CDF of X		4																														
197	2	The PDF of a continuous random variable X is $f(x) = \frac{1}{2}e^{- x }$ Find CDF F(x).		3																														
198	2	A man draws two balls from a bag containing 3 white and 5 black balls. If he is to receive 14 Rs for every white ball which he draws and 7 Rs for every black balls, what is his expectations.		4																														
199	2	For the continuous random variable having PDF $f(x) = \begin{cases} 4x^3 & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$ Find mean and variance of X.		4																														
200	2	For the continuous random variable having pdf $f(x) = \begin{cases} x, & 0 < x \leq 1 \\ 2 - x, & 1 < x \leq 2 \\ 0, & \text{otherwise} \end{cases}$ Find mean and variance of X.		3	LJU-2022 LJU-2024 LJU-2025																													
201	2	The joint probability distribution of two random variables X and Y is given below: <table border="1"><tr><td></td><td>Y</td><td>1</td><td>2</td><td>3</td></tr><tr><td>X</td><td></td><td></td><td></td><td></td></tr><tr><td>2</td><td></td><td>1/8</td><td>1/24</td><td>1/12</td></tr><tr><td>4</td><td></td><td>1/4</td><td>1/4</td><td>0</td></tr><tr><td>6</td><td></td><td>1/8</td><td>1/24</td><td>1/12</td></tr></table> Find $P(X < 6)$, $P(Y > 1)$, $P(X < 4/Y > 1)$, $P(2 \leq X \leq 5, Y > 1)$, $P(Y = 3/X = 2)$, $P(Y = 2/X = 1)$, Marginal PMF of X and Y.		Y	1	2	3	X					2		1/8	1/24	1/12	4		1/4	1/4	0	6		1/8	1/24	1/12		4	LJU-2024 LJU-2025				
	Y	1	2	3																														
X																																		
2		1/8	1/24	1/12																														
4		1/4	1/4	0																														
6		1/8	1/24	1/12																														
202	2	Consider Two random Variables X and Y With Joint PMF given in the table. (1) Find $P(X \leq 2, Y \leq 4)$ (2) Find the Marginal PMFs of X and Y. (3) Find $P(Y = 2 X = 1)$ <table border="1"><tr><td></td><td>Y=2</td><td>Y=4</td><td>Y= 5</td></tr><tr><td>X=1</td><td>$\frac{1}{12}$</td><td>$\frac{1}{24}$</td><td>$\frac{1}{24}$</td></tr><tr><td>X= 2</td><td>$\frac{1}{6}$</td><td>$\frac{1}{12}$</td><td>$\frac{1}{8}$</td></tr><tr><td>X= 3</td><td>$\frac{1}{4}$</td><td>$\frac{1}{8}$</td><td>$\frac{1}{12}$</td></tr></table>		Y=2	Y=4	Y= 5	X=1	$\frac{1}{12}$	$\frac{1}{24}$	$\frac{1}{24}$	X= 2	$\frac{1}{6}$	$\frac{1}{12}$	$\frac{1}{8}$	X= 3	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{12}$		4														
	Y=2	Y=4	Y= 5																															
X=1	$\frac{1}{12}$	$\frac{1}{24}$	$\frac{1}{24}$																															
X= 2	$\frac{1}{6}$	$\frac{1}{12}$	$\frac{1}{8}$																															
X= 3	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{12}$																															
203	2	From the following joint distribution of X and Y. Find the marginal distribution. <table border="1"><tr><td>X \ Y</td><td>0</td><td>1</td><td>2</td></tr><tr><td>0</td><td>3/28</td><td>9/28</td><td>3/28</td></tr><tr><td>1</td><td>3/14</td><td>3/14</td><td>0</td></tr><tr><td>2</td><td>1/28</td><td>0</td><td>0</td></tr></table>	X \ Y	0	1	2	0	3/28	9/28	3/28	1	3/14	3/14	0	2	1/28	0	0		4														
X \ Y	0	1	2																															
0	3/28	9/28	3/28																															
1	3/14	3/14	0																															
2	1/28	0	0																															

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Sr. No.	unit_ number	Question text	Answer	Marks	Previous Year	Option 1	Option 2	Option 3	Option 4																
204	2	The joint probability distribution of two random variables X and Y is given below: <table border="1"><tr><td>X \ Y</td><td>0</td><td>1</td><td>2</td></tr><tr><td>2</td><td>1/20</td><td>1/12</td><td>1/5</td></tr><tr><td>4</td><td>1/15</td><td>1/k</td><td>1/10</td></tr><tr><td>6</td><td>1/12</td><td>1/6</td><td>7/60</td></tr></table> Find (i) k, (ii) $P(0 < X \leq 1)$ and (iii) $P(Y > 2 / X \leq 1)$	X \ Y	0	1	2	2	1/20	1/12	1/5	4	1/15	1/k	1/10	6	1/12	1/6	7/60		4	LJU-2023				
X \ Y	0	1	2																						
2	1/20	1/12	1/5																						
4	1/15	1/k	1/10																						
6	1/12	1/6	7/60																						
205	3	If mean and mode of some data are 4 and 10 respectively, its median will be:	6	1		1.5	5.4	16	6																
206	3	Find the median of the given set of numbers 2, 6, 6, 8, 4, 2, 7, 9.	6	1		6	8	4	5																
207	3	Mean of observations 5, 7, 8, 2, 6, x, 7, 9, 2, 5 is 5.4 then x is ____.	3	1	LJU-2025	1	2	3	4																
208	3	If the median of the distribution 6, 8, 10, 16, x, 22, 25, 30 is 18, then the value of x is,	20	1	LJU-2025	14	16	18	20																
209	3	Find the median of the data set: 6,3,8,2,9,1.	4.5	1		4.5	5.5	6	5																
210	3	A variable X takes values 2,9,3,7,5,4,3,2,10. What is the median?	4	1		2	4	7	8																
211	3	The mode of a distribution is 24 and the mean is 60. What is its median?	48	1	LJU-2024	50	48	45	51																
212	3	The following observations 8, 11, 12, x + 6, 17, 18, 23 are arranged in ascending order. The median of the data is 15. Find the value of x.	9	1		15	9	11	None of these																
213	3	The following table shows the marks of 130 students of class 10. Median of given data is: <table border="1"><tr><td>Mar ks</td><td>20-30</td><td>30-40</td><td>40-50</td><td>50-60</td><td>60-70</td><td>70-80</td></tr><tr><td>No. of Can did ates</td><td>0</td><td>4</td><td>18</td><td>60</td><td>33</td><td>15</td></tr></table>	Mar ks	20-30	30-40	40-50	50-60	60-70	70-80	No. of Can did ates	0	4	18	60	33	15	57.16	1		61.11	57.16	47.47	54.17		
Mar ks	20-30	30-40	40-50	50-60	60-70	70-80																			
No. of Can did ates	0	4	18	60	33	15																			
214	3	If the median of the distribution 10, 12, 13, 16, x, 20, 25, 30 is 18, then the value of x is,	20	1		24	22	23	20																
215	3	If the median of the data 13, 17, (p+1), (p+5), 24, 27 is 21, then p is equal to	18	1	LJU-2024	18	19	20	21																
216	3	The median of the following observations 46, 64, 87, 41, 58, 77, 35, 90, 55, 92, 33 is 58. If 92 is replaced by 99 and 41 by 43 in the above data. The new median is,	58	1		56	61	58	49																
217	3	The median of the set {x + 1, x + 2, x + 3, x + 4, x + 5, x + 6} is,	x + 3.5	1	LJU-2024 LJU-2025	x + 3	x + 4	x + 3.5	x + 4.5																
218	3	The median of a set of 9 distinct observations is 20.5. If each of the largest 4 observation of the set is increased by 2, then the median of the new set	remains the same	1		is increased by 2	is decreased by 2	is two times the original median	remains the same																
219	3	For the following distribution, N = 100 and median = 32. The values of x and y respectively are, <table border="1"><tr><td>Class</td><td>0-10</td><td>10-20</td><td>20-30</td><td>30-40</td><td>40-50</td><td>50-60</td><td>T o ta l</td></tr><tr><td>Frequ ency</td><td>10</td><td>x</td><td>25</td><td>30</td><td>y</td><td>10</td><td>100</td></tr></table>	Class	0-10	10-20	20-30	30-40	40-50	50-60	T o ta l	Frequ ency	10	x	25	30	y	10	100	9, 16	1		10, 15	9,16	11,14	8, 17
Class	0-10	10-20	20-30	30-40	40-50	50-60	T o ta l																		
Frequ ency	10	x	25	30	y	10	100																		

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Sr. No.	unit_ number	Question text	Answer	Marks	Previous Year	Option 1	Option 2	Option 3	Option 4																		
220	3	Find the mode of the following distribution: <table border="1"><tr><td>Cla ss</td><td>0- 20</td><td>20- 40</td><td>40- 60</td><td>60- 80</td><td>80- 100</td><td>100 -</td><td>120 -</td><td>140</td></tr><tr><td>f</td><td>6</td><td>8</td><td>10</td><td>12</td><td>6</td><td>5</td><td>3</td><td></td></tr></table>	Cla ss	0- 20	20- 40	40- 60	60- 80	80- 100	100 -	120 -	140	f	6	8	10	12	6	5	3		65	1	LJU-2022 LJU-2025	60	65	70	75
Cla ss	0- 20	20- 40	40- 60	60- 80	80- 100	100 -	120 -	140																			
f	6	8	10	12	6	5	3																				
221	3	Runs scored by batsman in 5 one day matches are 50, 60, 42, 68, and 20. The standard deviation is	16.54	1	LJU-2025	4.07	16.54	25.79	12.54																		
222	3	The mode of the following frequency distribution is 36. Find the missing frequency f. <table border="1"><tr><td>Cl ass</td><td>0- 10</td><td>10- 20</td><td>20- 30</td><td>30- 40</td><td>40- 50</td><td>50- 60</td><td>60- 70</td></tr><tr><td>f</td><td>8</td><td>10</td><td>f</td><td>16</td><td>12</td><td>6</td><td>7</td></tr></table>	Cl ass	0- 10	10- 20	20- 30	30- 40	40- 50	50- 60	60- 70	f	8	10	f	16	12	6	7	10	1		12	16	10	9		
Cl ass	0- 10	10- 20	20- 30	30- 40	40- 50	50- 60	60- 70																				
f	8	10	f	16	12	6	7																				
223	3	What is the mode of the observations 1, 2, 3, 5, 7, 5, 9, 5, 10, 10.	5	1		5	10	1	9																		
224	3	For a distribution with mean, mode and standard deviation 25, 26 and 5 respectively, Karl Pearson’s coefficient of skewness equals to:	−0.20	1		−0.20	0.20	1	−1																		
225	3	For four observations -1, 0, 1, 4, the measure of kurtosis equals	2	1		1	2	3	4																		
226	3	If the standard deviation of 0, 1, 2, 3,...,9 is k, then the standard deviation of 10, 11, 12, 13,...,19 will be	k	1		k+1	k+2	k	k+10																		
227	3	The mean and variance of 10 observations are given to be 4 and 2 respectively. If every observation is multiplied by 2, the mean and the variance of the new series will be respectively.	8 & 8	1		8 & 8	8 & 20	8 & 4	80 & 40																		
228	3	Runs scored by batsman in 5 one day matches are 50, 70, 82, 93, and 20. The standard deviation is _____	25.79	1	LJU-2024	25.79	25.49	25.29	25.69																		
229	3	The standard deviation of 1, 4, 7, 2, 6 is	2.28	1		3.38	0	2.28	1																		
230	3	The first four moments of a distribution about the value 4 of the variables are 1,4,10 and 45. What is the third moment about mean (μ_3)	0	1	LJU-2025	1	4	2	0																		
231	3	For a moderately skewed distribution of retail price for men’s shoes, it is found that the mean price is 20 Rs. and the median price is 17 Rs. If the coefficient of variation is 20%, then the Pearson’s coefficient of skewness is _____.	2.25	1	LJU-2023	1.5	2.25	1.75	4																		
232	3	From the marks scored by 100 students in Section A and 100 students in Section B of a class, the following measures were obtained: <table border="1"><tr><td>Section A</td><td>$\mu_A = 55$</td><td>$\sigma_A = 15.4$</td><td>Mode = 58.72</td></tr><tr><td>Section B</td><td>$\mu_B = 53$</td><td>$\sigma_A = 15.4$</td><td>Mode = 48.83</td></tr></table> Determine which distribution of marks is more skewed.	Section A	$\mu_A = 55$	$\sigma_A = 15.4$	Mode = 58.72	Section B	$\mu_B = 53$	$\sigma_A = 15.4$	Mode = 48.83		3															
Section A	$\mu_A = 55$	$\sigma_A = 15.4$	Mode = 58.72																								
Section B	$\mu_B = 53$	$\sigma_A = 15.4$	Mode = 48.83																								
233	3	The average grade of male students in the class was 6.2 and that of females was 7.3. The mean grade of all the students was 6.53. Find the percentage of male and female students.		3																							
234	3	Find the arithmetic mean for the following data: <table border="1"><tr><td>x</td><td>35</td><td>45</td><td>55</td><td>60</td><td>75</td><td>80</td></tr><tr><td>f</td><td>12</td><td>18</td><td>10</td><td>6</td><td>3</td><td>11</td></tr></table>	x	35	45	55	60	75	80	f	12	18	10	6	3	11		3									
x	35	45	55	60	75	80																					
f	12	18	10	6	3	11																					

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Sr. No.	unit_ number	Question text	Answer	Marks	Previous Year	Option 1	Option 2	Option 3	Option 4																				
235	3	Find the average wages for construction of the building from the wages paid to different workers. <table><tr><td>Wages:</td><td>100</td><td>200</td><td>300</td><td>400</td><td>500</td></tr><tr><td>No.of workers:</td><td>3</td><td>5</td><td>6</td><td>9</td><td>2</td></tr></table>	Wages:	100	200	300	400	500	No.of workers:	3	5	6	9	2		3													
Wages:	100	200	300	400	500																								
No.of workers:	3	5	6	9	2																								
236	3	Find the arithmetic mean of marks from the following data: <table><tr><td>Marks:</td><td>0 – 10</td><td>10 – 20</td><td>20 – 30</td><td>30 – 40</td><td>40 – 50</td><td>50 – 60</td></tr><tr><td>No. of student</td><td>12</td><td>18</td><td>27</td><td>20</td><td>15</td><td>8</td></tr></table>	Marks:	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60	No. of student	12	18	27	20	15	8		3											
Marks:	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60																							
No. of student	12	18	27	20	15	8																							
237	3	The daily earnings (in rupees) of employees working on a daily basis in a firm are <table><tr><td>Daily earnings (in rupees)</td><td>100</td><td>120</td><td>140</td><td>160</td><td>180</td><td>200</td><td>220</td></tr><tr><td>Number of employees</td><td>3</td><td>6</td><td>10</td><td>15</td><td>24</td><td>42</td><td>75</td></tr></table> Calculate the mean of daily earnings.	Daily earnings (in rupees)	100	120	140	160	180	200	220	Number of employees	3	6	10	15	24	42	75		3									
Daily earnings (in rupees)	100	120	140	160	180	200	220																						
Number of employees	3	6	10	15	24	42	75																						
238	3	Calculate mean for the following frequency distribution: <table><tr><td>Classes:</td><td>0 – 8</td><td>8 – 16</td><td>16 – 24</td><td>24 – 32</td><td>32 – 40</td><td>40 – 48</td></tr><tr><td>Frequency</td><td>8</td><td>7</td><td>16</td><td>24</td><td>15</td><td>7</td></tr></table>	Classes:	0 – 8	8 – 16	16 – 24	24 – 32	32 – 40	40 – 48	Frequency	8	7	16	24	15	7		3											
Classes:	0 – 8	8 – 16	16 – 24	24 – 32	32 – 40	40 – 48																							
Frequency	8	7	16	24	15	7																							
239	3	The following table gives the distribution of companies according to size of capital. Find the mean size of capital of a company. <table><tr><td>Capital (Rs. in lacs):</td><td>< 5</td><td>< 10</td><td>< 15</td><td>< 20</td><td>< 25</td><td>< 30</td></tr><tr><td>No. of companies:</td><td>20</td><td>27</td><td>29</td><td>38</td><td>48</td><td>53</td></tr></table>	Capital (Rs. in lacs):	< 5	< 10	< 15	< 20	< 25	< 30	No. of companies:	20	27	29	38	48	53		4											
Capital (Rs. in lacs):	< 5	< 10	< 15	< 20	< 25	< 30																							
No. of companies:	20	27	29	38	48	53																							
240	3	The following table gives the distribution of marks by 60 students in a mathematics test <table><tr><td>Marks</td><td>More than 0</td><td>More than 10</td><td>More than 20</td><td>More than 30</td><td>More than 40</td><td>More than 50</td></tr><tr><td>No. of students:</td><td>60</td><td>56</td><td>40</td><td>20</td><td>10</td><td>3</td></tr></table> Find Mean.	Marks	More than 0	More than 10	More than 20	More than 30	More than 40	More than 50	No. of students:	60	56	40	20	10	3		4											
Marks	More than 0	More than 10	More than 20	More than 30	More than 40	More than 50																							
No. of students:	60	56	40	20	10	3																							
241	3	Find the median of the data 2, 8, 4, 6, 10, 12, 4, 8, 14, 16.		2																									
242	3	Obtain the Median for the following distribution. <table><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td></tr><tr><td>f</td><td>8</td><td>10</td><td>11</td><td>16</td><td>20</td><td>25</td><td>15</td><td>9</td><td>6</td></tr></table>	x	1	2	3	4	5	6	7	8	9	f	8	10	11	16	20	25	15	9	6		2					
x	1	2	3	4	5	6	7	8	9																				
f	8	10	11	16	20	25	15	9	6																				

		L.J Institute of Engineering and Technology, Ahmedabad. <u>Introduction to Probability Theory and Stochastic Processes</u> Practice Book (Sem- III)																									
		Note: This practice book is only for reference purpose. LJU Test question paper may not be completely set from this practice book.																									
Sr. No.	unit_ number	Question text	Answer	Marks	Previous Year	Option 1	Option 2	Option 3	Option 4																		
243	3	Wages earned in Rupees per day by the labourers are given the table: <table><tr><td>Wag es in Rs.</td><td>10-20</td><td>20-30</td><td>30-40</td><td>40-50</td><td>50-60</td></tr><tr><td>No. of Lab.</td><td>5</td><td>8</td><td>13</td><td>10</td><td>8</td></tr></table> Find the median of the distribution.	Wag es in Rs.	10-20	20-30	30-40	40-50	50-60	No. of Lab.	5	8	13	10	8		3											
Wag es in Rs.	10-20	20-30	30-40	40-50	50-60																						
No. of Lab.	5	8	13	10	8																						
244	3	The following data represents the number of foreign visitors in a multinational company in every 10 days during last 2 months. Use the data to find median. <table><tr><td>X</td><td>0-10</td><td>10-20</td><td>20-30</td><td>30-40</td><td>40-50</td><td>50-60</td></tr><tr><td>No. of Visit ors</td><td>12</td><td>18</td><td>27</td><td>20</td><td>17</td><td>06</td></tr></table>	X	0-10	10-20	20-30	30-40	40-50	50-60	No. of Visit ors	12	18	27	20	17	06		3									
X	0-10	10-20	20-30	30-40	40-50	50-60																					
No. of Visit ors	12	18	27	20	17	06																					
245	3	Find the missing frequency when median is 24. <table><tr><td>Marks</td><td>0-10</td><td>10-20</td><td>20-30</td><td>30-40</td><td>40-50</td></tr><tr><td>No. of Stude nts</td><td>15</td><td>20</td><td>x</td><td>14</td><td>16</td></tr></table>	Marks	0-10	10-20	20-30	30-40	40-50	No. of Stude nts	15	20	x	14	16		3											
Marks	0-10	10-20	20-30	30-40	40-50																						
No. of Stude nts	15	20	x	14	16																						
246	3	The following table gives the marks obtained by 50 students in mathematics. Find the median. <table><tr><td>Marks</td><td>10-14</td><td>15-19</td><td>20-24</td><td>25-29</td><td>30-34</td><td>35-39</td><td>40-44</td><td>45-49</td></tr><tr><td>No. of student</td><td>4</td><td>6</td><td>10</td><td>5</td><td>7</td><td>3</td><td>9</td><td>6</td></tr></table>	Marks	10-14	15-19	20-24	25-29	30-34	35-39	40-44	45-49	No. of student	4	6	10	5	7	3	9	6		3					
Marks	10-14	15-19	20-24	25-29	30-34	35-39	40-44	45-49																			
No. of student	4	6	10	5	7	3	9	6																			
247	3	The following tables gives the distribution of daily wages of 900 workers. However, the frequencies of classes 40 – 50 and 60 – 70 are missing. If the median of the distribution is 59.25 Rs, find the missing frequencies. <table><tr><td>Wage s (in Rs.)</td><td>30-40</td><td>40-50</td><td>50-60</td><td>60-70</td><td>70-80</td></tr><tr><td>No. of Work ers</td><td>120</td><td>?</td><td>200</td><td>?</td><td>185</td></tr></table>	Wage s (in Rs.)	30-40	40-50	50-60	60-70	70-80	No. of Work ers	120	?	200	?	185		4											
Wage s (in Rs.)	30-40	40-50	50-60	60-70	70-80																						
No. of Work ers	120	?	200	?	185																						
248	3	Find the value of X & Y of following data whose median is 46 and total frequency is 229. <table><tr><td>Cla ss</td><td>10-20</td><td>20-30</td><td>30-40</td><td>40-50</td><td>50-60</td><td>60-70</td><td>70-80</td></tr><tr><td>Fre que ncy</td><td>12</td><td>30</td><td>X</td><td>65</td><td>Y</td><td>25</td><td>18</td></tr></table>	Cla ss	10-20	20-30	30-40	40-50	50-60	60-70	70-80	Fre que ncy	12	30	X	65	Y	25	18		4							
Cla ss	10-20	20-30	30-40	40-50	50-60	60-70	70-80																				
Fre que ncy	12	30	X	65	Y	25	18																				
249	3	Find the mean and median of the following data: <table><tr><td>Clas ses</td><td>0-30</td><td>30-60</td><td>60-90</td><td>90-120</td><td>120-150</td><td>150-180</td></tr><tr><td>f</td><td>8</td><td>13</td><td>22</td><td>27</td><td>18</td><td>7</td></tr></table>	Clas ses	0-30	30-60	60-90	90-120	120-150	150-180	f	8	13	22	27	18	7		4									
Clas ses	0-30	30-60	60-90	90-120	120-150	150-180																					
f	8	13	22	27	18	7																					

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		Note: This practice book is only for reference purpose. LJU Test question paper may not be completely set from this practice book.																																																													
Sr. No.	unit_ number	Question text	Answer	Marks	Previous Year	Option 1	Option 2	Option 3	Option 4																																																						
250	3	Find the mode of the following frequency distribution. <table><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>f</td><td>4</td><td>7</td><td>10</td><td>8</td></tr></table>	x	1	2	3	4	f	4	7	10	8		2																																																	
x	1	2	3	4																																																											
f	4	7	10	8																																																											
251	3	Find the mode for the following distribution: <table><tr><td>Class interv al</td><td>0-10</td><td>10-20</td><td>20-30</td><td>30-40</td><td>40-50</td></tr><tr><td>Frequ ency</td><td>45</td><td>20</td><td>14</td><td>7</td><td>3</td></tr></table>	Class interv al	0-10	10-20	20-30	30-40	40-50	Frequ ency	45	20	14	7	3		3	LJU-2024																																														
Class interv al	0-10	10-20	20-30	30-40	40-50																																																										
Frequ ency	45	20	14	7	3																																																										
252	3	Find the mode for the following distribution: <table><tr><td>Cla ss</td><td>0-10</td><td>10-20</td><td>20-30</td><td>30-40</td><td>40-50</td><td>50-60</td><td>60-70</td></tr><tr><td>F</td><td>4</td><td>7</td><td>8</td><td>12</td><td>25</td><td>18</td><td>10</td></tr></table>	Cla ss	0-10	10-20	20-30	30-40	40-50	50-60	60-70	F	4	7	8	12	25	18	10		3																																											
Cla ss	0-10	10-20	20-30	30-40	40-50	50-60	60-70																																																								
F	4	7	8	12	25	18	10																																																								
253	3	An incomplete frequency distribution is given as follow: <table><tr><td>C</td><td>0-40</td><td>40-80</td><td>80-120</td><td>120-160</td><td>160-200</td><td>200-240</td><td>240-280</td><td>280-320</td></tr><tr><td>l</td><td>40</td><td>80</td><td>120</td><td>160</td><td>200</td><td>240</td><td>280</td><td>320</td></tr><tr><td>a</td><td>0</td><td>80</td><td>120</td><td>160</td><td>200</td><td>240</td><td>280</td><td>320</td></tr><tr><td>s</td><td></td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>s</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>f</td><td>14</td><td>22</td><td>x</td><td>124</td><td>y</td><td>32</td><td>15</td><td>5</td></tr></table> Given that the mode value is 1376, and frequency total is 360. Calculate the missing frequencies.	C	0-40	40-80	80-120	120-160	160-200	200-240	240-280	280-320	l	40	80	120	160	200	240	280	320	a	0	80	120	160	200	240	280	320	s		0	0	0	0	0	0	0	s									f	14	22	x	124	y	32	15	5		4					
C	0-40	40-80	80-120	120-160	160-200	200-240	240-280	280-320																																																							
l	40	80	120	160	200	240	280	320																																																							
a	0	80	120	160	200	240	280	320																																																							
s		0	0	0	0	0	0	0																																																							
s																																																															
f	14	22	x	124	y	32	15	5																																																							
254	3	The frequency distribution of marks obtained by 60 students of a class in a college is given by <table><tr><td>Marks:</td><td>30-34</td><td>35-39</td><td>40-44</td><td>45-49</td><td>50-54</td><td>55-59</td><td>60-64</td></tr><tr><td>frequ ency:</td><td>3</td><td>5</td><td>12</td><td>18</td><td>14</td><td>6</td><td>2</td></tr></table> Find the mode of the distribution.	Marks:	30-34	35-39	40-44	45-49	50-54	55-59	60-64	frequ ency:	3	5	12	18	14	6	2		3																																											
Marks:	30-34	35-39	40-44	45-49	50-54	55-59	60-64																																																								
frequ ency:	3	5	12	18	14	6	2																																																								
255	3	Following data related to the number of telegraphic transfers per day by a bank branch for 300 working days <table><tr><td>No . of tel egr ap hic tra nsf er per da y</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr><tr><td>Nu mb er of da ys</td><td>10</td><td>35</td><td>45</td><td>95</td><td>64</td><td>32</td><td>10</td><td>9</td></tr></table> calculate mean, mode and median.	No . of tel egr ap hic tra nsf er per da y	0	1	2	3	4	5	6	7	Nu mb er of da ys	10	35	45	95	64	32	10	9		4																																									
No . of tel egr ap hic tra nsf er per da y	0	1	2	3	4	5	6	7																																																							
Nu mb er of da ys	10	35	45	95	64	32	10	9																																																							

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		Note: This practice book is only for reference purpose. LJU Test question paper may not be completely set from this practice book.																													
Sr. No.	unit_ number	Question text	Answer	Marks	Previous Year	Option 1	Option 2	Option 3	Option 4																						
256	3	The table below shows the scores (out of 60) of applicants in an aptitude test. Calculate the mean, median and mode of the distribution. <table><tr><td>Score s</td><td>1-10</td><td>11-20</td><td>21-30</td><td>31-40</td><td>41-50</td><td>51-60</td></tr><tr><td>Frequency</td><td>5</td><td>6</td><td>8</td><td>10</td><td>7</td><td>4</td></tr></table>	Score s	1-10	11-20	21-30	31-40	41-50	51-60	Frequency	5	6	8	10	7	4		3													
Score s	1-10	11-20	21-30	31-40	41-50	51-60																									
Frequency	5	6	8	10	7	4																									
257	3	Calculate the mean, median and mode of the distribution. <table><tr><td>Class</td><td>15-25</td><td>25-35</td><td>35-45</td><td>45-55</td><td>55-65</td><td>65-75</td></tr><tr><td>Frequency</td><td>6</td><td>10</td><td>17</td><td>14</td><td>1</td><td>2</td></tr></table>	Class	15-25	25-35	35-45	45-55	55-65	65-75	Frequency	6	10	17	14	1	2		3	LJU-2024												
Class	15-25	25-35	35-45	45-55	55-65	65-75																									
Frequency	6	10	17	14	1	2																									
258	3	An insurance company obtained the following data for accident claims (in thousand rupees) from a particular region. Find its mean, median and Mode. <table><tr><td>Amount</td><td>1-3</td><td>3-5</td><td>5-7</td><td>7-9</td><td>9-11</td><td>11-13</td></tr><tr><td>Frequency</td><td>6</td><td>47</td><td>75</td><td>46</td><td>18</td><td>8</td></tr></table>	Amount	1-3	3-5	5-7	7-9	9-11	11-13	Frequency	6	47	75	46	18	8		4													
Amount	1-3	3-5	5-7	7-9	9-11	11-13																									
Frequency	6	47	75	46	18	8																									
259	3	Find the mean, median and Mode for the following frequency distribution: <table><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td></tr><tr><td>f</td><td>4</td><td>7</td><td>8</td><td>10</td><td>6</td><td>6</td><td>4</td><td>2</td><td>2</td><td>1</td></tr></table>	x	1	2	3	4	5	6	7	8	9	10	f	4	7	8	10	6	6	4	2	2	1		4					
x	1	2	3	4	5	6	7	8	9	10																					
f	4	7	8	10	6	6	4	2	2	1																					
260	3	Calculate Median for data of test of 60 marks given below: <table><tr><td>Marks (More than)</td><td>0</td><td>10</td><td>20</td><td>30</td><td>40</td><td>50</td></tr><tr><td>Students</td><td>50</td><td>44</td><td>32</td><td>21</td><td>12</td><td>5</td></tr></table>	Marks (More than)	0	10	20	30	40	50	Students	50	44	32	21	12	5		4	LJU-2025												
Marks (More than)	0	10	20	30	40	50																									
Students	50	44	32	21	12	5																									
261	3	Calculate median. <table><tr><td>Marks (less than)</td><td>5</td><td>10</td><td>15</td><td>20</td><td>25</td><td>30</td><td>35</td><td>40</td><td>45</td></tr><tr><td>No. of students</td><td>29</td><td>224</td><td>465</td><td>582</td><td>634</td><td>644</td><td>650</td><td>653</td><td>655</td></tr></table>	Marks (less than)	5	10	15	20	25	30	35	40	45	No. of students	29	224	465	582	634	644	650	653	655		3							
Marks (less than)	5	10	15	20	25	30	35	40	45																						
No. of students	29	224	465	582	634	644	650	653	655																						

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		Note: This practice book is only for reference purpose. LJU Test question paper may not be completely set from this practice book.																											
Sr. No.	unit_ number	Question text	Answer	Marks	Previous Year	Option 1	Option 2	Option 3	Option 4																				
262	3	Find the mean, median and Mode for the following frequency distribution: <table><tr><td>Cl as s</td><td>50 - 53</td><td>53 - 56</td><td>56 - 59</td><td>59 - 62</td><td>62 - 65</td><td>65 - 68</td><td>68 - 71</td><td>71 - 74</td><td>74 - 77</td></tr><tr><td>Fr eq ue nc y</td><td>3</td><td>8</td><td>14</td><td>30</td><td>36</td><td>28</td><td>16</td><td>10</td><td>3</td></tr></table>	Cl as s	50 - 53	53 - 56	56 - 59	59 - 62	62 - 65	65 - 68	68 - 71	71 - 74	74 - 77	Fr eq ue nc y	3	8	14	30	36	28	16	10	3		5					
Cl as s	50 - 53	53 - 56	56 - 59	59 - 62	62 - 65	65 - 68	68 - 71	71 - 74	74 - 77																				
Fr eq ue nc y	3	8	14	30	36	28	16	10	3																				
263	3	Find the standard deviation for the following data: <table><tr><td>x</td><td>5</td><td>10</td><td>15</td><td>20</td><td>25</td></tr><tr><td>f</td><td>7</td><td>4</td><td>6</td><td>3</td><td>5</td></tr></table>	x	5	10	15	20	25	f	7	4	6	3	5		3													
x	5	10	15	20	25																								
f	7	4	6	3	5																								
264	3	Find standard deviation from the following data. <table><tr><td>Class</td><td>9-11</td><td>12-14</td><td>15-17</td><td>18-20</td></tr><tr><td>Frequ ency</td><td>2</td><td>3</td><td>4</td><td>1</td></tr></table>	Class	9-11	12-14	15-17	18-20	Frequ ency	2	3	4	1		3															
Class	9-11	12-14	15-17	18-20																									
Frequ ency	2	3	4	1																									
265	3	From the following data calculate moments about (i) Assumed mean 25 (ii) Actual mean (iii) Zero. <table><tr><td>Varia ble</td><td>0-10</td><td>10-20</td><td>20-30</td><td>30-40</td></tr><tr><td>Frequ ency</td><td>1</td><td>3</td><td>4</td><td>2</td></tr></table>	Varia ble	0-10	10-20	20-30	30-40	Frequ ency	1	3	4	2		5	LJU-2024														
Varia ble	0-10	10-20	20-30	30-40																									
Frequ ency	1	3	4	2																									
266	3	Find the first three moments about mean for the following frequency distribution: <table><tr><td>x</td><td>5</td><td>7</td><td>10</td><td>18</td><td>26</td></tr><tr><td>f</td><td>5</td><td>14</td><td>22</td><td>6</td><td>3</td></tr></table>	x	5	7	10	18	26	f	5	14	22	6	3		4													
x	5	7	10	18	26																								
f	5	14	22	6	3																								
267	3	Calculate the first four moments of the following distribution about the mean. <table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td></tr><tr><td>f</td><td>1</td><td>8</td><td>28</td><td>56</td><td>70</td><td>56</td><td>28</td><td>8</td><td>1</td></tr></table>	x	0	1	2	3	4	5	6	7	8	f	1	8	28	56	70	56	28	8	1		4					
x	0	1	2	3	4	5	6	7	8																				
f	1	8	28	56	70	56	28	8	1																				
268	3	Calculate the first four moments about mean of the following data: <table><tr><td>x</td><td>5</td><td>10</td><td>15</td><td>20</td><td>25</td></tr><tr><td>f</td><td>6</td><td>10</td><td>14</td><td>6</td><td>4</td></tr></table>	x	5	10	15	20	25	f	6	10	14	6	4		4													
x	5	10	15	20	25																								
f	6	10	14	6	4																								
269	3	Obtain first four moments about arbitrary origin from the following table: <table><tr><td>Score s</td><td>50 – 60</td><td>60 – 70</td><td>70 – 80</td><td>80 – 90</td><td>90 – 100</td></tr><tr><td>Playe rs</td><td>8</td><td>11</td><td>18</td><td>09</td><td>04</td></tr></table>	Score s	50 – 60	60 – 70	70 – 80	80 – 90	90 – 100	Playe rs	8	11	18	09	04		4													
Score s	50 – 60	60 – 70	70 – 80	80 – 90	90 – 100																								
Playe rs	8	11	18	09	04																								
270	3	Calculate the first four moments from the following data: <table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td></tr><tr><td>f</td><td>5</td><td>10</td><td>15</td><td>20</td><td>25</td><td>20</td><td>15</td><td>10</td><td>5</td></tr></table> Also calculate the values of β_1 and β_2 .	x	0	1	2	3	4	5	6	7	8	f	5	10	15	20	25	20	15	10	5		5					
x	0	1	2	3	4	5	6	7	8																				
f	5	10	15	20	25	20	15	10	5																				
271	3	The quantities of water (in litres) supplied by municipal corporation on ten consecutive days in certain area are shown below: 218.2, 199.7, 207.3, 185.4, 213.7, 184.7, 179.5, 194.4, 224.3, 203.5. Evaluate the mean & the first four central moments of the water (in litres) of that area.		4																									

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		Note: This practice book is only for reference purpose. LJU Test question paper may not be completely set from this practice book.																													
Sr. No.	unit_ number	Question text	Answer	Marks	Previous Year	Option 1	Option 2	Option 3	Option 4																						
272	3	Calculate Karl Pearson’s coefficient of skewness from the following data. <table><tr><td>X</td><td>40-50</td><td>50-60</td><td>60-70</td><td>70-80</td><td>80-90</td><td>90-100</td><td>100-110</td><td>110-120</td><td>120-130</td><td>130-140</td></tr><tr><td>f</td><td>5</td><td>6</td><td>8</td><td>10</td><td>25</td><td>30</td><td>36</td><td>50</td><td>60</td><td>70</td></tr></table>	X	40-50	50-60	60-70	70-80	80-90	90-100	100-110	110-120	120-130	130-140	f	5	6	8	10	25	30	36	50	60	70		4	LJU-2022 LJU-2023				
X	40-50	50-60	60-70	70-80	80-90	90-100	100-110	110-120	120-130	130-140																					
f	5	6	8	10	25	30	36	50	60	70																					
273	3	Find Karl Pearson's coefficient of skewness for the following data. <table><tr><td>M ar ks</td><td>20</td><td>21</td><td>22</td><td>23</td><td>24</td><td>25</td><td>26</td><td>27</td><td>28</td></tr><tr><td>N o. of st ud en ts</td><td>7</td><td>12</td><td>25</td><td>10</td><td>8</td><td>6</td><td>8</td><td>3</td><td>1</td></tr></table>	M ar ks	20	21	22	23	24	25	26	27	28	N o. of st ud en ts	7	12	25	10	8	6	8	3	1		3							
M ar ks	20	21	22	23	24	25	26	27	28																						
N o. of st ud en ts	7	12	25	10	8	6	8	3	1																						
274	3	Find Karl Pearson’s coefficient of skewness for the following data: <table><tr><td>x</td><td>0-10</td><td>10-20</td><td>20-30</td><td>30-40</td><td>40-50</td></tr><tr><td>f</td><td>13</td><td>20</td><td>30</td><td>25</td><td>12</td></tr></table>	x	0-10	10-20	20-30	30-40	40-50	f	13	20	30	25	12		4															
x	0-10	10-20	20-30	30-40	40-50																										
f	13	20	30	25	12																										
275	3	The following data relate to the profits of 1,000 companies: <table><tr><td>Pro fits Rs. in tho usa nds</td><td>100 - 120</td><td>120 - 140</td><td>140 - 160</td><td>160 - 180</td><td>180 - 200</td><td>200 - 220</td><td>220 - 240</td></tr><tr><td>No. of co mp anie s</td><td>17</td><td>53</td><td>199</td><td>194</td><td>327</td><td>208</td><td>02</td></tr></table> Calculate the coefficient of skewness.	Pro fits Rs. in tho usa nds	100 - 120	120 - 140	140 - 160	160 - 180	180 - 200	200 - 220	220 - 240	No. of co mp anie s	17	53	199	194	327	208	02		4											
Pro fits Rs. in tho usa nds	100 - 120	120 - 140	140 - 160	160 - 180	180 - 200	200 - 220	220 - 240																								
No. of co mp anie s	17	53	199	194	327	208	02																								
276	3	The first three moments of distribution about the value two of the variables are 1, 16 and −40. Show that the mean = 3, variance = 15 and $\mu_3 = -86$.		3	LJU-2024																										
277	3	The first four raw moments of distribution are 2, 136, 320 and 40000. Calculate the first four moments about mean. Also find skewness.		3	LJU-2022																										
278	3	First three moments of a distribution about 3 are 2, 70, 150 respectively. Find those moments about 4.		3	LJU-2023																										
279	3	In a distribution, the <i>mean</i> = 65, <i>median</i> = 70, <i>coefficient of skewness</i> = −0.6. find mode and coefficient of variation.		3																											
280	3	For a group of 10 items, $\sum x = 452$, $\sum x^2 = 24270$, and mode = 43.7. Find the Karl Pearson’s coefficient of skewness.		3																											
281	4	The coefficient of correlation between two variables x and y is 0.48. The covariance is 36. The variance of x is 16. The standard deviation of y is:	18.75	1	LJU-2022 LJU-2024	10.15	13.32	16.5	18.75																						

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		Note: This practice book is only for reference purpose. LJU Test question paper may not be completely set from this practice book.																					
Sr. No.	unit_ number	Question text	Answer	Marks	Previous Year	Option 1	Option 2	Option 3	Option 4														
282	4	Calculate the correlation coefficient between the following values: X: 3, 5, 1, 7, 5. Y: 4, 3, 0, 8, 2	0.8	1		0.9	0.8	1	0.6														
283	4	The values of correlation coefficient lie in the interval:	[-1, 1]	1		[0, 1]	[-1, 1]	[1, 2]	[-1, 0]														
284	4	What is the value of correlation coefficient which is not possible?	1.2	1	LJU-2025	0.2	-0.2	1.2	-0.8														
285	4	If six hand writings were ranked by two judges in a competition and the rankings are as follows: <table><tr><td>Judge 1</td><td>6</td><td>5</td><td>4</td><td>3</td><td>2</td><td>1</td></tr><tr><td>Judge 2</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr></table>	Judge 1	6	5	4	3	2	1	Judge 2	1	2	3	4	5	6	-1	1	LJU-2022	-0.5	1	-1	0.5
Judge 1	6	5	4	3	2	1																	
Judge 2	1	2	3	4	5	6																	
286	4	Calculate the rank correlation coefficient, if two judges in a beauty contest ranked the entries as follows <table><tr><td>Judge X</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>Judge Y</td><td>5</td><td>4</td><td>3</td><td>2</td><td>1</td></tr></table>	Judge X	1	2	3	4	5	Judge Y	5	4	3	2	1	-1	1		-0.5	1	-1	0.5		
Judge X	1	2	3	4	5																		
Judge Y	5	4	3	2	1																		
287	4	If the sum of the squares of difference of ranks of 6 candidates in two criteria is 21, the rank correlation coefficient is_____.	0.4	1	LJU-2025	0.1	0.2	0.3	0.4														
288	4	The two lines of regression are $8x - 10y = 66$ and $40x - 18y = 214$, and variance of x series is 9. What is the standard deviation of y series?	4	1	LJU-2023	3	4	6	8														
289	4	The two regression lines are given by $x - y + 1 = 0$ and $2x - y + 4 = 0$. The two regression lines pass through the point:	$(-3, -2)$	1		$(-4, -3)$	$(-6, -5)$	$(3, -2)$	$(-3, -2)$														
290	4	If two regression coefficients are -0.1 and -0.9, then correlation coefficient is,	-0.3	1		0.3	-0.3	-0.9	0.9														
291	4	If two regression coefficients are -0.4 and -0.9, then the correlation coefficient is,	-0.6	1	LJU-2024 LJU-2025	0.6	-0.6	0.3	0														
292	4	If two regression coefficients are -0.8 and -0.2, what would be the value of coefficient of correlation?	-0.4	1		0.4	-0.4	0.16	-0.16														
293	4	The two regression lines of a sample are $x + 6y = 6$ and $3x + 2y = 10$. Then coefficient of correlation between x and y is	-1/3	1		-1/3	2/3	-2/3	3/4														
294	4	The two regression lines x and y always intersect at points	$(\bar{x}, \bar{y})$	1		$(\bar{x}, \bar{y})$	(x, y)	(0,0)	(-1,1)														
295	4	If $r = 0$ then the regression coefficients are	0	1		0	1	-1	90														
296	4	The regression lines of a sample are $x + 6y = 6$ and $3x + 2y = 10$. Find the sample means $\bar{x}$ and $\bar{y}$.	(3,0.5)	1		$(-3, -0.5)$	(3,0.5)	$(-3, 0.5)$	$(3, -0.5)$														
297	4	If the two lines of regression are $4x - 5y + 30 = 0$ and $20x - 9y - 107 = 0$ which of these are lines of regression of x on y?	$20x - 9y - 107 = 0$	1		$20x - 9y - 107 = 0$	$4x - 5y + 30 = 0$	$-20x + 9y + 107 = 0$	$4x + 5y - 30 = 0$														
298	4	From the following data calculate the regression coefficient b_{yx} ? <table><tr><td>X</td><td>6</td><td>2</td><td>10</td><td>4</td><td>8</td></tr><tr><td>Y</td><td>9</td><td>11</td><td>5</td><td>8</td><td>7</td></tr></table>	X	6	2	10	4	8	Y	9	11	5	8	7	-0.65	1	LJU-2025	0.65	-0.65	1.3	-1.3		
X	6	2	10	4	8																		
Y	9	11	5	8	7																		
299	4	If the difference between the rank of the 4 observations is 2.5, 0.5, -1.5, -1.5 then Spearman's rank correlation coefficient equals to _____.	None of these	1	LJU-2023	0.23	0.1	-0.1	None of these														

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		Note: This practice book is only for reference purpose. LJU Test question paper may not be completely set from this practice book.																													
Sr. No.	unit_ number	Question text	Answer	Marks	Previous Year	Option 1	Option 2	Option 3	Option 4																						
300	4	Obtain the two regression lines from the following data and hence find the correlation coefficient. <table><tr><td>x</td><td>6</td><td>2</td><td>10</td><td>4</td><td>8</td></tr><tr><td>y</td><td>9</td><td>11</td><td>5</td><td>8</td><td>7</td></tr></table>	x	6	2	10	4	8	y	9	11	5	8	7		4															
x	6	2	10	4	8																										
y	9	11	5	8	7																										
301	4	Find the line of regression of y on x. <table><tr><td>x</td><td>1.53</td><td>1.78</td><td>2.60</td><td>2.95</td><td>3.42</td></tr><tr><td>y</td><td>33.5</td><td>36.3</td><td>40.0</td><td>45.8</td><td>53.5</td></tr></table>	x	1.53	1.78	2.60	2.95	3.42	y	33.5	36.3	40.0	45.8	53.5		3															
x	1.53	1.78	2.60	2.95	3.42																										
y	33.5	36.3	40.0	45.8	53.5																										
302	4	Calculate the coefficient of correlation and obtain the lines of regression for the following: <table><tr><td>X</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td></tr><tr><td>Y</td><td>9</td><td>8</td><td>10</td><td>12</td><td>11</td><td>13</td><td>14</td><td>16</td><td>15</td></tr></table>	X	1	2	3	4	5	6	7	8	9	Y	9	8	10	12	11	13	14	16	15		4							
X	1	2	3	4	5	6	7	8	9																						
Y	9	8	10	12	11	13	14	16	15																						
303	4	Find correlation coefficient for the data given below. <table><tr><td>x</td><td>4</td><td>5</td><td>9</td><td>14</td><td>18</td><td>22</td><td>24</td></tr><tr><td>y</td><td>16</td><td>22</td><td>11</td><td>16</td><td>7</td><td>3</td><td>17</td></tr></table>	x	4	5	9	14	18	22	24	y	16	22	11	16	7	3	17		3											
x	4	5	9	14	18	22	24																								
y	16	22	11	16	7	3	17																								
304	4	Calculate the co-efficient of correlation between the given series of data for x and y in the following table: <table><tr><td>x</td><td>54</td><td>57</td><td>55</td><td>57</td><td>56</td><td>52</td><td>59</td></tr><tr><td>y</td><td>36</td><td>35</td><td>32</td><td>34</td><td>36</td><td>38</td><td>35</td></tr></table>	x	54	57	55	57	56	52	59	y	36	35	32	34	36	38	35		3											
x	54	57	55	57	56	52	59																								
y	36	35	32	34	36	38	35																								
305	4	Let $3x + 2y = 26$ and $6x + y = 31$, be the two regression lines. (i) Find the mean value and correlation coefficient between x and y (ii) if the variance of y is 4 find the standard deviation of x.		4																											
306	4	Find the coefficient of correlation from the data: x = 7, 8, 9, 11, 10, 13, 12 y = 1, 2, 3, 4, 5, 6, 7		3																											
307	4	Given $n = 10, \sigma_x = 5.4, \sigma_y = 6.2$ and the sum of the product of the deviations from the mean of x and y is 66. Find correlation coefficient.		3																											
308	4	The coefficient of rank correlation of the marks obtained by 10 students in physics and chemistry was found to be 0.5. It was later discovered that the difference in ranks in the two subjects obtained by one of the students was wrongly taken as 3 instead of 7. Find the correct coefficient of the rank correlation.		4																											
309	4	Obtain the correlation coefficient for the following data: <table><tr><td>x</td><td>100</td><td>98</td><td>78</td><td>85</td><td>110</td><td>93</td><td>80</td></tr><tr><td>y</td><td>85</td><td>90</td><td>70</td><td>72</td><td>95</td><td>81</td><td>74</td></tr></table>	x	100	98	78	85	110	93	80	y	85	90	70	72	95	81	74		4											
x	100	98	78	85	110	93	80																								
y	85	90	70	72	95	81	74																								
310	4	Calculate the correlation coefficient between x and y from the following data: $n = 10, \sum x = 140, \sum y = 150,$ $\sum (x - 10)^2 = 180, \sum (y - 15)^2 = 215, \sum (x - 10)(y - 15) = 60.$		4	LJU-2023																										
311	4	The following table shows how 10 students were ranked according to their achievements in both the laboratory and lecture portions of a python course. Find the coefficient of rank correlation. <table><tr><td>Labo rator y</td><td>8</td><td>3</td><td>9</td><td>2</td><td>7</td><td>10</td><td>4</td><td>6</td><td>1</td><td>5</td></tr><tr><td>Lect ure</td><td>9</td><td>5</td><td>10</td><td>1</td><td>8</td><td>7</td><td>3</td><td>4</td><td>2</td><td>6</td></tr></table>	Labo rator y	8	3	9	2	7	10	4	6	1	5	Lect ure	9	5	10	1	8	7	3	4	2	6		3					
Labo rator y	8	3	9	2	7	10	4	6	1	5																					
Lect ure	9	5	10	1	8	7	3	4	2	6																					

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Sr. No.	unit_ number	Question text	Answer	Marks	Previous Year	Option 1	Option 2	Option 3	Option 4																																	
312	4	Determine the correlation coefficient between temperature (in °C) and ice cream sales (in \$). <table><tr><td>Temperature</td><td>14.2</td><td>16.4</td><td>11.9</td><td>15.2</td><td>18.5</td><td>22.1</td><td>19.4</td><td>25.1</td><td>23.4</td><td>18.1</td><td>22.6</td><td>17.2</td></tr><tr><td>Icecream sales</td><td>215</td><td>325</td><td>185</td><td>332</td><td>406</td><td>522</td><td>412</td><td>614</td><td>544</td><td>421</td><td>445</td><td>408</td></tr></table>	Temperature	14.2	16.4	11.9	15.2	18.5	22.1	19.4	25.1	23.4	18.1	22.6	17.2	Icecream sales	215	325	185	332	406	522	412	614	544	421	445	408		3												
Temperature	14.2	16.4	11.9	15.2	18.5	22.1	19.4	25.1	23.4	18.1	22.6	17.2																														
Icecream sales	215	325	185	332	406	522	412	614	544	421	445	408																														
313	4	Calculate the coefficient of correlation for the following pairs of x and y: <table><tr><td>x</td><td>17</td><td>19</td><td>21</td><td>26</td><td>20</td><td>28</td><td>26</td><td>27</td></tr><tr><td>y</td><td>23</td><td>27</td><td>25</td><td>26</td><td>27</td><td>25</td><td>30</td><td>33</td></tr></table>	x	17	19	21	26	20	28	26	27	y	23	27	25	26	27	25	30	33		4																				
x	17	19	21	26	20	28	26	27																																		
y	23	27	25	26	27	25	30	33																																		
314	4	Find the regression coefficient of y on x for the following data: <table><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>y</td><td>160</td><td>180</td><td>140</td><td>180</td><td>200</td></tr></table>	x	1	2	3	4	5	y	160	180	140	180	200		4																										
x	1	2	3	4	5																																					
y	160	180	140	180	200																																					
315	4	The ranks of same 16 students in Maths and MOS are as follows: <table><tr><td>Ma ths</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td><td>11</td><td>12</td><td>13</td><td>14</td><td>15</td><td>16</td></tr><tr><td>M OS</td><td>1</td><td>10</td><td>3</td><td>4</td><td>5</td><td>7</td><td>2</td><td>6</td><td>8</td><td>11</td><td>15</td><td>9</td><td>14</td><td>12</td><td>6</td><td>3</td></tr></table> Calculate the rank correlation coefficient for proficiencies of this group in given subjects.	Ma ths	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	M OS	1	10	3	4	5	7	2	6	8	11	15	9	14	12	6	3		4				
Ma ths	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16																										
M OS	1	10	3	4	5	7	2	6	8	11	15	9	14	12	6	3																										
316	4	Obtain the correlation coefficient for the following data: <table><tr><td>x</td><td>58</td><td>64</td><td>51</td><td>74</td><td>88</td><td>91</td></tr><tr><td>y</td><td>12</td><td>18</td><td>15</td><td>41</td><td>46</td><td>52</td></tr></table>	x	58	64	51	74	88	91	y	12	18	15	41	46	52		4	LJU-2025																							
x	58	64	51	74	88	91																																				
y	12	18	15	41	46	52																																				
317	4	The following table gives the marks obtained by 11 students in Mathematics and Physics translation, Find the rank correlation coefficient. <table><tr><td>Mathema tics</td><td>40</td><td>46</td><td>54</td><td>60</td><td>70</td><td>80</td><td>82</td><td>85</td><td>85</td><td>90</td><td>95</td></tr><tr><td>Physics</td><td>45</td><td>45</td><td>50</td><td>43</td><td>40</td><td>75</td><td>55</td><td>72</td><td>65</td><td>42</td><td>70</td></tr></table>	Mathema tics	40	46	54	60	70	80	82	85	85	90	95	Physics	45	45	50	43	40	75	55	72	65	42	70		3														
Mathema tics	40	46	54	60	70	80	82	85	85	90	95																															
Physics	45	45	50	43	40	75	55	72	65	42	70																															
318	4	The number of bacterial cells(y) per unit volume in a culture at different hours(x) is given below: <table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td></tr><tr><td>y</td><td>43</td><td>46</td><td>82</td><td>98</td><td>123</td><td>167</td><td>199</td><td>213</td><td>245</td><td>272</td></tr></table> Fit a line of regression of y on x and x on y. estimate the number of bacterial cells after 15 hours.	x	0	1	2	3	4	5	6	7	8	9	y	43	46	82	98	123	167	199	213	245	272		4	LJU-2023															
x	0	1	2	3	4	5	6	7	8	9																																
y	43	46	82	98	123	167	199	213	245	272																																
319	4	Find the coefficient of correlation by spearman’s method from the following data: <table><tr><td>IQ X_i</td><td>106</td><td>86</td><td>100</td><td>101</td><td>99</td><td>103</td><td>97</td><td>113</td><td>112</td><td>110</td></tr><tr><td>Hou rs Y_i</td><td>7</td><td>0</td><td>27</td><td>50</td><td>28</td><td>29</td><td>20</td><td>12</td><td>6</td><td>17</td></tr></table> The above data shows the correlation between the IQ of a person and number of hours spent in front of the TV per week by person.	IQ X _i	106	86	100	101	99	103	97	113	112	110	Hou rs Y _i	7	0	27	50	28	29	20	12	6	17		4																
IQ X _i	106	86	100	101	99	103	97	113	112	110																																
Hou rs Y _i	7	0	27	50	28	29	20	12	6	17																																
320	4	Obtain the rank correlation coefficient from the following data: <table><tr><td>x</td><td>10</td><td>12</td><td>10</td><td>15</td><td>13</td><td>12</td><td>10</td></tr><tr><td>y</td><td>14</td><td>13</td><td>12</td><td>10</td><td>13</td><td>12</td><td>11</td></tr></table>	x	10	12	10	15	13	12	10	y	14	13	12	10	13	12	11		4	LJU-2025																					
x	10	12	10	15	13	12	10																																			
y	14	13	12	10	13	12	11																																			

		L.J Institute of Engineering and Technology, Ahmedabad. <u>Introduction to Probability Theory and Stochastic Processes</u> Practice Book (Sem- III)																																								
		Note: This practice book is only for reference purpose. LJU Test question paper may not be completely set from this practice book.																																								
Sr. No.	unit_ number	Question text	Answer	Marks	Previous Year	Option 1	Option 2	Option 3	Option 4																																	
321	4	Following are the scores of ten students in class and their IQ: <table><tr><td>Score</td><td>35</td><td>40</td><td>25</td><td>55</td><td>85</td><td>90</td><td>65</td><td>55</td><td>45</td><td>50</td></tr><tr><td>IQ</td><td>100</td><td>100</td><td>101</td><td>104</td><td>105</td><td>103</td><td>100</td><td>102</td><td>104</td><td>101</td></tr><tr><td></td><td>00</td><td>00</td><td>00</td><td>00</td><td>00</td><td>00</td><td>00</td><td>00</td><td>00</td><td>00</td></tr></table> Calculate the rank correlation co-efficient between the score and IQ.	Score	35	40	25	55	85	90	65	55	45	50	IQ	100	100	101	104	105	103	100	102	104	101		00	00	00	00	00	00	00	00	00	00		4					
Score	35	40	25	55	85	90	65	55	45	50																																
IQ	100	100	101	104	105	103	100	102	104	101																																
	00	00	00	00	00	00	00	00	00	00																																
322	4	Nine competitors in the reality show India’s Got Talent ranked by two judges in the following order. Calculate the Spearman’s rank correlation coefficient. <table><tr><td>Competitors</td><td>A</td><td>B</td><td>C</td><td>D</td><td>E</td><td>F</td><td>G</td><td>H</td><td>I</td></tr><tr><td>Kiran Kher</td><td>1</td><td>6</td><td>2</td><td>7</td><td>3</td><td>8</td><td>4</td><td>9</td><td>5</td></tr><tr><td>Karan Johar</td><td>5</td><td>2</td><td>6</td><td>8</td><td>7</td><td>1</td><td>3</td><td>9</td><td>4</td></tr></table>	Competitors	A	B	C	D	E	F	G	H	I	Kiran Kher	1	6	2	7	3	8	4	9	5	Karan Johar	5	2	6	8	7	1	3	9	4		3	LJU-2024							
Competitors	A	B	C	D	E	F	G	H	I																																	
Kiran Kher	1	6	2	7	3	8	4	9	5																																	
Karan Johar	5	2	6	8	7	1	3	9	4																																	
323	4	Find the correlation coefficient from the following data: <table><tr><td>X</td><td>50</td><td>50</td><td>55</td><td>60</td><td>65</td><td>65</td><td>65</td><td>60</td><td>60</td><td>50</td></tr><tr><td>Y</td><td>11</td><td>13</td><td>14</td><td>16</td><td>16</td><td>15</td><td>15</td><td>14</td><td>13</td><td>13</td></tr></table>	X	50	50	55	60	65	65	65	60	60	50	Y	11	13	14	16	16	15	15	14	13	13		4	LJU-2022															
X	50	50	55	60	65	65	65	60	60	50																																
Y	11	13	14	16	16	15	15	14	13	13																																
324	4	Raw material used in the production of a synthetic fibre is stored in a place which has no humidity control. Measurements of the relative humidity in the storage place and the moisture content of a sample of the raw material (both in %) on 7 days yielded the following results: <table><tr><td>Humidity (x):</td><td>42</td><td>35</td><td>50</td><td>43</td><td>48</td><td>62</td><td>31</td></tr><tr><td>Moisture content (y):</td><td>12</td><td>8</td><td>14</td><td>9</td><td>11</td><td>16</td><td>7</td></tr></table> Find the lines of regression of y on x and x on y.	Humidity (x):	42	35	50	43	48	62	31	Moisture content (y):	12	8	14	9	11	16	7		4																						
Humidity (x):	42	35	50	43	48	62	31																																			
Moisture content (y):	12	8	14	9	11	16	7																																			
325	4	Compute the coefficient of correlation between X and Y using the following data: <table><tr><td>X</td><td>2</td><td>4</td><td>5</td><td>6</td><td>8</td><td>11</td></tr><tr><td>Y</td><td>18</td><td>12</td><td>10</td><td>8</td><td>7</td><td>5</td></tr></table>	X	2	4	5	6	8	11	Y	18	12	10	8	7	5		3																								
X	2	4	5	6	8	11																																				
Y	18	12	10	8	7	5																																				
326	4	Find the Correlation coefficient and lines of regression from the following data: <table><tr><td>x</td><td>57</td><td>58</td><td>59</td><td>59</td><td>60</td><td>61</td><td>62</td><td>64</td></tr><tr><td>y</td><td>67</td><td>68</td><td>65</td><td>68</td><td>72</td><td>72</td><td>69</td><td>71</td></tr></table> Find the value of y when x = 66	x	57	58	59	59	60	61	62	64	y	67	68	65	68	72	72	69	71		4																				
x	57	58	59	59	60	61	62	64																																		
y	67	68	65	68	72	72	69	71																																		
327	4	Obtain both the lines of regression for the following data and hence find the correlation coefficient. <table><tr><td>x</td><td>60</td><td>34</td><td>40</td><td>50</td><td>45</td><td>41</td><td>22</td><td>43</td></tr><tr><td>y</td><td>75</td><td>32</td><td>34</td><td>40</td><td>45</td><td>33</td><td>12</td><td>30</td></tr></table>	x	60	34	40	50	45	41	22	43	y	75	32	34	40	45	33	12	30		4																				
x	60	34	40	50	45	41	22	43																																		
y	75	32	34	40	45	33	12	30																																		
328	4	Obtain both the lines of regression for the following data and hence find the correlation coefficient. <table><tr><td>x</td><td>25</td><td>28</td><td>35</td><td>32</td><td>31</td><td>36</td><td>29</td><td>38</td><td>34</td><td>32</td></tr><tr><td>y</td><td>43</td><td>46</td><td>49</td><td>41</td><td>36</td><td>32</td><td>31</td><td>30</td><td>33</td><td>39</td></tr></table>	x	25	28	35	32	31	36	29	38	34	32	y	43	46	49	41	36	32	31	30	33	39		4	LJU-2024															
x	25	28	35	32	31	36	29	38	34	32																																
y	43	46	49	41	36	32	31	30	33	39																																

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		Note: This practice book is only for reference purpose. LJU Test question paper may not be completely set from this practice book.																																								
Sr. No.	unit_ number	Question text	Answer	Marks	Previous Year	Option 1	Option 2	Option 3	Option 4																																	
329	4	Obtain the two lines of regression for the following data: <table><tr><td>Sal es</td><td>190</td><td>240</td><td>250</td><td>300</td><td>310</td><td>335</td><td>300</td></tr><tr><td>Ad vert isin g exp end itur e</td><td>5</td><td>10</td><td>15</td><td>20</td><td>20</td><td>30</td><td>30</td></tr></table>	Sal es	190	240	250	300	310	335	300	Ad vert isin g exp end itur e	5	10	15	20	20	30	30		4																						
Sal es	190	240	250	300	310	335	300																																			
Ad vert isin g exp end itur e	5	10	15	20	20	30	30																																			
330	4	The following data gives the age and blood pressure (BP) of 10 sports persons. <table><tr><td>N a m e</td><td>A</td><td>B</td><td>C</td><td>D</td><td>E</td><td>F</td><td>G</td><td>H</td><td>I</td><td>J</td></tr><tr><td>A ge (X)</td><td>42</td><td>36</td><td>55</td><td>58</td><td>35</td><td>65</td><td>60</td><td>50</td><td>48</td><td>51</td></tr><tr><td>B P (Y)</td><td>98</td><td>93</td><td>11 0</td><td>85</td><td>10 5</td><td>10 8</td><td>82</td><td>10 2</td><td>11 8</td><td>99</td></tr></table> (1) Find the regression equation of Y on X and X on Y. (2) Find the correlation coefficient.	N a m e	A	B	C	D	E	F	G	H	I	J	A ge (X)	42	36	55	58	35	65	60	50	48	51	B P (Y)	98	93	11 0	85	10 5	10 8	82	10 2	11 8	99		4	LJU-2022				
N a m e	A	B	C	D	E	F	G	H	I	J																																
A ge (X)	42	36	55	58	35	65	60	50	48	51																																
B P (Y)	98	93	11 0	85	10 5	10 8	82	10 2	11 8	99																																
331	4	Find the regression equation showing the capacity utilization on production from the following data: <table><tr><td></td><td>Average</td><td>Standard Deviation</td></tr><tr><td>Production (in lakh units)</td><td>35.6</td><td>10.5</td></tr><tr><td>Capacity utilization (in %)</td><td>84.8</td><td>8.5</td></tr><tr><td>Correlation Coefficient</td><td colspan="2">r = 0.62</td></tr></table> Estimate the production when capacity utilization is 70%.		Average	Standard Deviation	Production (in lakh units)	35.6	10.5	Capacity utilization (in %)	84.8	8.5	Correlation Coefficient	r = 0.62			4																										
	Average	Standard Deviation																																								
Production (in lakh units)	35.6	10.5																																								
Capacity utilization (in %)	84.8	8.5																																								
Correlation Coefficient	r = 0.62																																									
332	4	From the following results, obtain the two regression equations and estimate the yield when the rainfall is 29 cm and the rainfall, when the yield is 600 kg: <table><tr><td></td><td>Yield in kg.</td><td>Rainfall in cm</td></tr><tr><td>Mean</td><td>508.4</td><td>26.7</td></tr><tr><td>SD</td><td>36.8</td><td>4.6</td></tr></table> The coefficient of correlation between yield and rainfall is 0.52.		Yield in kg.	Rainfall in cm	Mean	508.4	26.7	SD	36.8	4.6		4	LJU-2024																												
	Yield in kg.	Rainfall in cm																																								
Mean	508.4	26.7																																								
SD	36.8	4.6																																								
333	4	Given that $n = 25, \sum X = 125, \sum X^2 = 650, \sum Y = 100, \sum Y^2 = 460$ and $\sum XY = 508$. It was later discovered at the time of checking that he had copied down two pairs as (6,14) and (8,6) while the correct pairs were (8,12) and (6,8). Obtain the correct value of the correlation coefficient.		4																																						

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		Note: This practice book is only for reference purpose. LJU Test question paper may not be completely set from this practice book.																													
Sr. No.	unit_ number	Question text	Answer	Marks	Previous Year	Option 1	Option 2	Option 3	Option 4																						
334	4	From the following data of the marks obtained by 8 students in Computer Networking (CN) and Compiler Design (CD) papers, compute rank coefficient of correlation. <table><tr><td>CN</td><td>15</td><td>20</td><td>28</td><td>12</td><td>40</td><td>60</td><td>20</td><td>80</td></tr><tr><td>CD</td><td>40</td><td>30</td><td>50</td><td>30</td><td>20</td><td>10</td><td>30</td><td>60</td></tr></table>	CN	15	20	28	12	40	60	20	80	CD	40	30	50	30	20	10	30	60		4									
CN	15	20	28	12	40	60	20	80																							
CD	40	30	50	30	20	10	30	60																							
335	4	The coefficient of rank correlation of marks obtained by 10 students in English and Economics was found to be 0.6. It was later discovered that the difference in ranks in the two subjects obtained by one of the students was wrongly taken as 7 instead of 1. Find the correct coefficient of rank correlation.		4																											
336	4	Compute the coefficient of rank correlation between Economics marks and Statistics marks as given below: <table><tr><td>Econ omic s mark s</td><td>80</td><td>56</td><td>50</td><td>48</td><td>50</td><td>62</td><td>60</td></tr><tr><td>Stati stics mark s</td><td>90</td><td>75</td><td>75</td><td>65</td><td>65</td><td>50</td><td>65</td></tr></table>	Econ omic s mark s	80	56	50	48	50	62	60	Stati stics mark s	90	75	75	65	65	50	65		4											
Econ omic s mark s	80	56	50	48	50	62	60																								
Stati stics mark s	90	75	75	65	65	50	65																								
337	4	In partially destroyed laboratory record of an analysis of correlation data, the following results are eligible. - Variance of x, $\sigma_x^2 = 9$ - Two line of regressions: $8x - 10y + 66 = 0$, $40x - 18y = 214$. From the above obtain mean values of x and y , the standard deviation of y and correlation coefficient.		4																											
338	4	Psychological tests of intelligence and of engineering ability were applied to 10 students as per the following data. Find the coefficient of correlation. <table><tr><td>In te lli ge nc e</td><td>105</td><td>104</td><td>102</td><td>101</td><td>100</td><td>99</td><td>98</td><td>96</td><td>93</td><td>92</td></tr><tr><td>ab ili ty</td><td>101</td><td>100</td><td>100</td><td>98</td><td>95</td><td>96</td><td>100</td><td>92</td><td>97</td><td>94</td></tr></table>	In te lli ge nc e	105	104	102	101	100	99	98	96	93	92	ab ili ty	101	100	100	98	95	96	100	92	97	94		4					
In te lli ge nc e	105	104	102	101	100	99	98	96	93	92																					
ab ili ty	101	100	100	98	95	96	100	92	97	94																					
339	4	In a college, IT department has arranged one competition for IT students to develop an efficient program to solve a problem. Ten students took part in the competition and ranked by two judges given in the following table. Find the degree of agreement between the two judges using Rank correlation coefficient. (J=Judge) <table><tr><td>J-1</td><td>3</td><td>5</td><td>8</td><td>4</td><td>7</td><td>10</td><td>2</td><td>1</td><td>6</td><td>9</td></tr><tr><td>J-2</td><td>6</td><td>4</td><td>9</td><td>8</td><td>1</td><td>2</td><td>3</td><td>10</td><td>5</td><td>7</td></tr></table>	J-1	3	5	8	4	7	10	2	1	6	9	J-2	6	4	9	8	1	2	3	10	5	7		3					
J-1	3	5	8	4	7	10	2	1	6	9																					
J-2	6	4	9	8	1	2	3	10	5	7																					

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		Note: This practice book is only for reference purpose. LJU Test question paper may not be completely set from this practice book.																																								
Sr. No.	unit_ number	Question text	Answer	Marks	Previous Year	Option 1	Option 2	Option 3	Option 4																																	
340	4	Calculate the Co-efficient of correlation from the following data: <table><tr><td>x</td><td>12</td><td>9</td><td>8</td><td>10</td><td>11</td><td>13</td><td>7</td></tr><tr><td>y</td><td>14</td><td>8</td><td>6</td><td>9</td><td>11</td><td>12</td><td>3</td></tr></table>	x	12	9	8	10	11	13	7	y	14	8	6	9	11	12	3		3	LJU-2024																					
x	12	9	8	10	11	13	7																																			
y	14	8	6	9	11	12	3																																			
341	4	Calculate the correlation coefficient between the following values of demand and the corresponding price of a commodity: <table><tr><td>De ma nd</td><td>65</td><td>66</td><td>67</td><td>67</td><td>68</td><td>69</td><td>70</td><td>72</td></tr><tr><td>Pri ce</td><td>67</td><td>68</td><td>65</td><td>68</td><td>72</td><td>72</td><td>69</td><td>71</td></tr></table>	De ma nd	65	66	67	67	68	69	70	72	Pri ce	67	68	65	68	72	72	69	71		4																				
De ma nd	65	66	67	67	68	69	70	72																																		
Pri ce	67	68	65	68	72	72	69	71																																		
342	4	Ten competitors in a musical test were ranked by the three judges A, B and C in the following order: <table><tr><td>R an k b y A</td><td>1</td><td>6</td><td>5</td><td>10</td><td>3</td><td>2</td><td>4</td><td>9</td><td>7</td><td>8</td></tr><tr><td>R an k b y B</td><td>3</td><td>5</td><td>8</td><td>4</td><td>7</td><td>10</td><td>2</td><td>1</td><td>6</td><td>9</td></tr><tr><td>R an k b y C</td><td>6</td><td>4</td><td>9</td><td>8</td><td>1</td><td>2</td><td>3</td><td>10</td><td>5</td><td>7</td></tr></table> Using the rank correlation method, find which pair of judges has the nearest approach to common liking in music.	R an k b y A	1	6	5	10	3	2	4	9	7	8	R an k b y B	3	5	8	4	7	10	2	1	6	9	R an k b y C	6	4	9	8	1	2	3	10	5	7		3					
R an k b y A	1	6	5	10	3	2	4	9	7	8																																
R an k b y B	3	5	8	4	7	10	2	1	6	9																																
R an k b y C	6	4	9	8	1	2	3	10	5	7																																
343	4	Obtain the rank correlation coefficient from the following data: <table><tr><td>x</td><td>10</td><td>12</td><td>18</td><td>18</td><td>15</td><td>40</td></tr><tr><td>y</td><td>12</td><td>18</td><td>25</td><td>25</td><td>50</td><td>25</td></tr></table>	x	10	12	18	18	15	40	y	12	18	25	25	50	25		3																								
x	10	12	18	18	15	40																																				
y	12	18	25	25	50	25																																				
344	4	If the two lines of regression are $4x - 5y + 30 = 0$ and $20x - 9y - 107 = 0$, which of these are lines of regression of x on y and y on x ? Find r and σ_y when $\sigma_x = 3$.		4																																						
345	5	If ' p ', ' q ' and ' n ' are probability of success, failure and number of trials respectively in a Binomial Distribution, what is its Standard Deviation?	$\sqrt{npq}$	1		$\sqrt{np}$	$\sqrt{pq}$	$(np)^2$	$\sqrt{npq}$																																	
346	5	If ' m ' is the mean of a Poisson Distribution, then variance is given by _____.	m	1		m^2	$\sqrt{m}$	m	$\frac{m}{2}$																																	
347	5	If ' λ ' is the mean of Poisson Distribution, the $P(0)$ is given by _____.	$e^{-\lambda}$	1		$e^{-\lambda}$	e^{λ}	e	λ^{-e}																																	
348	5	For a Poisson Distribution, if mean = 1, then $P(1)$ is?	$\frac{1}{e}$	1		$\frac{1}{e}$	e	$\frac{e}{2}$	Indeterminate form																																	
349	5	Consider a random variable to which a Poisson distribution is best fitted. It happens that $P(X = 1) = \frac{2}{3} P(X = 2)$ on this distribution plot. The variance of this distribution will be _____.	3	1	LJU-2023	3	1	2	$\frac{2}{3}$																																	
350	5	If X is a poisson random variate with mean 3, then $P(X - 3 < 1)$ will be _____.	$\frac{9}{2}e^{-3}$	1	LJU-2022 LJU-2025	$\frac{9}{2}e^{-3}$	$3e^{-3}$	$\frac{e^{-3}}{2}$	$\left(\frac{99}{8}\right)e^{-3}$																																	

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		Note: This practice book is only for reference purpose. LJU Test question paper may not be completely set from this practice book.							
Sr. No.	unit_number	Question text	Answer	Marks	Previous Year	Option 1	Option 2	Option 3	Option 4
351	5	A lot has 10 % defective items. Ten items are chosen randomly from this lot. The probability that exactly 2 of the chosen items are defective is	0.1937	1		0.0036	0.1937	0.2234	0.3874
352	5	In a binomial distribution the mean is 15 and variance is 10, then parameter n is	45	1	LJU-2024	28	16	45	25
353	5	Consider an unbiased cubic dice with opposite faces colored identically and each face-colored red, blue or green such that each color appears only two times on the dice. If the dice thrown thrice, the probability of getting red color on top face of the dice at least twice is _____.	$\frac{7}{27}$	1		$\frac{7}{27}$	$\frac{10}{127}$	$\frac{19}{27}$	$\frac{1}{3}$
354	5	The number of tosses of a coin that are needed so that the probability of getting at least one head being 0.875 is _____.	3	1		2	3	4	5
355	5	If 20% of the bolts produced by a machine are defective, the probability that out of 4 bolts chosen, at most 2 bolts will be defective is,	0.9728	1		0.9728	0.2897	0.4096	0.1536
356	5	A fair coin is tossed independently four times. The probability of the event “the number of times heads show up is more than the number of times tails show up”	$\frac{5}{16}$	1		$\frac{5}{16}$	$\frac{1}{4}$	$\frac{1}{16}$	$\frac{7}{16}$
357	5	A fair coin is tossed 6 times. The probability of obtaining at least 5 heads is,	0.1094	1		0.1094	0.4019	0	0.9410
358	5	A dice is thrown 6 times. If getting an odd number is a success, the probability of 5 successes is,	3/32	1		3/32	1/32	63/64	7/64
359	5	A dice is thrown 6 times. If getting an odd number is a success, the probability of at least 5 successes is,	7/64	1		3/32	1/32	7/64	63/64
360	5	A dice is thrown 6 times. If getting an odd number is a success, the probability of at the most 5 successes is,	63/64	1		3/32	1/32	7/64	63/64
361	5	If the probability of a defective bolt is 0.1. What will be the mean of the distribution of defective bolts in a total of 400?	40	1		50	60	40	70
362	5	Find the expectation for how many bacteria there are per field if there are 2350 bacteria are randomly distributed over 340 fields (all having the same size) next to each other.	6.91	1		4.98	3.875	6.91	7.37
363	5	A student arrives late for a class 40% of the time. Class meets five times each week. The probability if student being late for at least three classes in a given week is,	0.317	1		0.317	0	0.5	0.517
364	5	A student arrives late for a class 40% of the time. Class meets five times each week. The probability of student will not be late at all during a given week is,	0.0778	1		0	0.0778	0.0887	0.0227
365	5	The mean and variance of a binomial variate are 8 and 6 then parameter n is	32	1	LJU-2025	22	28	32	35
366	5	In a company, there are 250 workers. The probability of a worker remain absent on any day is 0.02. The probability that on a day seven workers are absent is,	0.104	1		0.104	0	0.905	0.401
367	5	In the inspection of tin plate produced by a continuous electrolytic process, 0.2 imperfections spotted per minute, on average. The probability of spotting one imperfection in 3 minutes is,	0.329	1		0.329	0	0.923	0.239
368	5	In the inspection of tin plate produced by a continuous electrolytic process, 0.2 imperfections spotted per minute, on average. The probability of spotting at least two imperfections in 5 minutes is,	0.2644	1	LJU-2025	0.4426	0	0.2644	0.3678
369	5	In the inspection of tin plate produced by a continuous electrolytic process, 0.2 imperfections	0.199	1		0.199	0	0.2644	0.3678

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Sr. No.	unit_number	Question text	Answer	Marks	Previous Year	Option 1	Option 2	Option 3	Option 4
		spotted per minute, on average. The probability of spotting at most one imperfection in 15 minutes is,							
370	5	The number of defects in a thin copper wire follows Poisson distribution with mean 2.3 defects per millimeter. Then the probability of exactly two defects per millimeter of wire is,	0.2652	1	LJU-2024	0	0.23	0.2652	0.5226
371	5	If $P(1) = P(5)$ in Poisson distribution, then the value of mean is,	3.31	1		3.31	3.38	5.38	6.38
372	5	In an experiment, positive and negative values are equally likely to occur. The probability of obtaining at most one negative value in five trials is _____.	$\frac{6}{32}$	1	LJU-2022	$\frac{1}{32}$	$\frac{6}{32}$	$\frac{5}{32}$	$\frac{7}{32}$
373	5	If '16' is the mean of a Poisson Distribution, then variance is given by _____	16	1	LJU-2024	2	4	12	16
374	5	A student takes an 18-question multiple choice exam, with four choice per question. Suppose one of the choices is obviously incorrect, and the student makes an "educated" guess of the remaining choices, then the expected number of the correct answer is	6	1		9	10	8	6
375	5	The overall percentage of failure in an examination is 30. What is the probability that in a group of 5 students at least 4 passed the examination?	0.5282	1	LJU-2023	0.3216	0.4658	0.5282	0.5916
376	5	Obtain the binomial distribution for which mean is 10 and variance is 5.		2					
377	5	The mean and variance of a binomial variate are 8 and 6. Find $P(X \geq 2)$.		2					
378	5	The probability that a patient will be cured of corona virus when injected with the new vaccine is 0.8. Find the probability that exactly 3 out of the 8 corona virus patients will be cured on being injected with the vaccine.		3					
379	5	The average percentage of failure in a certain examination is 40. What is the probability that out of a group of 6 candidates, at least 4 passed in examination?		3					
380	5	If 3 of 12 car drivers do not carry driving license, what is the probability that a traffic inspector who randomly checks 3 car drivers, will catch 1 for not carrying driving license. (use binomial dist.)		3					
381	5	4 coins are tossed simultaneously. What is the probability of getting (i) 2 heads? (ii) at least 2 heads? (iii) at most 2 heads?		3					
382	5	A multiple-choice test consists of 8 questions with 3 answers to each question (of which only one is correct). A student answers each question by rolling a balanced die and checking the first answer if he gets 1 or 2, the second answer if he gets 3 or 4, and the third answer if he gets 5 or 6. To get a distinction, the student must secure at least 75% correct answers. If there is no negative marks, what is the probability that the student secures a distinction?		3					
383	5	The probability of a man hitting a target is $\frac{1}{3}$. (i) If he fires 5 times, what is the probability of his hitting the target at least twice? (ii) How many times must he fire so that the probability of his hitting the target at least once is more than 90%?		3					
384	5	If hens of certain breed lays eggs on 5 days a week on an average, find how many days during a season		3					

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		of 100 days a will poultry keeper with 5 hens of this breed expect to receive at least 4 eggs.																									
385	5	An irregular 6 faced die is thrown such that the probability that it gives 3 even numbers in 5 throws is twice the probability that it gives 2 even numbers in 5 throws. How many sets of exactly 5 trials can be expected to give no even number out of 2500 sets?		3																							
386	5	Out of 800 families with 5 children each, how many would you expect to have (i) 3 boys? (ii) 5 girls? (iii) either 2 or 3 boys? (iv) at least one boy? Assume equal probabilities for boys and girls.		4	LJU-2024																						
387	5	Out of 1000 families with 4 children each, how many would you expect to have (i) 2 boys and 2 girls? (ii) at least one boy? (iii) no girl? (iv) at most two girls?		4																							
388	5	Fit a binomial distribution to the following data: <table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>f</td><td>2</td><td>14</td><td>20</td><td>34</td><td>22</td><td>8</td></tr></table>	x	0	1	2	3	4	5	f	2	14	20	34	22	8		4									
x	0	1	2	3	4	5																					
f	2	14	20	34	22	8																					
389	5	Seven unbiased coins are tossed 128 times and the number of heads obtained is noted as given below: <table><tr><td>No . of heads</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr><tr><td>Frequency</td><td>7</td><td>6</td><td>19</td><td>35</td><td>30</td><td>23</td><td>7</td><td>1</td></tr></table> Fit a binomial distribution to the data.	No . of heads	0	1	2	3	4	5	6	7	Frequency	7	6	19	35	30	23	7	1		3	LJU-2023 LJU-2024				
No . of heads	0	1	2	3	4	5	6	7																			
Frequency	7	6	19	35	30	23	7	1																			
390	5	It is known from past records that 80% of the students in a school do their homework. Find the probability that during a random check of 10 students (i) all have done their homework (ii) at the most two have not done their homework and (iii) at least one has not done the homework.		4																							
391	5	Fit a binomial distribution to the following data: <table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>f</td><td>12</td><td>66</td><td>109</td><td>59</td><td>10</td></tr></table>	x	0	1	2	3	4	f	12	66	109	59	10		4											
x	0	1	2	3	4																						
f	12	66	109	59	10																						
392	5	In a binomial distribution, the sum and product of the mean and variance are $\frac{25}{3}$ and $\frac{50}{3}$ respectively. Determine the distribution.		3																							
393	5	The mean and variance of a binomial distribution are 3 and 1.2 respectively. Find n, p and $P(X < 4)$.		2																							
394	5	If a publisher of nontechnical books takes great pains to ensure that its books are free of typographical errors, so that the probability of any given page containing at least one such error is 0.005 and errors are independent from page to page, what is the probability that one of its 400-page novels will contain (i) exactly one page with errors? (ii) At most three pages with errors?		3																							
395	5	Assume that on the average one telephone number out of fifteen called between 1 p.m. and 2 p.m. on week days is busy. What is the probability that if 6 randomly selected telephone numbers are called (i) not more than three, (ii) at least three of them would be busy?		3																							

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396	5	In eight throws of a die, 5 or 6 is considered as a success. Find the mean number of success and the standard deviation.		2					
397	5	If 10% of the screws produced by a machine are defective, find the probability that out of 5 screws chosen at random, (i) none is defective, (ii) one is defective, and (iii) at most two are defective.		3					
398	5	In sampling a large number of parts manufactured by a machine, the mean number of defectives in a sample of 20 is 2. Out of 1000 such samples, how many would be expected to contain exactly two defective parts?		3					
399	5	A multiple-choice test in mathematics with 40 questions, each having 5 options, is given to a student. If the student guesses all questions, what are the mean and standard deviation of the number of correct answers? (Use Binomial Distribution)		3					
400	5	Suppose that a central university has to form a committee of 5 members from a list of 20 candidates out of whom 12 are teachers and 8 are students. If the members of the committee are selected at random. What is the probability that the majority of the committee members are students?		3					
401	5	In a binomial distribution consisting of 5 independent trials, the probability of 1 and 2 successes are 0.4096 and 0.2048 respectively. Find the parameter p of the distribution.		3					
402	5	Assume that half the population is vegetarian so that the chance of an individual being vegetarian is $\frac{1}{2}$. Assuming that 100 investigators each take sample of 10 individuals to see whether they are vegetarians. How many investigators would you expect to report that three people or less were vegetarian?		3	LJU-2022				
403	5	If 10% of toys produced by a machine are defective. Determine the probability that out of 10 toys, chosen at random (I) 1 (II) none (III) at most 2 toys will be defective.		4					
404	5	A book contains 100 misprints distributed randomly throughout its 100 pages. What is the probability that a page observed at random contains at least two misprints? Assume Poisson Distribution.		3					
405	5	Suppose a random book of 1045 pages contains 35 printing errors. If these errors are distributed as Poisson throughout the book, what is the probability that 60 pages, selected at random, (1) will be free from errors and (2) will have 2 errors?		4	LJU-2025				
406	5	If 2% of light bulbs are defective, find the probability that (1) at least one is defective, and (2) exactly 7 are defective. Also, find $P(1 < X < 8)$ in a sample of 100. Assume Poisson Distribution.		3	LJU-2024				
407	5	A car hire firm has two cars which it hires out day to day. The number of demands for a car on each day is distributed as Poisson variate with mean 1.5. Calculate the proportion of days on which (i) neither car is used, and (ii) some demand is refused. ($e^{-1.5} = 0.2231$).		3	LJU-2024				
408	5	The number of flaws in a fiber optic cable follows a Poisson process with an average of 0.6 per 100 feet.		3	LJU-2023				

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Sr. No.	unit_number	Question text	Answer	Marks	Previous Year	Option 1	Option 2	Option 3	Option 4														
		(i) Find the probability of exactly 2 flaws in a 200-foot cable. (ii) Find the probability of exactly 1 flaw in the first 100 feet and exactly 1 flaw in the second 100 feet.																					
409	5	Potholes on a highway can be a serious problem. The past experience suggests that there are, on the average, 2 potholes per mile after a certain amount of usage. It is assumed that the Poisson process applies to the random variable “number of potholes.” What is the probability that no more than 4 potholes will occur in a given section of 5 miles?		3																			
410	5	If the mean of a Poisson variable is 1.8, find (i) $P(X > 1)$ (ii) $P(X = 5)$ (iii) $P(0 < X < 5)$.		3																			
411	5	If X is a Poisson variate such that $3P(X = 4) = \frac{1}{2}P(X = 2) + P(X = 0)$. Find (i) the mean of X (ii) $P(X \leq 2)$.		3																			
412	5	An insurance company insured 4000 people against loss of both eyes in a car accident. Based on previous data, the rates were computed on the assumption that on the average, 10 persons in 100000 will have car accidents each year that result in this type of injury. What is the probability that more than 3 of the insured will collect on their policy in a given year?		3																			
413	5	Assume that the probability of an individual coal miner being killed in a mine accident during a year is 1/2400. Use the Poisson distribution to calculate the probability that in a mine employing 200 miners, there will be at least one fatal accident in a year.		4																			
414	5	Suppose a book of 585 pages contains 43 typographical errors. If these errors are randomly distributed throughout the book, what is the probability that 10 pages, selected at random, will be free from errors?		3																			
415	5	The number of accidents in a year attributed to taxi drivers in a city follows Poisson distribution with a mean of 3. Out of 1000 taxi drivers, find approximately the number of drivers with (i) no accidents in a year (ii) more than 3 accidents in a year.		3	LJU-2022																		
416	5	Assuming that the typing mistakes per page committed by a typist follows a Poisson distribution, find the expected frequencies for the following distribution of typing mistakes: <table border="1"><tr><td>Number of mistakes per page</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>Number of pages</td><td>40</td><td>30</td><td>20</td><td>15</td><td>10</td><td>5</td></tr></table>	Number of mistakes per page	0	1	2	3	4	5	Number of pages	40	30	20	15	10	5		4					
Number of mistakes per page	0	1	2	3	4	5																	
Number of pages	40	30	20	15	10	5																	
417	5	A manufacturer of electric bulbs sends out 500 lots each consisting of 100 bulbs. If 5 % bulbs are defective, in how many lots can we expect (i) 97 or more good bulbs? (ii) less than 96 good bulbs?		3																			
418	5	In a certain factory producing certain articles, the probability that an article is defective is $\frac{1}{500}$. The articles are supplied in packets of 20. Find approximately the number of packets containing no		4																			

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		defective, one defective, two defectives in a consignment of 20000 packets.																											
419	5	Fit a Poisson distribution to the following data: <table> <tr> <td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td></tr> <tr> <td>f</td><td>56</td><td>156</td><td>132</td><td>92</td><td>37</td><td>22</td><td>4</td><td>0</td><td>1</td></tr> </table>	x	0	1	2	3	4	5	6	7	8	f	56	156	132	92	37	22	4	0	1		4					
x	0	1	2	3	4	5	6	7	8																				
f	56	156	132	92	37	22	4	0	1																				
420	5	If X is a Poisson variate such that $P(X = 0) = P(X = 1)$, find $P(X = 0)$ and using recurrence relation formula, find the probabilities at $x = 1, 2, 3, 4$ and 5.		3																									
421	5	If two cards are drawn from a pack of 52 cards which are diamonds, using Poisson distribution, find the probability of getting two diamonds at least 3 times in 51 consecutive trials of two cards drawing each time.		3																									
422	5	A manufacturer of blades knows that 5% of his product is defective. If he sells blades in boxes of 100, and guarantees that not more than 10 blades will be defective, what is the probability that a box will fail to meet the guaranteed quality?		3																									
423	5	Air Corporation having had just 2 air crashes during its first fifty years of existence wants to make the next decade “air crash-free”. Assuming that the same trend will continue, what is the probability of the corporation meeting the target?		3																									
424	5	If a random variable has a Poisson distribution such that $P(X = 1) = P(X = 2)$, find (i) the mean of the distribution, (ii) $P(X = 4)$, (iii) $P(X \geq 1)$, and (iv) $P(1 < X < 4)$.		4																									
425	5	If X is a Poisson variate such that $P(X = 2) = 9P(X = 4) + 90 P(X = 6)$. Find (i) the means of X , (ii) the variance of X , (iii) $P(X < 2)$, (iv) $P(X > 4)$, and (v) $P(X \geq 1)$.		5																									
426	5	If the probability that an individual suffers a bad reaction from a particular injection is 0.001, determine the probability that out of 2000 individuals (i) exactly three, and (ii) more than two individuals suffer a bad reaction.		3																									
427	5	It is known from past experience that in a certain plant, there are on the average 4 industrial accidents per year. Find the probability that in a given year, there will be less than 4 accidents. Assume Poison distribution.		3																									
428	5	Suppose that a local appliances shop has found from experience that the demand for tub lights is roughly distributed as Poisson with a mean of 4 tube lights per week. If the shop keeps 6 tube lights during a particular week, what is the probability that the demand will exceed the supply during that week?		3																									
429	5	A manufacturer, who produce medicine bottles, find that 0.1% of the bottles are defective. The bottles are packed in boxes containing 500 bottles. A drug manufacturer buys 100 boxes from the producer of bottles. Using Poisson distribution, find how many boxes will contain (i) No defective bottles and (ii) at least 2 defective bottles.		4																									

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430	6	A mobile conversation follows an exponential distribution $f(x) = \frac{1}{3}e^{-\frac{x}{3}}$. What is the probability that the conversation takes more than 5 minutes?	$e^{-\frac{5}{3}}$	1	LJU-2025	$e^{-\frac{5}{3}}$	e^{-15}	$5e^{-15}$	$\frac{e^{-5}}{3}$
431	6	A random variable X has an exponential distribution with probability distribution function is given by $f(x) = \begin{cases} 3e^{-3x} & \text{for } x > 0 \\ 0 & \text{otherwise} \end{cases}$ Find probability that X is not less than 2.	e^{-6}	1		e^{-3}	$e^{-6} - 3$	e^{-6}	$e^{-6} - 1$
432	6	The mean and variance of the density function $f(x) = 2e^{-2x}$ are	$\frac{1}{2}$ and $\frac{1}{4}$	1	LJU-2023	$\frac{1}{2}$ and $\frac{1}{4}$	$\frac{1}{4}$ and $\frac{1}{2}$	2 and 4	4 and 2
433	6	If X is random variable which follows an exponential distribution with parameter λ with $P(X \leq 1) = P(X > 1)$, what is $Var(x)$?	$\frac{1}{(\ln 2)^2}$	1	LJU-2022 LJU-2025	$\frac{1}{(\ln 2)^2}$	$\frac{1}{2}$	$\frac{1}{(\ln 2)}$	$\frac{1}{\lambda}$
434	6	The annual precipitation data of a city is normally distributed with mean and standard deviation as 1000 mm and 200 mm, respectively. The probability that the annual precipitation will be more than 1200 mm is,	<50%	1	LJU-2023	>50%	<50%	>75%	NONE
435	6	The normal probability density function curve is symmetrical about the mean, μ , i.e., the area to the right of the mean is the same as the area to the left of the mean. This means that $P(X < \mu) = P(X > \mu)$ is equal to:	0.5	1		0	1	0.5	0.25
436	6	The range of normal distribution is:	$-\infty$ to ∞	1		0 to n	0 to ∞	-1 to 1	$-\infty$ to ∞
437	6	If $X \sim N(100, 64)$, then standard deviation σ is ____.	8	1		100	64	8	36
438	6	If $X \sim N(200, 49)$, then standard deviation σ is ____.	7	1	LJU-2025	7	49	0	100
439	6	In a standard normal distribution, the area to the left of $Z = 1$ is:	0.8413	1		0.6413	0.7413	0.8413	0.3413
440	6	If Z is a standard normal variate, then $P(-2.33 \leq Z \leq 2.33)$ is equal to ____.	0.9802	1		0.4901	0.6827	0.9545	0.9802
441	6	If Z is a standard normal variate, then $P(-1.65 \leq Z \leq 1.65)$ is equal to ____.	0.901	1	LJU-2025	0.901	0.9520	0.9810	0.99
442	6	If Z is a standard normal variate, then $P(-2.58 \leq Z \leq 2.58)$ is equal to ____.	0.9902	1	LJU-2025	0.9951	0.9902	0.4951	0.4949
443	6	If Z is a standard normal variate, then $P(Z < 1.96)$ is equal to ____.	0.95	1		0.0250	0.4750	0.95	0.9750
444	6	For a normal distribution with $\mu = 10, \sigma = 2$, then probability of a value greater than 10 is ____.	0.5000	1		0.1915	0.3085	0.6915	0.5000
445	6	If x is normally distributed with mean 1 and variance 4, then obtain k if $P(x \leq k) = 0.90$	3.56	1	LJU-2025	3.56	2.56	1.56	1.645
446	6	The time to pass through a security screening at an airport follows an exponential distribution. The mean time to pass through the security screening is 15 minutes. To catch the flight, a passenger must clear the security screening within 15 minutes. The probability that the passenger will miss the flight is ____.	0.368	1	LJU-2024	0.368	1.921	0.863	None of these
447	6	A variable X is exponentially distributed for $x \geq 0$ with a mean of 1. The probability that the value of X will lie between 1 and 2 is ____.	$e^{-1} - e^{-2}$	1	LJU-2025	$e^{-1} - e^{-2}$	$e^1 - e^{-2}$	$e^{-1} - e^2$	$e^{-2} - e^{-1}$
448	6	A random variable P follows exponential distribution with mean value 0.5. The expectation of P^2 will be	0.50	1	LJU-2024 LJU-2025	0.50	$\frac{1}{50}$	0.30	$\frac{1}{30}$
449	6	Normal Distribution is symmetric about.....	Mean	1		Mean	Median	Standard Deviation	Variance

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450	6	A personal computer has the length of time between charges of the battery is normally distributed with a mean of 66 hours and a standard deviation of 20 hours. What is the probability when the length of time will be between 58 and 75 hours?	0.329	1		0.595	0.329	0.0443	1.98
451	6	Approximately what area is covered under the normal distribution curve between ± 3 standard deviation?	99.74%	1		88%	68.28%	99.74%	99.45%
452	6	An approximate area covered within two standard-deviation of the mean by standard normal variate is	95.45%	1	LJU-2023	65.25%	68.27%	95.45%	99.74%
453	6	Let X be a normal random variable with mean zero and variance 9. If $a = P(X \geq 3)$, then $P(X \leq 3)$ equal to....	$1 - 2a$	1	LJU-2022 LJU-2025	a	$2a$	$1 - a$	$1 - 2a$
454	6	Let X be normal variate with mean 0 and SD 5. If $P(X < 5) = k$ then $P(X \geq 5)$ will be	2-2k	1	LJU-2023	1-2k	2-2k	1-k	NONE
455	6	The mean of a normal distribution is the average of the first ten natural numbers then what is the probability of variable between 4 and 7 if $P(X \leq 4) = 1/5$	3/5	1		4/5	2/5	9/10	3/5
456	6	For a standard normal variate, the value of mean is?	0	1		1	0	∞	Not defined
457	6	The standard normal curve is symmetric about the value....	0	1		0.5	1	∞	0
458	6	For a standard normal probability distribution, the mean (μ) and the standard deviation (s)	$\mu=0$ and $s=1$	1		$\mu=0$ and $s=1$	$\mu=16$, $s=4$	$\mu=25$, $s=5$	$\mu=100$, $s=10$
459	6	If $Y = 5X + 10$ and X is $N(10,25)$, then mean of Y is?	60	1	LJU-2022 LJU-2024 LJU-2025	135	50	70	60
460	6	In a company, amount of light bills follows normal distribution with $\sigma = 60$. 11.31% of customers pay bill less than 260.The average amount of light bill is _____.	332.60	1	LJU-2025	132.60	223.60	332.60	232.60
461	6	If the distribution of a normal variable is shown as $N(20,4)$ then which of the following intervals includes 99.73% observations?	(14, 26)	1	LJU-2024	(13, 39)	(18, 22)	(16, 24)	(14, 26)
462	6	For Normal distribution with mean 2 and variance 9. Find the value of x such that the probability of the interval $[2, x]$ is 0.4115	6.05	1	LJU-2025	1.35	6.05	3.56	1.98
463	6	If X is a normal variate with a mean of 30 and an SD of 5, find (i) $P(26 \leq X \leq 40)$ (ii) $P(X \geq 45)$.		2					
464	6	If X is normally distributed with a mean of 2 and an SD of 0.1, find $P(X - 2 \geq 0.01)$?		2					
465	6	Assume that the mean height of Indian soldiers is 68.22 inches with a variance of 10.8 inches. How many soldiers in a regiment of 1000 would you expect to be over 6 feet tall?		3					
466	6	The marks obtained by students in a college are normally distributed with a mean of 65 and a variance of 25. If 3 students are selected at random from this college, what is the probability that at least one of them would have scored more than 75 marks?		3					
467	6	If X is normally distributed with a mean and standard deviation of 4, find (i) $P(5 \leq X \leq 10)$ (ii) $P(X \geq 15)$ (iii) $P(10 \leq X \leq 15)$ (iv) $P(X \leq 5)$.		4					
468	6	The compressive strength of samples of cement can be modelled by a normal distribution with a mean 6000 kilograms per square centimeter a standard deviation of 100 kilograms per square centimeter.		4					

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Sr. No.	unit_ number	Question text	Answer	Marks	Previous Year	Option 1	Option 2	Option 3	Option 4																
		(i)What is the probability that a sample's strength is less than 6250 Kg/cm²? (ii) What is the probability if sample strength is between 5800 and 5900 Kg/cm²? (iii) What strength is exceeded by 95% of the samples?																							
469	6	In a photographic process, the developing time of prints may be looked upon as a random variable having the normal distribution with a mean of 16.28 seconds and a standard deviation of 0.12 second. Find the probability that it will take (i) anywhere from 16.00 to 16.50 seconds to develop one of the prints; (ii) at least 16.20 seconds to develop one of the prints; (iii) at most 16.35 seconds to develop one of the prints.		4																					
470	6	Weights of 500 students of a college are normally distributed with average weight 95 lbs and standard deviation 7.5. find how many students have the weight between 100 and 110 lbs.		3																					
471	6	Let X be random variable with pdf $f(x) = \begin{cases} \frac{1}{5}e^{-\frac{x}{5}} & ; \ x > 0 \\ 0 & ; \ x \leq 0 \end{cases}$. Find (i) $P(X > 5)$ (ii) $P(3 \leq X \leq 6)$ (iii) mean (iv) variance		3																					
472	6	If the weights of 300 students are normally distributed with a mean of 68 kg and a standard deviation of 3 kg, how many students have weights (i) greater than 72 kg? (ii) less than or equal to 64 kg? (iii) between 65 kg and 71 kg inclusive?		4																					
473	6	What is the probability that a standard normal variate Z will be (i) greater than 1.09 ? (ii) less than -1.65 ? (iii) lying between -1 and 1.96? (iv) lying between 1.25 and 2.75?		4																					
474	6	The continuous random variable Z has a standard normal distribution. Calculate the probability of the following. (a) $Z < 1.3$ (b) $Z > 1.3$ (c) $Z > -1.3$ (d) $Z < -1.3$ (e) $-1.37 \leq Z \leq 2.01$ (f) $ Z \leq 0.5$ (g) $-1.79 \leq Z \leq -0.54$		5																					
475	6	<table border="1"><tr><td>Cla ss Inte rval</td><td>11 – 20</td><td>21 – 30</td><td>31 – 40</td><td>41 – 50</td><td>51 – 60</td><td>61 – 70</td><td>71 – 80</td></tr><tr><td>Fre que ncy</td><td>20</td><td>28</td><td>40</td><td>60</td><td>32</td><td>20</td><td>8</td></tr></table> Fit a normal curve from the following distribution. It is given that the mean of the distribution is 43.7 and its standard deviation is 14.8.	Cla ss Inte rval	11 – 20	21 – 30	31 – 40	41 – 50	51 – 60	61 – 70	71 – 80	Fre que ncy	20	28	40	60	32	20	8		4					
Cla ss Inte rval	11 – 20	21 – 30	31 – 40	41 – 50	51 – 60	61 – 70	71 – 80																		
Fre que ncy	20	28	40	60	32	20	8																		
476	6	The income distribution of officers of a certain company was found to follow normal distribution. The average income of an officer was Rs. 15,000. The standard deviation of the income of officers was Rs. 5,000. If there were 242 officers drawing salary above Rs. 18,500, how many officers were there in the company?		3																					

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Sr. No.	unit_number	Question text	Answer	Marks	Previous Year	Option 1	Option 2	Option 3	Option 4																														
477	6	The life -time in hours of a certain electrical equipment has the normal distribution with mean = 80 and standard deviation = 16. (i) What is the probability that the equipment lasts at least 100hours? (ii) If the equipment has already lasted 88 hours, what is the conditional probability that it will last at least another 12 hours?		3																																			
478	6	Find the area under the normal curve in each of the cases (I) In between $z = -0.68$ and $z = 0$; (II) In between $z = 0.81$ and $z = 1.94$; and (III) Right of $z = -1.28$.		3																																			
479	6	Fit a normal distribution to the following data. It is given that mean is 165.5 and standard deviation is 15.26. <table border="1"> <tr> <td>x</td><td>12</td><td>13</td><td>14</td><td>15</td><td>16</td><td>17</td><td>18</td><td>195</td><td>205</td></tr> <tr> <td></td><td>5</td><td>5</td><td>5</td><td>5</td><td>5</td><td>5</td><td>5</td><td></td><td></td></tr> <tr> <td>f</td><td>1</td><td>1</td><td>14</td><td>22</td><td>25</td><td>19</td><td>13</td><td>3</td><td>2</td></tr> </table>	x	12	13	14	15	16	17	18	195	205		5	5	5	5	5	5	5			f	1	1	14	22	25	19	13	3	2		3					
x	12	13	14	15	16	17	18	195	205																														
	5	5	5	5	5	5	5																																
f	1	1	14	22	25	19	13	3	2																														
480	6	A random variable having the normal distribution with $\mu = 18.2, \sigma = 1.25$, find the probabilities that it will take on a value (i) less than 16.5, (ii) Between 16.5 and 18.8.		3																																			
481	6	Distribution of height of 1000 soldiers is normally distributed with mean 165 cms and standard deviation 15 cms. How many soldiers are of height (i) less than 138 cms (ii) more than 198 cms (iii) between 138 and 198 cms.		3																																			
482	6	In a normal distribution, 31% item are under 45 and 8% are over 64. Find the mean and SD. Find also, the percentage of items lying between 30 and 75.		5																																			
483	6	Of a large group of men, 5% are under 60 inches in height and 40% are between 60 and 65 inches. Assuming a normal distribution, find the mean and standard deviation of distribution.		4																																			
484	6	In a large group of 400 females, 28% have a weight less than 55 kg, and 67% have a weight between 55 and 62 kg. Assuming a normal distribution, find the mean and standard deviation of the distribution. Also, determine how many females have a weight greater than 65 kg.		4	LJU-2025																																		
485	6	Of a large group of men, 15% are under 45 inches in height and 70% are between 45 and 65 inches. Assuming a normal distribution, find the mean and standard deviation of distribution.		4	LJU-2023																																		
486	6	The marks of 1000 students of a university are found to be normally distributed with a mean of 70 and a standard of deviation 5. Estimate the number of students whose marks will be (i) between 60 and 75, (ii) more than 75, and (iii) less than 68.		4																																			
487	6	The lifetime of a certain batteries has a mean life of 400 hours and the standard deviation as 45 hours. Assuming the distribution of lifetime to be normal, find (i) the percentage of batteries with a lifetime of at least 470 hours, (ii) the proportion of batteries with a lifetime between 385 and 415 hours, and (iii) the minimum life of the best 5% of batteries.		4																																			
488	6	The marks obtained by students in an examination follow a normal distribution. If 30% of the students got marks below 35 and 10% got marks above 60, find the mean and percentage of students who got marks between 40 and 50.		5	LJU-2022																																		

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Sr. No.	unit_ number	Question text	Answer	Marks	Previous Year	Option 1	Option 2	Option 3	Option 4																		
489	6	Find the mean and standard deviation in which 7% of items are under 35 and 89% are under 63.		4																							
490	6	If X is a normal variate with a mean of 120 and a standard deviation of 10, find c such that (1) $P(X > c) = 0.02$ and (2) $P(X < c) = 0.05$		4																							
491	6	If X is a normal variate with a mean of 25 and SD of 5, Find the value of $X = x_1$ such that $P(X \leq x_1) = 0.01$		4																							
492	6	If X is normal variate with a mean of 30 and an SD of 6, find the value of $X = x_1$ such that $P(X \leq x_1) = 0.05$.		4	LJU-2025																						
493	6	Fit a normal distribution to the following data: <table><tr><td>Cla ss</td><td>60-65</td><td>65-70</td><td>70-75</td><td>75-80</td><td>80-85</td><td>85-90</td><td>90-95</td><td>95-100</td></tr><tr><td>Fre qu enc y</td><td>3</td><td>21</td><td>150</td><td>335</td><td>326</td><td>135</td><td>26</td><td>4</td></tr></table>	Cla ss	60-65	65-70	70-75	75-80	80-85	85-90	90-95	95-100	Fre qu enc y	3	21	150	335	326	135	26	4		4					
Cla ss	60-65	65-70	70-75	75-80	80-85	85-90	90-95	95-100																			
Fre qu enc y	3	21	150	335	326	135	26	4																			
494	6	In an examination, it is laid down that a student passes if he secures 40% or more. He is placed in the first, second and third division according to whether he secures 60% or more marks, between 50% and 60% marks and between 40% and 50% marks respectively. He gets a distinction in case he secures 75% or more. It is noticed from the result that 10% of the students failed in the examination, whereas 5% of them obtained distinction. Calculate the percentage of students placed in the second division. (Assume Normal Distribution of marks.)		4																							
495	6	A random variable has pdf $f(x) = ce^{-2x}, x > 0$. Find (i) $P(X > 2)$ (ii) $P\left(X < \frac{1}{c}\right)$.		3																							
496	6	The mileage which car owners get with a certain kind of radial tire is a random variable having an exponential distribution with mean 4000 km. Find the probabilities that one of these tires will last (i) at least 2000 km (ii) at most 3000 km.		3																							
497	6	The average time it takes to serve a customer at a petrol pump is 6 minutes. The service time follows exponential distribution. Calculate the probability that (i) A customer will take less than 2 minutes to complete the service. (ii) A customer will take between 4 and 5 minutes to get the service. (iii) A customer will take more than 10 minutes for his service.		4																							
498	6	The daily consumption of milk in excess of 20000 gallons is approximately exponentially distributed $\lambda = \frac{1}{3000}$. The city has a daily stock of 35000 gallons. What is the probability that of 2 days selected at random, the stock is insufficient for both the days?		4	LJU-2025																						
499	6	The lifetime T of an alkline battery is exponentially distributed with $\lambda = 0.05$ per hour. (a) What are the mean are standard deviation of the battery’s lifetime? (b) What are the probabilities for the battery to last between 10 and 15 hours and to last more than 20 hours?		4																							

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Sr. No.	unit_number	Question text	Answer	Marks	Previous Year	Option 1	Option 2	Option 3	Option 4
500	6	The time between breakdown of a particular machine follows an exponential distribution with a mean of 17 days. Calculate the probability that a machine breaks down in a 15 day period.		3					
501	6	The time (in hours) required to repair a machine is exponentially distributed with parameter $\lambda = \frac{1}{2}$. (i) What is the probability that the repair time exceed 2 hours? (ii) What is the conditional probability that a repair takes at least 11 hours given that its direction exceeds 8 hours?		3					
502	6	The Life length X of an electronic component follows an exponential distribution. These are 2 Processes by which the component may be manufactured. The Expected life length of component is 100 hrs if process I is used to manufacture while it is 150 hrs if process II is used. The cost of manufacturing a single component by process I is Rs. 10, while is Rs.20 for Process II. Moreover, if the component lasts less than the guaranteed life of 200 hrs, a loss of Rs. 50 is to be borne by the manufacturer. Which process is Advantageous to the manufacturer?		5	LJU-2022 LJU-2023				
503	6	The length of the shower on a tropical island during rainy season has an exponential distribution, with parameter $\lambda = 2$, time being measured in minutes. What is the probability that a shower will last more than three minutes? If a shower has already lasted for 2 minutes, what is the probability that it will last for at least one more minute?		3					
504	6	The life of an electronic component follows exponential distribution with a mean of 4 years. The manufacturer of this component gives a replacement warranty of 3 years. (a) What proportion of components will be replaced in the period of warranty? (b) What is the probability that a randomly selected component will have life within two standard deviations of the mean life?		4	LJU-2025				
505	6	If X is an exponentially distributed random variable with parameter λ , find the value of k such that $P(X > k) = aP(X \leq k)$		3					
506	6	The length of time X to complete a job is exponentially distributed with $E(X) = \mu = 10$ hours. (i) Compute the probability of job completion between two consecutive jobs exceeding 20 hours. (ii) The cost of job completion is given by $C = 4 + 2X + 4X^2$. Find the expected value of C.		3	LJU-2024				
507	6	In a newly constructed township, 2000 electric lamps are installed with an average life of 1000 burning hours and standard deviation of 200 hours. Assuming the life of the lamps follows normal distribution, find (i) The number of lamps expected to fail during the first 700 hours.(ii) In which period of burning hours 10% of the lamps fail.		3	LJU-2024				
508	7	A quality control expert is required to estimate the mean thickness of aluminum sheets used in the production of airframes. A sample of 100 sheets reveal a mean of 0.048 inches with a standard deviation of 0.01 inches. The 99 per cent confidence interval is _____.	(0.0454, 0.0506)	1	LJU-2025	(0.0454, 0.0506)	(0.1464, 0.1516)	(0.6545, 0.8478)	(0.0641, 0.0825)

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Sr. No.	unit_number	Question text	Answer	Marks	Previous Year	Option 1	Option 2	Option 3	Option 4
509	7	The point where the Null Hypothesis gets rejected is called as?	Critical Value	0.5		Significant Value	Rejection Value	Acceptance Value	Critical Value
510	7	In a random sample of 180 workers exposed to a certain amount of radiation, 19 experienced some ill effects. 99% confidence interval for the corresponding true percentage is_____	(0.0466, 0.1646)	1	LJU-2025	(0.0512, 0.1732)	(0.0466, 0.1646)	(0.4660, 0.1646)	(0.5120, 0.1732)
511	7	A random sample of 300 shoppers at a supermarket includes 204 who regularly uses cents off coupons. In another sample of 500 shoppers at a supermarket includes 75 who regularly uses cents off coupons. Obtain 95% confidence limits for the difference in the population proportions.	(0.469, 0.591)	1	LJU-2025	(0.469, 0.591)	(0.053, 0.968)	(0.591, 0.947)	(0.875, 1.435)
512	7	In a random sample of 160 Worker exposed to a certain amount of radiation, 24 experienced some ill effects. What is a 95% confidence limit interval for corresponding true percentage?	(0.0947, 0.2053)	1	LJU-2022 LJU-2025	(0.0947, 0.2053)	(0.947, 0.0205)	(0.15, 0.85)	(1.96, 2.58)
513	7	Before an increase in excise duty on tea, 800 people out of a sample of 1000 were consumers Of tea. After an increase in excise duty, 800 people were consumers of tea in a sample of 1200 persons. On the claim of significant decrease in the consumption of tea after the Increase in duty what is your alternative hypothesis and value of Z (Test statistics)?	$P_1 > P_2$, $Z = 6.84$	1	LJU-2022 LJU-2025	$p_1 = p_2$, $Z = 2.64$	$P_1 = P_2$, $Z = 6.84$	$p_1 < p_2$, $Z = 6.84$	$P_1 > P_2$, $Z = 6.84$
514	7	A random sample of 400 items is found to have a mean of 82 and standard deviation of 18. Confidence limit of 95% is_____	(80.236 , 83.764)	1	LJU-2024	(69.852 , 71.253)	(80.236 , 83.764)	(91.264 , 94.862)	(22.963 , 26.454)
515	7	A buyer of electric bulbs bought 100 bulbs each of two famous brands A and B. Upon testing both these samples, he found that brand A had a mean life of 1500 hours with a standard deviation of 50 hours whereas brand B had an average life of 1530 hours with a standard deviation of 60 hours. In this test of hypothesis, what is the value of the test statistics?	$Z = -3.84$	1	LJU-2025	$Z = -1.87$	$Z = 1.56$	$Z = -3.84$	$Z = 2.22$
516	7	A coin was tossed 960 times and returned heads 183 times. Test the hypothesis that the coin is unbiased. Use a 0.05 level of significance.		4					
517	7	A dice is tossed 960 times and it falls with 5 upwards 184 times. Is the dice unbiased at a level of significance of 0.01?		4					
518	7	A die is thrown 600 times and the digit 2 or 4 is considered as success. Digit 2 or 4 are obtained for 212 times. Is a die unbiased?		4	LJU-2025				
519	7	A manufacturer claims that at least 95% of the equipment which he supplied to a factory conformed to specification. An examination of a sample of 200 pieces of equipment revealed that 18 were faulty. Test his claim at 5% level of significance.		4					
520	7	In a hospital 480 female and 520 male babies were born in a week. Do these figures confirm the hypothesis that males and female were born in equal numbers?		4					
521	7	In a study designed to investigate whether certain detonators used with explosive in a coal mining meet the requirement that at least 90% will ignite the explosive when charged. It is found that 174 of 200 detonators function properly. Test the null hypothesis $P = 0.9$ against the alternative hypothesis $P < 0.9$ atthe 0.05 level of significance.		4					

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Sr. No.	unit_number	Question text	Answer	Marks	Previous Year	Option 1	Option 2	Option 3	Option 4
522	7	A salesman in a departmental store claim that at most 60 percent of the shoppers entering the store leave without making a purchase. A random sample 50 shoppers showed that 35 of them left without making a purchase. Are these sample results consistent with the claim of the salesman? Use a level of significance of 0.05.		4					
523	7	The fatality rate of typhoid patients is believed to be 17.26%. In a certain year 640 patients suffering from typhoid were treated in a metropolitan hospital and only 63 patients died. Can you consider the hospital efficient at 1% level of significance?		4					
524	7	In a big city, 325 men out of 600 were found to be smokers. Does this information support the conclusion that the majority of men in this city are smokers?		4					
525	7	A manufacturer claims at least 95% of the items he produces are failure free. Examinations of a random sample of 600 items showed 39 to be defective. Test the claim at a significance level of 0.05.		4					
526	7	In a sample of 1000 in Karnataka, 540 are rice eaters and the rest are wheat eaters. Can we assume that both rice and wheat are equally popular in this state at 1% level of significance?		4					
527	7	Marketers believe that 92% of adults in the United States own a cell phone. A cell phone manufacturer believes that number is actually lower. 200 American adults are surveyed, of which, 174 report having cell phones. Use a 5% level of significance. State the null and alternative hypothesis, state your conclusion.		4					
528	7	A sample of 600 persons selected at random from a large city shows that the percentage of male in the sample is 53%. It is believed that male to the total population ratio in the city is $\frac{1}{2}$. Test whether this belief is confirmed by the observation.		4					
529	7	A survey claims that 9 out of 10 doctors recommend aspirin for their patients with headaches. To test this claim, a random sample of 100 doctors is obtained. Of these 100 doctors, 82 indicate that they recommend aspirin. Is this claim accurate at 5% level of significance?		4					
530	7	Random samples of 400 men and 600 women were asked whether they would like to have a flyover near their residence 200 men and 325 women were in favor of the proposal. Test the hypothesis that proportions of men and women in favor of the proposal are same at 5% level of significance.		4					
531	7	In a city A, 20% of a random sample of 900 school boys has a certain slight physical defect. In another city B, 18.5% of a random sample of 1600 school boys has the same defect. Is the difference between the proportions significant at 0.05 level of significance?		4					
532	7	In a college, a teacher observed that on Saturdays, 72% of students wear black clothing. To verify this observation, a sample of 20 classes with 35 students each was taken, and it was found that 68.6% of the students wore black clothing. Test the		4	LJU-2025				

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Sr. No.	unit_ number	Question text	Answer	Marks	Previous Year	Option 1	Option 2	Option 3	Option 4						
		hypothesis at the 1% level of significance that the teacher’s observation is too high.													
533	7	Before an increase in excise duty on tea, 800 people out of a sample of 1000 were consumers of tea. After an increase in excise duty, 800 people were consumers of tea in a sample of 1200 persons. Find whether is significant decrease in the consumption of tea after the increase in duty.		4											
534	7	15.5% of a random sample of 1600 undergraduates’ smokers, whereas 20% of a random sample of 900 postgraduates were smokers in a state. Can we conclude that less number of undergraduates are smokers than the postgraduates?		4	LJU-2023										
535	7	A manufacturer of electronic equipment subject samples of two completing brands of transistors to an accelerated performance test. If 45 of 180 transistors of the first kind and 34 of 120 transistors of second kind fail the test, what can be conclude at the level of significance $\alpha = 0.05$ about the difference between the corresponding sample proportion?		4	LJU-2025										
536	7	A machine produced 20 defective articles in a batch of 400. After overhauling it produced 10 defective articles in a batch of 300. Has the machine improved?		4											
537	7	Time magazine reported the result of a telephone poll of 800 adult Americans. The question posed of the Americans who were surveyed was: "Should the federal tax on cigarettes be raised to pay for health care reform?" The results of the survey were:<table><tr><td>Non- Smokers</td><td>Smokers</td></tr><tr><td>$n_1 = 605$</td><td>$n_2 = 195$</td></tr><tr><td>$y_1 = 351$ said ‘yes’</td><td>$y_2 = 41$ said ‘yes’</td></tr></table>Is there sufficient evidence at the $\alpha = 0.05$, say, to conclude that the two populations — smokers and non-smokers — differ significantly with respect to their opinions?	Non- Smokers	Smokers	$n_1 = 605$	$n_2 = 195$	$y_1 = 351$ said ‘yes’	$y_2 = 41$ said ‘yes’		4					
Non- Smokers	Smokers														
$n_1 = 605$	$n_2 = 195$														
$y_1 = 351$ said ‘yes’	$y_2 = 41$ said ‘yes’														
538	7	In two large populations, there are 30% and 25% fair haired people respectively. Is this difference likely to be hidden in samples of 1200 and 900 respectively from the two populations?		4											
539	7	A company has the head office at Kolkata and a branch at Mumbai. The personnel director wanted to know if the workers at the two places would like the introduction of a new plan of work and a survey was conducted for this purpose. Out of a sample of 500 workers at Kolkata, 62% favoured the new plan. At Mumbai, out of a sample of 400 workers, 41% were against the new plan. Is there any significant difference between the two groups in their attitude towards the new plan at 5% level?		4											
540	7	On the basis of their total scores, 200 candidates of a civil service examination are divided into two groups, the upper 30% and the remaining 70%. Consider the first question of the examination. Among the first group ,40 had the correct answer, whereas among the second group,80 had the correct answer. On the basis of this question likely to be useful for discriminating the ability of the type being tested?		4											

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Sr. No.	unit_number	Question text	Answer	Marks	Previous Year	Option 1	Option 2	Option 3	Option 4
541	7	A teacher believes that 85% of students in the class will want to go on a field trip to the local zoo. She performs a hypothesis test to determine if the percentage is the same or different from 85%. The teacher samples 50 students and 39 reply that they would want to go to the zoo. For the hypothesis test, use a 5% level of significance.		4					
542	7	A random sample of 100 Indians has an average life span of 71.8 years with standard deviation of 8.9 years. Can it be concluded that the average life span of an Indian is 70 years?		4					
543	7	A random sample of boots worn by 36 soldiers in desert region showed an average life of 1.08 years with a standard deviation of 0.6 years. Under the standard conditions, the boots are known to have an average life of 1.28 years. Is there a reason to assert, at 1% level of significance, that use in desert caused the mean life of such boots to decrease? What will be your conclusion if the level of significance is 5%? Assume that the life of boots is normally distributed.		4	LJU-2025				
544	7	The mean life time of sample of 100 fluorescent light bulbs produced by a company is computed to be 1570 hours with a standard deviation of 120 hours. The company claims that the average life of the bulbs produced by it is 1600 hours. Using the level of significance of 0.05, is the claim acceptable?		4					
545	7	A random sample of 50 items gives the mean 6.2 and variance 10.24. can it be regarded as drawn from a normal population with mean 5.4 at 5% level of significance?		4					
546	7	A random sample of 400 members is found to have a mean of 4.45 cm. can it be reasonably regarded as a sample from a large population whose mean is 5 cm and variance is 4 cm?		4					
547	7	A sample of 900 members has a mean of 3.4 cm and SD 2.61 cm. Is the sample from a large population of mean 3.25 cm and SD 2.61 cm? If the population is normal and its mean is unknown, find the 95% fiducial limits of its true mean.		4					
548	7	A type company claims that the lives of tyres have mean 42000 km with s.d. of 4000 km. A change in the production process is believed to result in better product. A test sample of 81 new tyres has a mean life of 42500 km. test at 5% level of significance that the new product is significantly better than the old one.		4					
549	7	The mean length of the lumber is supposed to be 8.5 feet. A builder wants to check whether the shipment of lumber she receives has a mean length different from 8.5 feet. If the builder observes that the sample mean of 61 pieces of lumber is 8.3 feet with a sample standard deviation of 1.2 feet. What will she conclude? Is 8.3 very different from 8.5?		4	LJU-2022				
550	7	The mean IQ of a sample of 1600 children was 99. Is it likely that this was a random sample from a population with mean IQ 100 and SD 15.		4					
551	7	An insurance agent claims that the average age of the policy-holders who insure through him is less than the average age for all the agents, which is		4	LJU-2025				

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		Note: This practice book is only for reference purpose. LJU Test question paper may not be completely set from this practice book.																			
Sr. No.	unit_ number	Question text	Answer	Marks	Previous Year	Option 1	Option 2	Option 3	Option 4												
		30.5 years. A random sample of 100 policy-holders who had insured through him gave the adjoining age distribution <table><tr><td>Age Last Birthday(years)</td><td>Number of persons</td></tr><tr><td>16-20</td><td>12</td></tr><tr><td>21-25</td><td>22</td></tr><tr><td>26-30</td><td>20</td></tr><tr><td>31-35</td><td>30</td></tr><tr><td>36-40</td><td>16</td></tr></table> Calculate the mean and standard deviation of the sample and use it to test the claim that the average is less than the figure for all the agents at 5% level of significance.	Age Last Birthday(years)	Number of persons	16-20	12	21-25	22	26-30	20	31-35	30	36-40	16							
Age Last Birthday(years)	Number of persons																				
16-20	12																				
21-25	22																				
26-30	20																				
31-35	30																				
36-40	16																				
552	7	The mean breaking strength of cables supplied by a manufacturer is 1800 with standard deviation 100. By a new technique in the manufacturing process, it is claimed that the breaking strength of the cable has increased. In order to test the claim a sample of 50 cables is tested. It is found that the mean breaking strength is 1850. Can we support the claim at 1% level of significance?		4																	
553	7	It is claimed that a random sample of 49 types have a mean life of 15200 km. This sample was drawn from a population whose mean is 15150 km and a standard deviation of 1200 km. Test the significance at 0.05 level.		4																	
554	7	Test the significance of the difference between the means of two normal populations with the same standard deviation from the following data: <table><tr><td></td><td>Size</td><td>Mean</td><td>SD</td></tr><tr><td>Sample I</td><td>100</td><td>64</td><td>6</td></tr><tr><td>Sample II</td><td>200</td><td>67</td><td>8</td></tr></table>		Size	Mean	SD	Sample I	100	64	6	Sample II	200	67	8		4					
	Size	Mean	SD																		
Sample I	100	64	6																		
Sample II	200	67	8																		
555	7	The mean of simple samples of sizes 1000 and 2000 are 67.5 and 68 cm respectively. can the samples be regarded as drawn from the same population of S.D. 2.5 cm.		4																	
556	7	The average of marks scored by 32 boys is 72 with standard deviation 8 while that of 36 girls is 70 with standard deviation 6. Test at 1% level of significance whether the boys perform better than the girls.		4																	
557	7	A simple sample of heights of 6400 English men has a mean of 170 cm and a s.d. of 6.4 cm, while a simple sample of heights of 1600 Americans has a mean of 172 cm and a s.d. of 6.3 cm. do the data indicate that American are, on the average, taller than the English men?		4																	
558	7	In a certain factory there are two different processes of manufacturing the same item. The average weight in a sample of 250 items produced from one process is found to be 120 gm with a s.d. of 12 gm; the corresponding figures in a sample of 400 items from the other process are 124 gm and 14 gm. Is this difference between the two sample means significant?		4																	

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Sr. No.	unit_ number	Question text	Answer	Marks	Previous Year	Option 1	Option 2	Option 3	Option 4									
559	7	The mean height of 50 male students who participate in sports is 68.2 inches with a s.d. of 2.5 inches. The mean height of 50 male students who have not participated in sport is 67.2 inches with a s.d. of 2.8 inches. Test the hypothesis that the height of students who have participated in sports is more than the students who have not participated in sports.		4	LJU-2024													
560	7	A researcher wants to know the intelligence of students in a school. He selected two groups of students. In the first group, there are 150 students having mean IQ of 75 with a SD of 15. In the second group there are 250 students having mean IQ of 70 with SD of 20. Test at 1% level of significance whether the difference between the mean IQs of the two groups is significant.		4														
561	7	In order to make a survey of the buying habits, two markets <i>A</i> and <i>B</i> are chosen at two different parts of a city. 400 women shoppers are chosen at random in market <i>A</i> . Their average daily expenditure on food is found to be <i>Rs.</i> 250 with standard deviation of <i>Rs.</i> 40. The figures are <i>Rs.</i> 220 and <i>Rs.</i> 55 respectively in the market <i>B</i> where also 400 women shoppers are chosen at random. Test at 1% level of significance whether the average daily food expenditure of the two populations of shoppers are equal.		4														
562	7	A buyer of electric bulbs bought 100 bulbs each of two famous brands A and B. Upon testing both these samples, he found that brand A had a mean life of 1500 hours with a standard deviation of 50 hours whereas brand B had an average life of 1530 hours with a standard deviation of 60 hours. Can it be concluded at 5% level of significance that the two brands differ significantly in quality?		3														
563	7	The SD of a random samples of 1000 is found to be 2.6 and the SD of another random sample of 500 is 2.7. Assuming the samples to be independent, find whether the two samples could have come from populations with the same SD.		4														
564	7	Random samples drawn from two countries gave the following data relating to the heights of adult males: <table><tr><td></td><td>Country A</td><td>Country B</td></tr><tr><td>Standard deviation (in inches)</td><td>2.58</td><td>2.50</td></tr><tr><td>Number in samples</td><td>1000</td><td>1200</td></tr></table> Is the difference between the standard deviation significant?		Country A	Country B	Standard deviation (in inches)	2.58	2.50	Number in samples	1000	1200		4					
	Country A	Country B																
Standard deviation (in inches)	2.58	2.50																
Number in samples	1000	1200																
565	7	The SD of a random sample of 900 members is 4.6 and that of another independent sample of 1600 members is 4.8. Examine if the two samples could have been drawn from a population with SD 4?		4	LJU-2022 LJU-2025													
566	7	The variance of a random sample of 125 members is 11.56 and that of another independent sample of 175 members is 14.44. Test at 1% level of significance whether the difference between the two population SD is significant.		4	LJU-2023 LJU-2024													

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Sr. No.	unit_ number	Question text	Answer	Marks	Previous Year	Option 1	Option 2	Option 3	Option 4									
567	7	Explain whether the two samples for which the data are given in the following table could have been drawn from populations with the same SD. <table><tr><td></td><td>Size</td><td>SD</td></tr><tr><td>Sample I</td><td>100</td><td>5</td></tr><tr><td>Sample II</td><td>200</td><td>7</td></tr></table>		Size	SD	Sample I	100	5	Sample II	200	7		4					
	Size	SD																
Sample I	100	5																
Sample II	200	7																
568	7	A college claims that its average class size is 35 students. A random sample of 64 classes has a mean size of 37 students with a standard deviation of 6 students. Test at the $\alpha = 0.05$ level of significance if the claimed value is too low.		4														
569	7	A political party claims that 45% of the voters in an election district prefer its candidate. A sample of 200 voters include 80 who prefer this candidate. Test if the claims is valid at the 5% significance level.		4														
570	7	A stenographer claims that she can type at the rate of 120 words per minute. She demonstrated, on the basis of 100 trials, an average speed of 116 words with a standard deviation of 15 words. Does this enable us to reject the null hypothesis $\mu = 120$ against the alternative hypothesis $\mu < 120$ at the 0.05 level of significance?		4														
571	7	If 57 out of 150 patients suffering from certain disease are cured by allopathy and 33 out Of 100 patients with the same disease are cured by homeopathy. Is there reason to believe that allopathy is better than homeopathy at 0.05 level of significance?		4														
572	7	The mean yield of two sets of plots and their variability are as given below. Examine whether the difference in the mean yields of the two sets of plots is significant at 5% level of significance. <table><tr><td></td><td>Set of 40 plots</td><td>Set of 60 plots</td></tr><tr><td>Mean yield per plot</td><td>1258 lb</td><td>1243 lb</td></tr><tr><td>Standard Deviation per plot</td><td>34</td><td>28</td></tr></table>		Set of 40 plots	Set of 60 plots	Mean yield per plot	1258 lb	1243 lb	Standard Deviation per plot	34	28		4					
	Set of 40 plots	Set of 60 plots																
Mean yield per plot	1258 lb	1243 lb																
Standard Deviation per plot	34	28																
573	7	500 units from a factory are inspected and 12 are found to be defective, 800 units from another factory are inspected and 12 are found to be defective. Can it be concluded at 5% level of significance that production at second factory is better that in first factory.		4														
574	7	In an advertisement, a pizza shop claims that the mean delivery is less than 30 minutes. A random selection of 36 delivery times a sample mean of 28.5 minutes and a standard deviation of 3.5 minutes. Is there enough evidence to support the claim at 5%?		4	LJU-2023													
575	7	A weighing machine without any display was used by an average of 320 persons per day with a standard deviation of 50 persons. After installing an attractive display, data were collected for 100 days, and it was observed that the average number of users increased by 15 persons per day. Using a 5% level of significance, test whether the installation of the attractive display significantly increased the average number of users.		4	LJU-2024													

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Sr. No.	unit_number	Question text	Answer	Marks	Previous Year	Option 1	Option 2	Option 3	Option 4														
576	7	Two groups <i>A</i> and <i>B</i> consist of 100 people each who have a disease. A serum is given to group <i>A</i> but not to group <i>B</i> . It is found that in groups <i>A</i> and <i>B</i> ,75 and 65 people, respectively, recover from the disease. Test the hypothesis that the serum helps to cure the disease. Also find 99% of confidence limit.		4	LJU-2024																		
577	8	A random sample of 10 pairs of observation has mean and variance are 160 and 16 respectively. What are the 95% confidence limits the population?	(156.984, 163.016)	1		(156.984, 163.016)	(120.371, 159.190)	(147.101, 150.924)	(150.283, 165.942)														
578	8	Arrange the following steps in the process of hypothesis testing in proper sequence (a) Select level of significance (b) Setup null and alternative hypothesis (c) Establish the decision rule (d) Performance computations (e) Select test statistics (f) Draw conclusion Choose the correct answer from the options given below	B,A,E,C, D,F	1		A,B,C,D,E,F	A,B,E,D,C,F	B,A,E,C,D, F	B,A,C,D,E,F														
579	8	In order to examine the significance of difference between any two small samples means, a researcher follows	T test	1	LJU-2023	F test	T test	Z test	Chi-square test														
580	8	What is the mean of a Chi Square distribution with 18 degrees of freedom?	18	1		18	9	17	10														
581	8	The variance of a Chi Square distribution with sample size 9 is ____.	16	1	LJU-2022 LJU-2023 LJU-2024 LJU-2025	2	4	10	16														
582	8	The variance of a Chi Square distribution with sample size 19 is ____.	36	1	LJU-2025	18	19	36	20														
583	8	A random sample of size 16 has mean 53. The sum of squares of deviations from mean is 150. 95% confidence limits for the mean is ____.	(51.315, 54.686)	1	LJU-2022	(49.314, 52.316)	(51.315, 54.686)	(51.982, 53.789)	(50.123, 54.987)														
584	8	A dice was thrown 264 times and the following frequencies were observed: <table border="1"><tr><td>No obtained</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>Frequency</td><td>40</td><td>32</td><td>28</td><td>50</td><td>54</td><td>60</td></tr></table> Calculate the value of chi-square at 5% level of significance is	No obtained	1	2	3	4	5	6	Frequency	40	32	28	50	54	60	18.36	1	LJU-2025	18.36	14.36	15.36	17.36
No obtained	1	2	3	4	5	6																	
Frequency	40	32	28	50	54	60																	
585	8	A random sample of size 20 has mean 63. The sum of squares of deviations from mean is 180. Then 99% confidence limits for the mean is ____.	(61.5595 , 64.4405)	1	LJU-2024 LJU-2025	(51.3151, 54.6862)	(49.3143, 52.3165)	(61.5595 , 74.4405)	(61.5595 , 64.4405)														
586	8	A sample of 18 high-performance batteries is tested and the mean lifetime is found to be 135.6 hours with a variance of 9.2 hours. The manufacturer claims that the average lifetime of their batteries is 153.6 hours. What is the value of the test statistic?	t = - 24.468	1	LJU-2025	t = -13.213	t = -24.468	t = -10.312	t = 9.234														
587	8	A sample of 26 bulbs gives a mean life of 990 hrs with a SD of 20 hrs. the manufacturer claims that the mean life of bulbs is 1000 hrs. Is the sample not up to standard?		4																			
588	8	The average breaking strength of the steel rods is specified to be 18.5 thousand pounds. To test this, sample of 14 rods were tested. The mean and SD obtained were 17.85 and 1.955 respectively. Is the result of experiment significant?		3																			

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Sr. No.	unit_number	Question text	Answer	Marks	Previous Year	Option 1	Option 2	Option 3	Option 4
589	8	A soap manufacturing company was distributing a particular brand of soap through a large number of retail shops. Before a heavy advertisement campaign, the mean sales per week per shop were 140 dozen. After the campaign, a sample of 26 shops was taken and the mean sale was found to be 147 dozen with standard deviation 16. Can you consider the advertisement effective?		4					
590	8	A sample of 20 items has mean 42 units and standard deviation 5 units, test the hypothesis that it is a random sample from a normal population with mean 45 units.		4					
591	8	A random sample of size 16 from a normal population showed a mean of 103.75 cm and sum of square of deviation from the mean 843.75 cm^2 can we say that the population has a mean of 108.75 cm.		4					
592	8	A random sample of six steels beams has mean compressive strength of 58392 psi (pounds per square inch) with a SD of 648 psi use this information and level of significance $\alpha = 0.05$ to test whether the true average compressive strength of the steel from which this sample came is 58000 psi. Assume normality.		4					
593	8	A machine is designed to produce insulating washers for electrical devices of average thickness of 0.025 cm. A random sample of 10 washers was found to have an average thickness of 0.024 cm with S.D of 0.002 cm. test the significance of the deviation.		4					
594	8	The mean lifetime of a sample of 25 bulbs is found as 1550 hrs with a SD of 120 hrs. The company manufacturing the bulbs claim that the average life of their bulbs is 1600hrs is the claim acceptance at 5% level of significance?		4					
595	8	A random sample from a company's very extensive files shows that orders for a certain piece of machinery were filled, respectively, in 10, 12, 19, 14, 15, 18, 11 and 13 days. Use the level of significance $\alpha = 0.01$ to test the claim that on average such orders are filled in 10.5 days. Choose the alternative hypothesis so that rejection of the null hypothesis $\mu = 10.5$ implies that it takes longer than indicated. Assume normality.		4					
596	8	A random sample of 10 boys had the following IQs: - 70,120,110,101,88,83,95,98,107 and 100. (a) Do these data support the assumption of a population mean IQ of 100? (b) Find 95% confidence limits for the mean IQ?		4					
597	8	Ten objects are chosen at random from a large population and their weight are found to be in gms:- 63,63,64,65,66, 69, 69, 70, 70, 71 discuss the suggestion that mean weight is 65 gm.		4					
598	8	The height of 10 males of a given locality are 70, 67, 62, 68, 61, 68, 70, 64, 64, 66 inches is it reasonable to believe that the average height is greater than 64 inches. Test at 5% significance level.		4					
599	8	A courier service advertises that its average delivery time is less than 5 hrs. for local deliveries. A random sample of 10 for the amount of time this		4					

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Sr. No.	unit_ number	Question text	Answer	Marks	Previous Year	Option 1	Option 2	Option 3	Option 4																						
		courier service takes to deliver packages to an address across town produced the following times: 8, 3,4,7,10,5,6,4,5,8. Is this evidence support the claim of the courier service at 5% level of significance?																													
600	8	Ten bearing made by a certain process have a mean diameter of 0.506 cm and a SD 0.004 cm Assuming that the data may be looked upon as a random variable from a normal population construct a 95% confidence interval for the actual average diameter of bearing made by this process.		4																											
601	8	The height of 10 males of a given locality are found to be 175, 168, 155, 170, 152, 170, 175, 160, 160 and 165 cm. Based on this sample of 10 items, Test the hypothesis that the mean height of males is 170 cm. Also find 95% confidence limits for the height of males in that locality.		3																											
602	8	Two types of batteries are tested for their length of life and the following data is obtained. <table border="1"><tr><td></td><td>No of samples</td><td>Mean life (hours)</td><td>Variance</td></tr><tr><td>Type A</td><td>9</td><td>500</td><td>100</td></tr><tr><td>Type B</td><td>8</td><td>540</td><td>121</td></tr></table> Is there a significant difference in the two means?		No of samples	Mean life (hours)	Variance	Type A	9	500	100	Type B	8	540	121		3															
	No of samples	Mean life (hours)	Variance																												
Type A	9	500	100																												
Type B	8	540	121																												
603	8	Samples of size 10 & 14 were taken from two normal populations with standard deviation 3.5 & 5.2 the sample means were found to be 20.3 & 18.6. Test whether the mean of two population are same at 5% level.		4																											
604	8	Two HEAD samples of 8 & 7 items respectively had the following values of the variable (weight in kg.) <table border="1"><tr><td>Sa mp le I</td><td>9</td><td>11</td><td>13</td><td>11</td><td>15</td><td>9</td><td>12</td><td>14</td></tr><tr><td>Sa mp leII</td><td>10</td><td>12</td><td>10</td><td>14</td><td>9</td><td>8</td><td>10</td><td></td></tr></table> Do the two estimates of population variance differ significantly?	Sa mp le I	9	11	13	11	15	9	12	14	Sa mp leII	10	12	10	14	9	8	10			4									
Sa mp le I	9	11	13	11	15	9	12	14																							
Sa mp leII	10	12	10	14	9	8	10																								
605	8	Two samples of size 9 & 8 give the sum of squares of deviations from their respective means equal 160 inches & 91 inches square respectively. Can they be regarded as drawn from two normal populations with the same variance.		4																											
606	8	A random sample of 10 nations gives a correlation coefficient of 0.5 between literacy rate and political stability. Is the relationship significant?		4																											
607	8	The following figures refer to observations in live independent samples <table border="1"><tr><td>Sa mp le I</td><td>25</td><td>30</td><td>28</td><td>34</td><td>24</td><td>20</td><td>13</td><td>32</td><td>22</td><td>38</td></tr><tr><td>Sa mp le II</td><td>40</td><td>34</td><td>22</td><td>20</td><td>31</td><td>40</td><td>30</td><td>23</td><td>36</td><td>17</td></tr></table> Analyses whether the samples have been drawn from the population of equal means test whether the means of two populations are same at 5% level.	Sa mp le I	25	30	28	34	24	20	13	32	22	38	Sa mp le II	40	34	22	20	31	40	30	23	36	17		4					
Sa mp le I	25	30	28	34	24	20	13	32	22	38																					
Sa mp le II	40	34	22	20	31	40	30	23	36	17																					

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Sr. No.	unit_ number	Question text	Answer	Marks	Previous Year	Option 1	Option 2	Option 3	Option 4																																																																																								
608	8	To examine the hypothesis that the husbands are more intelligent than wives, an investigator took sample of two couples and administered them a test which measures the IQ. The results are as follows: <table><tr><td>H</td><td>1</td><td>1</td><td>9</td><td>1</td><td>1</td><td>1</td><td>8</td><td>7</td><td>1</td><td>1</td></tr><tr><td>us</td><td>1</td><td>0</td><td>7</td><td>0</td><td>2</td><td>0</td><td>6</td><td>8</td><td>0</td><td>0</td></tr><tr><td>ba</td><td>7</td><td>5</td><td></td><td>5</td><td>3</td><td>9</td><td></td><td></td><td>3</td><td>7</td></tr><tr><td>n</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>ds</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>w</td><td>1</td><td>9</td><td>8</td><td>1</td><td>1</td><td>9</td><td>9</td><td>6</td><td>1</td><td>8</td></tr><tr><td>iv</td><td>0</td><td>8</td><td>7</td><td>0</td><td>1</td><td>5</td><td>0</td><td>9</td><td>0</td><td>5</td></tr><tr><td>es</td><td>6</td><td></td><td></td><td>4</td><td>6</td><td></td><td></td><td></td><td>8</td><td></td></tr></table> Test the hypothesis with reasonable test level of significance of 0.05	H	1	1	9	1	1	1	8	7	1	1	us	1	0	7	0	2	0	6	8	0	0	ba	7	5		5	3	9			3	7	n											ds											w	1	9	8	1	1	9	9	6	1	8	iv	0	8	7	0	1	5	0	9	0	5	es	6			4	6				8			4					
H	1	1	9	1	1	1	8	7	1	1																																																																																							
us	1	0	7	0	2	0	6	8	0	0																																																																																							
ba	7	5		5	3	9			3	7																																																																																							
n																																																																																																	
ds																																																																																																	
w	1	9	8	1	1	9	9	6	1	8																																																																																							
iv	0	8	7	0	1	5	0	9	0	5																																																																																							
es	6			4	6				8																																																																																								
609	8	A random sample of 27 pairs of observations from a bivariate normal population gives a correlation coefficient of 0.42. Is it likely that the variable are uncorrelated in the population?		4																																																																																													
610	8	Find the least value of r in a sample of 27 pairs from a bivariate normal population which is significant at 5% level.		3																																																																																													
611	8	Find the least value of r in samples of 18 pairs of observations from a bivariate normal population, which is significant at 5% level		3																																																																																													
612	8	The means of two random samples of sizes 9 and 7 are 196.42 and 198.82 respectively. The sums of squares of the deviation from the mean are 26.94 and 18.73 respectively. Can the sample be considered to have been drawn from the same population?		3																																																																																													
613	8	A random sample of 18 pairs of observation from a bivariate normal population gives a correlation coefficient of 0.3 is it likely that variables are uncorrelated in the population?		3																																																																																													
614	8	A certain injection administered to 12-patients resulted in the following changes of blood pressure: 5, 2, 8, -1, 3, 0, 6, -2, 1, 5, 0, 4. Can it be concluded that the injection will be in general accompanied by an increase in blood pressure?		3																																																																																													
615	8	The mean height and SD height of 8 randomly chosen soldiers are 166.9 cm and 8.29 cm respectively. The corresponding values of 6 randomly chosen sailors are 170.3 cm and 8.50 cm respectively. Based on this data, can we conclude that soldiers are, in general, shorter than sailors?		4																																																																																													
616	8	The sales in 6 shops before and after “the more sale campaign” are shown below: <table><tr><td>Shop</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>Sale s befo re Cam paig n</td><td>53</td><td>28</td><td>32</td><td>48</td><td>50</td><td>42</td></tr><tr><td>Sale s after Cam paig n</td><td>58</td><td>32</td><td>30</td><td>50</td><td>56</td><td>45</td></tr></table>	Shop	1	2	3	4	5	6	Sale s befo re Cam paig n	53	28	32	48	50	42	Sale s after Cam paig n	58	32	30	50	56	45		3	LJU-2023																																																																							
Shop	1	2	3	4	5	6																																																																																											
Sale s befo re Cam paig n	53	28	32	48	50	42																																																																																											
Sale s after Cam paig n	58	32	30	50	56	45																																																																																											

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Sr. No.	unit_ number	Question text	Answer	Marks	Previous Year	Option 1	Option 2	Option 3	Option 4																										
		Can we say that the campaign is successful?																																	
617	8	Two samples are drawn from two normal populations. From the following data test whether the two samples have the same variance at 5% level? <table><tr><td>Sa m pl e I</td><td>60</td><td>65</td><td>71</td><td>74</td><td>76</td><td>82</td><td>85</td><td>87</td><td></td><td></td></tr><tr><td>Sa m pl e II</td><td>61</td><td>66</td><td>67</td><td>85</td><td>78</td><td>63</td><td>85</td><td>86</td><td>88</td><td>91</td></tr></table>	Sa m pl e I	60	65	71	74	76	82	85	87			Sa m pl e II	61	66	67	85	78	63	85	86	88	91		4									
Sa m pl e I	60	65	71	74	76	82	85	87																											
Sa m pl e II	61	66	67	85	78	63	85	86	88	91																									
618	8	Two nicotine contents in two random samples of tobacco are given below: <table><tr><td>Sam ple I</td><td>21</td><td>24</td><td>25</td><td>26</td><td>27</td><td></td></tr><tr><td>Sam ple II</td><td>22</td><td>27</td><td>28</td><td>30</td><td>31</td><td>36</td></tr></table> Can we say that two samples came from the same population?	Sam ple I	21	24	25	26	27		Sam ple II	22	27	28	30	31	36		4	LJU-2022 LJU-2025																
Sam ple I	21	24	25	26	27																														
Sam ple II	22	27	28	30	31	36																													
619	8	Two random samples are drawn from two populations and the following results were obtained: <table><tr><td>Sa mp le I</td><td>16</td><td>17</td><td>18</td><td>19</td><td>20</td><td>21</td><td>22</td><td>24</td><td>26</td><td>27</td><td></td><td></td></tr><tr><td>Sa mp le II</td><td>19</td><td>22</td><td>25</td><td>25</td><td>26</td><td>28</td><td>29</td><td>30</td><td>31</td><td>32</td><td>35</td><td>36</td></tr></table> Find the variances of the two samples and test whether the two populations have the same variances.	Sa mp le I	16	17	18	19	20	21	22	24	26	27			Sa mp le II	19	22	25	25	26	28	29	30	31	32	35	36		4					
Sa mp le I	16	17	18	19	20	21	22	24	26	27																									
Sa mp le II	19	22	25	25	26	28	29	30	31	32	35	36																							
620	8	The time taken by workers in performing a job by method I and method II is given below <table><tr><td>Met hod I</td><td>20</td><td>16</td><td>26</td><td>27</td><td>22</td><td></td><td></td></tr><tr><td>Met hod II</td><td>27</td><td>33</td><td>42</td><td>35</td><td>32</td><td>34</td><td>38</td></tr></table> Do the data show that the variance of time distribution in a population from which these samples are drawn do not differ significantly?	Met hod I	20	16	26	27	22			Met hod II	27	33	42	35	32	34	38		4															
Met hod I	20	16	26	27	22																														
Met hod II	27	33	42	35	32	34	38																												
621	8	A group of 10 rats fed on diet A and another group of 8 rats fed o diet B recorded following increase in weight: <table><tr><td>Di et A</td><td>5</td><td>6</td><td>8</td><td>11</td><td>12</td><td>4</td><td>3</td><td>9</td><td>6</td><td>10</td></tr><tr><td>Di et B</td><td>2</td><td>3</td><td>6</td><td>8</td><td>11</td><td>10</td><td>2</td><td>8</td><td></td><td></td></tr></table> Find, if the variances are significantly different?	Di et A	5	6	8	11	12	4	3	9	6	10	Di et B	2	3	6	8	11	10	2	8				4									
Di et A	5	6	8	11	12	4	3	9	6	10																									
Di et B	2	3	6	8	11	10	2	8																											
622	8	In a test given to two groups of students drawn from two normal populations, the marks obtained were as follows:		4																															

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		Note: This practice book is only for reference purpose. LJU Test question paper may not be completely set from this practice book.																																
Sr. No.	unit_ number	Question text										Answer	Marks	Previous Year	Option 1	Option 2	Option 3	Option 4																
		Gr ou p I	18	20	36	50	49	36	34	49	41																							
		Gr ou p II	29	28	26	35	30	44	46																									
		Examine at 5% level, whether the two populations have the same variances.																																
623	8	The standard deviations calculated from two random samples of sizes 9 & 13 are 2.1 & 1.8 respectively. Test whether there is a significant difference between the variances of the two populations at 5% level of significance?											4																					
624	8	In two independent samples of sizes 8 & 10, the sum of square of deviations of samples values from the respective means was 84.4 and 102.6. Test whether the difference of variances of the population is significant or not. Use a 0.05 level of significance.											4																					
625	8	In a laboratory experiment two samples gave the following results: <table><tr><td>Sample</td><td>size</td><td>Sample Mean</td><td>Sum of square of deviation from the mean</td></tr><tr><td>1</td><td>10</td><td>15</td><td>90</td></tr><tr><td>2</td><td>12</td><td>14</td><td>108</td></tr></table> Test the equality of sample variances at 5% level of significance.										Sample	size	Sample Mean	Sum of square of deviation from the mean	1	10	15	90	2	12	14	108		4									
Sample	size	Sample Mean	Sum of square of deviation from the mean																															
1	10	15	90																															
2	12	14	108																															
626	8	Two random samples gave the following data: <table><tr><td></td><td>size</td><td>mean</td><td>variance</td></tr><tr><td>Sample I</td><td>8</td><td>9.6</td><td>1.2</td></tr><tr><td>Sample II</td><td>11</td><td>16.5</td><td>2.5</td></tr></table> Can we conclude that the two samples have been drawn from the same normal population?											size	mean	variance	Sample I	8	9.6	1.2	Sample II	11	16.5	2.5		4									
	size	mean	variance																															
Sample I	8	9.6	1.2																															
Sample II	11	16.5	2.5																															
627	8	Following results were obtained from two samples, each drawn from two different population A & B <table><tr><td>Population</td><td>A</td><td>B</td></tr><tr><td>Sample</td><td>I</td><td>II</td></tr><tr><td>Sample size</td><td>15</td><td>17</td></tr><tr><td>Sample SD</td><td>3</td><td>2</td></tr></table> Test the equality of sample variances at 5% level of significance.										Population	A	B	Sample	I	II	Sample size	15	17	Sample SD	3	2		4									
Population	A	B																																
Sample	I	II																																
Sample size	15	17																																
Sample SD	3	2																																
628	8	Five dice are thrown 192 times and the number of times 4,5 or 6 are obtained are as follows: <table><tr><td>No. of dice showing 4,5,6</td><td>5</td><td>4</td><td>3</td><td>2</td><td>1</td><td>0</td></tr><tr><td>Frequency</td><td>6</td><td>46</td><td>70</td><td>48</td><td>20</td><td>2</td></tr></table> Calculate χ^2										No. of dice showing 4,5,6	5	4	3	2	1	0	Frequency	6	46	70	48	20	2		4							
No. of dice showing 4,5,6	5	4	3	2	1	0																												
Frequency	6	46	70	48	20	2																												

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		Note: This practice book is only for reference purpose. LJU Test question paper may not be completely set from this practice book.																															
Sr. No.	unit_ number	Question text	Answer	Marks	Previous Year	Option 1	Option 2	Option 3	Option 4																								
629	8	A pair of dice are thrown 360 times and frequency of each sum is given below: <table border="1"><tr><td>Su m</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td><td>11</td><td>12</td></tr><tr><td>Fr eq</td><td>8</td><td>24</td><td>35</td><td>37</td><td>44</td><td>65</td><td>51</td><td>42</td><td>26</td><td>14</td><td>14</td></tr></table> Would you say that the dice are fair on the basis of chi-square test at 0.05 level of significance?	Su m	2	3	4	5	6	7	8	9	10	11	12	Fr eq	8	24	35	37	44	65	51	42	26	14	14		4					
Su m	2	3	4	5	6	7	8	9	10	11	12																						
Fr eq	8	24	35	37	44	65	51	42	26	14	14																						
630	8	The following mistakes per page were observed in a book: <table border="1"><tr><td>No. of mista kes per page</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>No. of pages</td><td>211</td><td>90</td><td>19</td><td>5</td><td>0</td></tr></table> Fit a Poisson distribution and test the goodness of fit.	No. of mista kes per page	0	1	2	3	4	No. of pages	211	90	19	5	0		4																	
No. of mista kes per page	0	1	2	3	4																												
No. of pages	211	90	19	5	0																												
631	8	The number of car accidents in a metropolitan city was found to be 20, 17, 12, 6, 7, 15, 8, 5, 16 and 14 per month respectively. Use χ^2 test to check whether these frequencies are in agreement with the belief that the occurrence of accidents was the same during 10 months periods. Test at occurrence of accidents was the same during 10 months periods. Test at 5% level of significance.		4																													
632	8	200 digits were chosen at random from a set of tables, the frequency of the digits are shown below: <table border="1"><tr><td>Digits</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td></tr><tr><td>frequenc y</td><td>18</td><td>19</td><td>23</td><td>21</td><td>16</td><td>25</td><td>22</td><td>20</td><td>21</td><td>15</td></tr></table> Use the χ^2 -test to access the correctness of the hypothesis that the digits were distributed in equal number in the tables from which these were chosen.	Digits	0	1	2	3	4	5	6	7	8	9	frequenc y	18	19	23	21	16	25	22	20	21	15		4							
Digits	0	1	2	3	4	5	6	7	8	9																							
frequenc y	18	19	23	21	16	25	22	20	21	15																							
633	8	A die is thrown 276 times and the results of these throws are given below: - <table border="1"><tr><td>Num ber appe ared on the die</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>Freq uenc y</td><td>40</td><td>32</td><td>29</td><td>59</td><td>57</td><td>59</td></tr></table> Test whether the dice is unbiased.	Num ber appe ared on the die	1	2	3	4	5	6	Freq uenc y	40	32	29	59	57	59		4															
Num ber appe ared on the die	1	2	3	4	5	6																											
Freq uenc y	40	32	29	59	57	59																											
634	8	A die is thrown 132 times and the results of these throws are given below: - <table border="1"><tr><td>Num ber appe ared on the die</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr></table>	Num ber appe ared on the die	1	2	3	4	5	6		4																						
Num ber appe ared on the die	1	2	3	4	5	6																											

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		Note: This practice book is only for reference purpose. LJU Test question paper may not be completely set from this practice book.																																											
Sr. No.	unit_ number	Question text							Answer	Marks	Previous Year	Option 1	Option 2	Option 3	Option 4																														
		f	15	20	25	15	29	28																																					
		Test whether the dice is unbiased.																																											
635	8	4 coins are tossed 160 times and the following results were obtained. <table border="1"><tr><td>No. of heads</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>Observed frequencies</td><td>17</td><td>52</td><td>54</td><td>31</td><td>6</td></tr></table> Under the assumption that coin is balanced, find the expected frequencies of 0, 1,2,3 or 4 heads, and test the goodness of fit.							No. of heads	0	1	2	3	4	Observed frequencies	17	52	54	31	6		4																							
No. of heads	0	1	2	3	4																																								
Observed frequencies	17	52	54	31	6																																								
636	8	Fit a Poisson distribution to the following data and its goodness of fit at level of significance 0.05: <table border="1"><tr><td><i>X</i></td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td><i>f</i></td><td>419</td><td>352</td><td>154</td><td>56</td><td>19</td></tr></table>							<i>X</i>	0	1	2	3	4	<i>f</i>	419	352	154	56	19		4																							
<i>X</i>	0	1	2	3	4																																								
<i>f</i>	419	352	154	56	19																																								
637	8	Fit the equation of the best fitting normal curve to the following data: <table border="1"><tr><td>X</td><td>13</td><td>14</td><td>15</td><td>16</td><td>17</td><td>18</td><td>19</td><td>20</td><td>Total</td></tr><tr><td></td><td>5</td><td>5</td><td>5</td><td>5</td><td>5</td><td>5</td><td>5</td><td>5</td><td></td></tr><tr><td>f</td><td>2</td><td>14</td><td>22</td><td>25</td><td>19</td><td>13</td><td>3</td><td>2</td><td>100</td></tr></table> Compare the theoretical and observed frequencies. Using χ^2 -test find goodness of fit. Given that $\mu = 165.6$ and $\sigma = 15.02$.							X	13	14	15	16	17	18	19	20	Total		5	5	5	5	5	5	5	5		f	2	14	22	25	19	13	3	2	100		4					
X	13	14	15	16	17	18	19	20	Total																																				
	5	5	5	5	5	5	5	5																																					
f	2	14	22	25	19	13	3	2	100																																				
638	8	The following table gives the number of accidents in a city during a week. Find the accidents are uniformly distributed over a week. Use χ^2 test. <table border="1"><tr><td>Day</td><td>sun</td><td>mon</td><td>tue</td><td>wed</td><td>thu</td><td>fri</td><td>sat</td><td>total</td></tr><tr><td>No. of accidents</td><td>13</td><td>15</td><td>9</td><td>11</td><td>12</td><td>10</td><td>14</td><td>84</td></tr></table>							Day	sun	mon	tue	wed	thu	fri	sat	total	No. of accidents	13	15	9	11	12	10	14	84		4																	
Day	sun	mon	tue	wed	thu	fri	sat	total																																					
No. of accidents	13	15	9	11	12	10	14	84																																					
639	8	The distribution of defects in printed circuit board is hypothesized to follow Poisson distribution. A random sample of 60 printed boards shows the following data: <table border="1"><tr><td>No. of defects</td><td>0</td><td>1</td><td>2</td><td>3</td></tr><tr><td>Observed frequency</td><td>32</td><td>15</td><td>9</td><td>4</td></tr></table> Does the hypothesis of Poisson distribution appropriate?							No. of defects	0	1	2	3	Observed frequency	32	15	9	4		4																									
No. of defects	0	1	2	3																																									
Observed frequency	32	15	9	4																																									
640	8	Theory predicts that the proportion of beans in the four group A, B, C, D should be 9:3:3:1 In an experiment among 1600 beans, the numbers in the four groups were 882, 313, 287 and 118. Does the experimental result support the theory?								4																																			
641	8	Weights in kilograms of 10 students are given below: 38, 40, 45, 53, 47,43, 55, 48, 52, 49 can we say that the mean of the distribution from which the above sample is drawn is 20 kg?								4																																			

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Sr. No.	unit_ number	Question text	Answer	Marks	Previous Year	Option 1	Option 2	Option 3	Option 4																								
642	8	Can vaccination be regarded as preventive measure of smallpox as evidenced by the following data of 1482 persons exposed to smallpox in a locality. 368 in all were attacked of these 1482 persons and 343 were vaccinated and of these only 35 were attacked.		4	LJU-2022																												
643	8	A set of five similar coins is tossed 320 times and result is obtained as follows: <table><tr><td>No of head s</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>Freq uenc y</td><td>6</td><td>27</td><td>72</td><td>112</td><td>71</td><td>32</td></tr></table> Test the hypothesis that the data follow a Binomial distribution.	No of head s	0	1	2	3	4	5	Freq uenc y	6	27	72	112	71	32		4															
No of head s	0	1	2	3	4	5																											
Freq uenc y	6	27	72	112	71	32																											
644	8	From the following data, find whether there is any significant linking in the habit of taking soft drinks among the categories of employees. <table><tr><td rowspan="2">Soft drink</td><td colspan="3">Employees</td></tr><tr><td>clerks</td><td>Teachers</td><td>Officers</td></tr><tr><td>Pepsi</td><td>10</td><td>25</td><td>65</td></tr><tr><td>Thums up</td><td>15</td><td>30</td><td>65</td></tr><tr><td>Fanta</td><td>50</td><td>60</td><td>30</td></tr></table>	Soft drink	Employees			clerks	Teachers	Officers	Pepsi	10	25	65	Thums up	15	30	65	Fanta	50	60	30		4										
Soft drink	Employees																																
	clerks	Teachers	Officers																														
Pepsi	10	25	65																														
Thums up	15	30	65																														
Fanta	50	60	30																														
645	8	A random sample of 500 students were classified according to economic condition of their family and also according to merit as shown below: <table><tr><td rowspan="2">merit</td><td colspan="3">Economic condition</td><td rowspan="2">total</td></tr><tr><td>rich</td><td>Middle class</td><td>poor</td></tr><tr><td>meritori ous</td><td>42</td><td>137</td><td>61</td><td>240</td></tr><tr><td>Not-meritori ous</td><td>58</td><td>113</td><td>89</td><td>260</td></tr><tr><td>total</td><td>100</td><td>250</td><td>150</td><td>500</td></tr></table> Test whether the two attributes merit and economic condition are associated or not.	merit	Economic condition			total	rich	Middle class	poor	meritori ous	42	137	61	240	Not-meritori ous	58	113	89	260	total	100	250	150	500		4	LJU-2024					
merit	Economic condition			total																													
	rich	Middle class	poor																														
meritori ous	42	137	61	240																													
Not-meritori ous	58	113	89	260																													
total	100	250	150	500																													
646	8	Two researchers adopted different sampling techniques while investigating some group of students falling into different intelligence level. The result are as follows: <table><tr><td>resea rcher s</td><td>Below avera ge</td><td>avera ge</td><td>Abov e avera ge</td><td>geniu s</td><td>total</td></tr><tr><td>X</td><td>86</td><td>60</td><td>44</td><td>10</td><td>200</td></tr><tr><td>Y</td><td>40</td><td>33</td><td>25</td><td>2</td><td>100</td></tr><tr><td>total</td><td>126</td><td>93</td><td>69</td><td>12</td><td>300</td></tr></table> Would you say that the sampling techniques adopted by the two researchers are significantly different?	resea rcher s	Below avera ge	avera ge	Abov e avera ge	geniu s	total	X	86	60	44	10	200	Y	40	33	25	2	100	total	126	93	69	12	300		4					
resea rcher s	Below avera ge	avera ge	Abov e avera ge	geniu s	total																												
X	86	60	44	10	200																												
Y	40	33	25	2	100																												
total	126	93	69	12	300																												

		L.J Institute of Engineering and Technology, Ahmedabad. <u>Introduction to Probability Theory and Stochastic Processes</u> Practice Book (Sem- III)																											
		Note: This practice book is only for reference purpose. LJU Test question paper may not be completely set from this practice book.																											
Sr. No.	unit_ number	Question text	Answer	Marks	Previous Year	Option 1	Option 2	Option 3	Option 4																				
647	8	Records taken of the number of male and female births in 830 families having four children are as follows: <table><tr><td>Numb er of male births</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>Numb er of femal e births</td><td>4</td><td>3</td><td>2</td><td>1</td><td>0</td></tr><tr><td>Numb er of famili es</td><td>32</td><td>178</td><td>290</td><td>236</td><td>94</td></tr></table> Test whether the data are consistent with the hypothesis that the Binomial law holds and the chance of male birth is equal to that of female birth, namely $p = q = 1/2$.	Numb er of male births	0	1	2	3	4	Numb er of femal e births	4	3	2	1	0	Numb er of famili es	32	178	290	236	94		4							
Numb er of male births	0	1	2	3	4																								
Numb er of femal e births	4	3	2	1	0																								
Numb er of famili es	32	178	290	236	94																								
648	8	A total of 3759 individuals were interviewed in a public opinion survey on a political proposal. Of them 1872 were men and the rest were women. A total of 2257 individuals were in favor of the proposal and 917 were opposed to it. A total of 243 men were undecided and 442 women were opposed to it. Do you justify or contradict the hypothesis that there is no association between gender and attitude at 5% level of significance?		4																									
649	8	A drug X claimed to be effective in curing colds. In an experiment on 500 persons with cold, half of them were given the drug X and half of them were given placebo (sugar pills). The patient’s reactions to the treatment are recorded in the following table: <table><tr><td>Treatm ent</td><td>Helped</td><td>Reactio n</td><td>No effect</td><td>Total</td></tr><tr><td>Drug</td><td>150</td><td>30</td><td>70</td><td>250</td></tr><tr><td>Placebo</td><td>130</td><td>40</td><td>80</td><td>250</td></tr><tr><td>Total</td><td>280</td><td>70</td><td>150</td><td>500</td></tr></table> Can it be concluded that there is a significant difference in the effect of drug X and placebo?	Treatm ent	Helped	Reactio n	No effect	Total	Drug	150	30	70	250	Placebo	130	40	80	250	Total	280	70	150	500		3	LJU-2023				
Treatm ent	Helped	Reactio n	No effect	Total																									
Drug	150	30	70	250																									
Placebo	130	40	80	250																									
Total	280	70	150	500																									
650	8	A random sample of 9 values from a normal population has a mean of 41.5 inches and sum of squares of deviations from the mean equal to 135 inches. Another sample of 12 values from an unknown population has a mean of 43 inches and sum of squares of deviations from their mean is equal to 171 inches. Show that the two samples may be regarded as coming from the same normal population.		4	LJU-2024																								

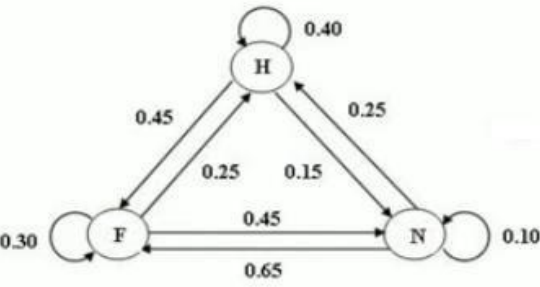
		L.J Institute of Engineering and Technology, Ahmedabad. <u>Introduction to Probability Theory and Stochastic Processes</u> Practice Book (Sem- III)																											
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Sr. No.	unit_ number	Question text	Answer	Marks	Previous Year	Option 1	Option 2	Option 3	Option 4																				
651	8	A survey was conducted on a total of 1223 individuals to examine the relationship between age group and exercise frequency. Among the respondents, 843 were aged between 18 and 30 years, 282 were between 31 and 50 years, and the remaining were aged 51 years or older. In terms of exercise habits, 560 people reported exercising regularly, while 48 indicated that they never exercise. Additional details reveal that, in the 51+ age group, 64 individuals exercise regularly and 13 never exercise. In the 31–50 age group, 230 people exercise occasionally and in the 18–30 age group, 35 individuals never exercise. Test at the 5% significance level, whether there is an association between age group and exercise frequency.		6	LJU-2025																								
652	8	A survey was conducted among 3600 employees of a large organization to study whether job designation is related to job satisfaction. The employees were classified into three groups as Junior, Middle-level, and Senior. Out of the total employees, 40% were Junior staff, 1200 were Middle-level staff, and the remaining were Senior staff. Overall, 1500 employees were satisfied with their jobs, 1100 employees were Neutral, and the remaining employees were dissatisfied. It was further observed that among Junior employees, the number of satisfied employees exceeded the number of dissatisfied employees by 300. Among Middle-level employees, 350 employees were Neutral. Among Senior employees, 320 employees were dissatisfied. The total number of satisfied employees in the Junior and Middle-level groups together was 1000. At the 5% level of significance, test whether job designation and job satisfaction are independent or whether there is an association between them.		5	LJU-2025																								
653	8	Table below shows the performances of students in Mathematics and Physics. Test the hypothesis that the performance in Mathematics is independent of performance in Physics. <table><tr><td>Grades in Physics</td><td colspan="3">Grades in Mathematics</td></tr><tr><td></td><td>High</td><td>Medium</td><td>Low</td></tr><tr><td>High</td><td>56</td><td>71</td><td>12</td></tr><tr><td>Medium</td><td>47</td><td>163</td><td>38</td></tr><tr><td>Low</td><td>14</td><td>42</td><td>81</td></tr></table>	Grades in Physics	Grades in Mathematics				High	Medium	Low	High	56	71	12	Medium	47	163	38	Low	14	42	81		6	LJU-2025				
Grades in Physics	Grades in Mathematics																												
	High	Medium	Low																										
High	56	71	12																										
Medium	47	163	38																										
Low	14	42	81																										
654	9	Fit the straight line to the following data: <table><tr><td>x</td><td>1</td><td>2</td><td>3</td></tr><tr><td>y</td><td>1</td><td>2</td><td>3</td></tr></table>	x	1	2	3	y	1	2	3	$y = x$	1	LJU-2023	$y = x$	$y = x + 1$	$y = 2x$	$y = 2x + 1$												
x	1	2	3																										
y	1	2	3																										
655	9	Fit the straight line to the following data: <table><tr><td>x</td><td>0</td><td>5</td></tr><tr><td>y</td><td>7</td><td>11</td></tr></table>	x	0	5	y	7	11	$y = 0.8x + 7$	1		$y = 0.94x + 6.6$	$y = 0.04x + 5$	$y = 0.8x + 7$	$y = 5.6x + 0.04$														
x	0	5																											
y	7	11																											
656	9	Fit a straight line for the given pairs of (x, y) which are (0,3), (1,6).	$y = 3 + 3x$	1		$y = 4 + 3x$	$y = 3.52 + 2.26x$	$y = 3 + 3x$	$y = 2.26x$																				
657	9	If the normal equations for a straight-line $y = ax + b$ are $26 = 4a + 6b$ and $34 = 6a + 4b$ then fit the above straight line.	$y = 5x + 1$	1	LJU-2022	$y = 5x - 1$	$y = 5x + 1$	$y = x + 5$	$y = x - 5$																				
658	9	If the normal equations for a straight-line $y = a + bx$ are $12 = 8a - 6b$ and $12 = -6a + 54b$ then fit the above straight line.	$y = 1.8182 + 0.4242x$	1	LJU-2025	$y = -1.8182 + 0.4242x$	$y = 1.8182 - 0.4242x$	$y = -1.8182 - 0.4242x$	$y = 1.8182 + 0.4242x$																				

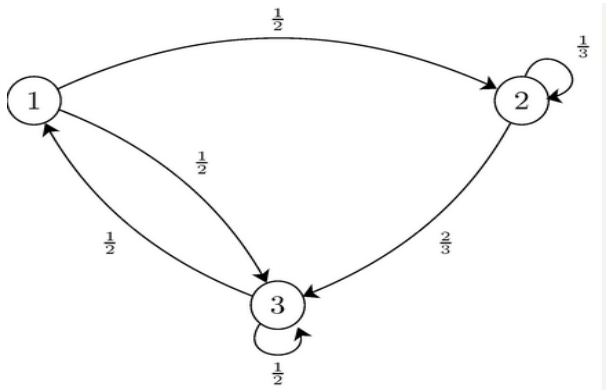
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Sr. No.	unit_ number	Question text	Answer	Marks	Previous Year	Option 1	Option 2	Option 3	Option 4																
659	9	Fit the parabola $y = a + bx + cx^2$, if their normal equations are $9a + 60c = 11$, $60b = 51$ and $60a + 708c = -9$.	$y = 3.0042 + 0.85x - 0.2673x^2$	1	LJU-2022	$y = -3.0042 + 0.85x - 0.2673x^2$	$y = 0.85 + 3.0042x - 0.2673x^2$	$y = -0.85 + 3.0042x - 0.2673x^2$	$y = 3.0042 + 0.85x - 0.2673x^2$																
660	9	What would be the value of the co-efficient of x when you fit a quadratic curve to the given data? <table><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>y</td><td>1.7</td><td>1.8</td><td>2.3</td><td>3.2</td></tr></table>	x	1	2	3	4	y	1.7	1.8	2.3	3.2	-0.5	1	LJU-2025	-0.5	2	1	0						
x	1	2	3	4																					
y	1.7	1.8	2.3	3.2																					
661	9	If the normal equations for a straight-line $y = a + bx$ are $135 = 6a + 21b$ and $561 = 21a + 91b$ then fit the above straight line.	$y = 4.8 + 5.05x$	1	LJU-2025	$y = 4.8 - 5.05x$	$y = 3.2 - 5x$	$y = 4.8 + 5.05x$	$y = -3.2 + 5x$																
662	9	Fit the parabola $y = a + bx + cx^2$, if their normal equations are $5a + 10b + 30c = 30$, $10a + 30b + 100c = 120$, $30a + 100b + 354c = 434$	$y = -4 + 2x + x^2$	1	LJU-2024	$y = 4 + 2x - x^2$	$y = 4 + x + 2x^2$	$y = -4 - 2x + x^2$	$y = -4 + 2x + x^2$																
663	9	If the normal equations for a straight-line $y = a + bx$ are $204 = 5a + 15b$ and $748 = 15a + 55b$ then fit the above straight line.	$y = 13.6x$	1	LJU-2025	$y = x$	$y = -13.6x$	$y = 13.6x$	$y = -13.6$																
664	9	What would be the coefficient of x^2 when you fit a quadratic curve to the following data? <table><tr><td>x</td><td>70</td><td>20</td><td>10</td></tr><tr><td>y</td><td>20</td><td>70</td><td>90</td></tr></table>	x	70	20	10	y	20	70	90	0.0167	1	LJU-2023	0.0167	113.3	-2.5	1.5								
x	70	20	10																						
y	20	70	90																						
665	9	Fit a straight line for the given pairs of (x, y) which are (1,15), (3,6)	$y = 19.5 - 4.5x$	1	LJU-2024	$y = 19.5 - 4.5x$	$y = -2.78 + 0.21x$	$y = 2.78 - 0.21x$	$y = 19.5 + 4.5x$																
666	9	The parabola equation $y = a + bx + cx^2$, if their normal equations are $5a + c = 81$, $5b + 6c = 11$ and $6a + b = 61$ then,	$a = 13$, $b = -17$, $c = 16$	1	LJU-2025	$a = 12$, $b = 12$, $c = 16$	$a = 13$, $b = -17$, $c = 16$	$a = 13$, $b = -12$, $c = 16$	$a = 13$, $b = 12$, $c = -11$																
667	9	If f(x) is a quadratic curve such that $f(0) = 1, f(1) = 2.5, f(2) = 6, f(3) = 10.5, f(4) = 16$, find the quadratic curve $f(x)$ using the least squares method. Then, predict the value of $f(8)$.	51.8601	1	LJU-2025	53.8617	51.8601	50.2533	51.6375																
668	9	Find the relation of the type $R = aV + b$, when some values of R and V obtained from an experiment are <table><tr><td>V</td><td>60</td><td>65</td><td>70</td><td>75</td><td>80</td><td>85</td><td>90</td></tr><tr><td>R</td><td>109</td><td>114</td><td>118</td><td>123</td><td>127</td><td>130</td><td>133</td></tr></table>	V	60	65	70	75	80	85	90	R	109	114	118	123	127	130	133		3					
V	60	65	70	75	80	85	90																		
R	109	114	118	123	127	130	133																		
669	9	The results of a measurement of electric resistance R of a copper bar at various temperatures $t^{\circ}C$ are listed below: <table><tr><td>$t^{\circ}C$</td><td>19</td><td>25</td><td>30</td><td>36</td><td>40</td><td>45</td><td>50</td></tr><tr><td>R</td><td>76</td><td>77</td><td>79</td><td>80</td><td>82</td><td>83</td><td>85</td></tr></table> Find a relation $R = a + bt$ where a and b are constants to be determined.	$t^{\circ}C$	19	25	30	36	40	45	50	R	76	77	79	80	82	83	85		3					
$t^{\circ}C$	19	25	30	36	40	45	50																		
R	76	77	79	80	82	83	85																		
670	9	If P is the pull required to lift a load W by means of a pulley block, find a linear law of the form $P = mW + C$ connecting P & W using the following data. <table><tr><td>P</td><td>12</td><td>15</td><td>21</td><td>25</td></tr><tr><td>W</td><td>50</td><td>70</td><td>100</td><td>120</td></tr></table> Where P & W are taken in kg – wt. Compute P when $W = 150\text{ kg}$	P	12	15	21	25	W	50	70	100	120		3											
P	12	15	21	25																					
W	50	70	100	120																					

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Sr. No.	unit_number	Question text	Answer	Marks	Previous Year	Option 1	Option 2	Option 3	Option 4																		
671	9	Fit a straight line to the given data regarding x as the independent variable. <table><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>y</td><td>1200</td><td>900</td><td>600</td><td>200</td><td>110</td><td>50</td></tr></table>	x	1	2	3	4	5	6	y	1200	900	600	200	110	50		3									
x	1	2	3	4	5	6																					
y	1200	900	600	200	110	50																					
672	9	If $f(x)$ is a linear curve such that $f(0) = -3$, $f(1) = 6$, $f(2) = 8$ then find $f(x)$ using least square method. Also find $f(8)$.		3																							
673	9	A simply supported beam carries a concentrated load $P(lb)$ at its midpoint. Corresponding to various values of P , the maximum deflection $Y(in)$ is measured. The data is given below. <table><tr><td>P</td><td>100</td><td>120</td><td>140</td><td>160</td><td>180</td><td>200</td></tr><tr><td>Y</td><td>0.45</td><td>0.55</td><td>0.60</td><td>0.70</td><td>0.80</td><td>0.85</td></tr></table> Find a law of the form $Y = a + bP$ using the least square method.	P	100	120	140	160	180	200	Y	0.45	0.55	0.60	0.70	0.80	0.85		3	LJU-2022 LJU-2025								
P	100	120	140	160	180	200																					
Y	0.45	0.55	0.60	0.70	0.80	0.85																					
674	9	Fit a straight line to the following data: <table><tr><td>x</td><td>71</td><td>68</td><td>73</td><td>69</td><td>67</td><td>65</td><td>66</td><td>67</td></tr><tr><td>y</td><td>69</td><td>72</td><td>70</td><td>70</td><td>68</td><td>67</td><td>68</td><td>64</td></tr></table>	x	71	68	73	69	67	65	66	67	y	69	72	70	70	68	67	68	64		3					
x	71	68	73	69	67	65	66	67																			
y	69	72	70	70	68	67	68	64																			
675	9	Fit a straight line to the following data taking x as the dependent variable. <table><tr><td>x</td><td>1</td><td>3</td><td>4</td><td>6</td><td>8</td><td>9</td><td>11</td><td>14</td></tr><tr><td>y</td><td>1</td><td>2</td><td>4</td><td>4</td><td>5</td><td>7</td><td>8</td><td>9</td></tr></table>	x	1	3	4	6	8	9	11	14	y	1	2	4	4	5	7	8	9		3					
x	1	3	4	6	8	9	11	14																			
y	1	2	4	4	5	7	8	9																			
676	9	Fit a straight line to the following data. Using a equation find the value of y when $x = 2.4$ <table><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr><tr><td>y</td><td>0.5</td><td>2.5</td><td>2.0</td><td>4.0</td><td>3.5</td><td>6.0</td><td>5.5</td></tr></table>	x	1	2	3	4	5	6	7	y	0.5	2.5	2.0	4.0	3.5	6.0	5.5		3	LJU-2025						
x	1	2	3	4	5	6	7																				
y	0.5	2.5	2.0	4.0	3.5	6.0	5.5																				
677	9	Fit a straight line to the data given below. Also estimate the value of y at $x = 2.5$ <table><tr><td>X</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>Y</td><td>1</td><td>1.8</td><td>3.3</td><td>4.5</td><td>6.3</td></tr></table>	X	0	1	2	3	4	Y	1	1.8	3.3	4.5	6.3		3											
X	0	1	2	3	4																						
Y	1	1.8	3.3	4.5	6.3																						
678	9	Using method of least squares, Find the best fitting second-degree curve to the following data: <table><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>y</td><td>6</td><td>11</td><td>18</td><td>27</td></tr></table>	x	1	2	3	4	y	6	11	18	27		4													
x	1	2	3	4																							
y	6	11	18	27																							
679	9	Fit a parabola to the following data: <table><tr><td>x</td><td>-2</td><td>-1</td><td>0</td><td>1</td><td>2</td></tr><tr><td>y</td><td>1.0</td><td>1.8</td><td>1.3</td><td>2.5</td><td>6.3</td></tr></table>	x	-2	-1	0	1	2	y	1.0	1.8	1.3	2.5	6.3		4											
x	-2	-1	0	1	2																						
y	1.0	1.8	1.3	2.5	6.3																						
680	9	Fit a second-degree polynomial using the least square method to the following data: <table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>y</td><td>1</td><td>1.8</td><td>1.3</td><td>2.5</td><td>6.3</td></tr></table>	x	0	1	2	3	4	y	1	1.8	1.3	2.5	6.3		4											
x	0	1	2	3	4																						
y	1	1.8	1.3	2.5	6.3																						
681	9	Fit straight line into following data: <table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>y</td><td>3</td><td>6</td><td>8</td><td>11</td><td>13</td><td>14</td></tr></table>	x	0	1	2	3	4	5	y	3	6	8	11	13	14		4									
x	0	1	2	3	4	5																					
y	3	6	8	11	13	14																					

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Sr. No.	unit_ number	Question text	Answer	Marks	Previous Year	Option 1	Option 2	Option 3	Option 4																				
682	9	Fit the second-degree parabola $y = a + bx^2$ to the following data: <table><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>y</td><td>1.8</td><td>5.1</td><td>8.9</td><td>14.1</td><td>19.8</td></tr></table>	x	1	2	3	4	5	y	1.8	5.1	8.9	14.1	19.8		4	LJU-2025												
x	1	2	3	4	5																								
y	1.8	5.1	8.9	14.1	19.8																								
683	9	Fit a parabola $y = a + bx + cx^2$ to the following data: <table><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>5</td><td>6</td></tr><tr><td>y</td><td>1.1</td><td>5.8</td><td>17.5</td><td>55.9</td><td>86.7</td></tr></table>	x	1	2	3	5	6	y	1.1	5.8	17.5	55.9	86.7		4	LJU-2024												
x	1	2	3	5	6																								
y	1.1	5.8	17.5	55.9	86.7																								
684	9	Fit a second-degree polynomial using the least square method to the following data: <table><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>y</td><td>1.7</td><td>1.8</td><td>2.3</td><td>3.2</td></tr></table>	x	1	2	3	4	y	1.7	1.8	2.3	3.2		4															
x	1	2	3	4																									
y	1.7	1.8	2.3	3.2																									
685	9	Fit a second-degree parabola to the data. <table><tr><td>X</td><td>1929</td><td>1930</td><td>1931</td><td>1932</td><td>1933</td><td>1934</td><td>1935</td></tr><tr><td>Y</td><td>352</td><td>356</td><td>357</td><td>358</td><td>360</td><td>361</td><td>361</td></tr></table>	X	1929	1930	1931	1932	1933	1934	1935	Y	352	356	357	358	360	361	361		4									
X	1929	1930	1931	1932	1933	1934	1935																						
Y	352	356	357	358	360	361	361																						
686	9	Fit a curve $y = ax + bx^2$ to the following data: <table><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>y</td><td>2.51</td><td>5.82</td><td>9.93</td><td>14.84</td><td>20.55</td><td>27.06</td></tr></table>	x	1	2	3	4	5	6	y	2.51	5.82	9.93	14.84	20.55	27.06		3											
x	1	2	3	4	5	6																							
y	2.51	5.82	9.93	14.84	20.55	27.06																							
687	9	Fit a curve $y = ax + bx^2$ to the following data: <table><tr><td>x</td><td>-2</td><td>-1</td><td>0</td><td>1</td><td>2</td></tr><tr><td>y</td><td>-72</td><td>-46</td><td>-12</td><td>35</td><td>93</td></tr></table>	x	-2	-1	0	1	2	y	-72	-46	-12	35	93		3													
x	-2	-1	0	1	2																								
y	-72	-46	-12	35	93																								
688	9	Fit a parabola $y = a + bx + cx^2$ to the following data: <table><tr><td>x</td><td>0</td><td>2</td><td>5</td><td>10</td></tr><tr><td>y</td><td>4</td><td>7</td><td>6.4</td><td>-6</td></tr></table>	x	0	2	5	10	y	4	7	6.4	-6		4															
x	0	2	5	10																									
y	4	7	6.4	-6																									
689	9	Fit a curve $y = a_0 + a_1x + a_2x^2$ for the given data: <table><tr><td>x</td><td>3</td><td>5</td><td>7</td><td>9</td><td>11</td><td>13</td></tr><tr><td>y</td><td>2</td><td>3</td><td>4</td><td>6</td><td>5</td><td>8</td></tr></table>	x	3	5	7	9	11	13	y	2	3	4	6	5	8		4											
x	3	5	7	9	11	13																							
y	2	3	4	6	5	8																							
690	9	Find the least square polynomial approximation of degree two to the data. <table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>y</td><td>-4</td><td>-1</td><td>4</td><td>11</td><td>20</td></tr></table>	x	0	1	2	3	4	y	-4	-1	4	11	20		4	LJU-2022												
x	0	1	2	3	4																								
y	-4	-1	4	11	20																								
691	9	Fit a second-degree parabola to the following data: <table><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>y</td><td>10</td><td>12</td><td>13</td><td>16</td><td>19</td></tr></table>	x	1	2	3	4	5	y	10	12	13	16	19		4													
x	1	2	3	4	5																								
y	10	12	13	16	19																								
692	9	Fit a second-degree parabola to the following data taking x as the independent variable. <table><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td></tr><tr><td>y</td><td>2</td><td>6</td><td>7</td><td>8</td><td>10</td><td>11</td><td>11</td><td>10</td><td>9</td></tr></table>	x	1	2	3	4	5	6	7	8	9	y	2	6	7	8	10	11	11	10	9		4	LJU-2025				
x	1	2	3	4	5	6	7	8	9																				
y	2	6	7	8	10	11	11	10	9																				
693	9	By the method of least squares, fit a parabola to the following data: <table><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>y</td><td>5</td><td>12</td><td>26</td><td>60</td><td>97</td></tr></table>	x	1	2	3	4	5	y	5	12	26	60	97		4													
x	1	2	3	4	5																								
y	5	12	26	60	97																								

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Sr. No.	unit_number	Question text	Answer	Marks	Previous Year	Option 1	Option 2	Option 3	Option 4																		
694	9	Fit a second-degree parabola $y = a_0 + a_1x + a_2x^2$ for the given data: <table><tr><td>x</td><td>1</td><td>1.5</td><td>2</td><td>2.5</td><td>3</td><td>3.5</td><td>4</td></tr><tr><td>y</td><td>1.2</td><td>1.4</td><td>1.9</td><td>2.4</td><td>2.8</td><td>3.3</td><td>4.2</td></tr></table>	x	1	1.5	2	2.5	3	3.5	4	y	1.2	1.4	1.9	2.4	2.8	3.3	4.2		4							
x	1	1.5	2	2.5	3	3.5	4																				
y	1.2	1.4	1.9	2.4	2.8	3.3	4.2																				
695	9	Fit a straight line to the following data. <table><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td><td>6</td><td>8</td></tr><tr><td>y</td><td>2.4</td><td>3</td><td>3.6</td><td>4</td><td>5</td><td>6</td></tr></table>	x	1	2	3	4	6	8	y	2.4	3	3.6	4	5	6		3									
x	1	2	3	4	6	8																					
y	2.4	3	3.6	4	5	6																					
696	9	Fit a straight line to the following data: <table><tr><td>X</td><td>103</td><td>186</td><td>99</td><td>100</td></tr><tr><td>Y</td><td>168</td><td>120</td><td>72</td><td>63</td></tr></table>	X	103	186	99	100	Y	168	120	72	63		3	LJU-2023												
X	103	186	99	100																							
Y	168	120	72	63																							
697	9	Fit a quadratic curve to the following data: <table><tr><td>x</td><td>2</td><td>4</td><td>6</td><td>8</td><td>10</td></tr><tr><td>y</td><td>0.1825</td><td>0.2019</td><td>0.2621</td><td>0.3721</td><td>0.5123</td></tr></table>	x	2	4	6	8	10	y	0.1825	0.2019	0.2621	0.3721	0.5123		4	LJU-2023										
x	2	4	6	8	10																						
y	0.1825	0.2019	0.2621	0.3721	0.5123																						
698	9	Fit a straight line $y = a + bx$ to the following data: <table><tr><td>x</td><td>1</td><td>5</td><td>7</td><td>9</td><td>12</td></tr><tr><td>y</td><td>10</td><td>15</td><td>12</td><td>15</td><td>21</td></tr></table>	x	1	5	7	9	12	y	10	15	12	15	21		3	LJU-2024										
x	1	5	7	9	12																						
y	10	15	12	15	21																						
699	9	Fit a quadratic curve to the following data: <table><tr><td>x</td><td>1</td><td>3</td><td>3.2</td><td>4</td><td>6.1</td><td>7</td><td>8</td></tr><tr><td>y</td><td>$\frac{3}{2}$</td><td>$\frac{5}{2}$</td><td>$\frac{18}{4}$</td><td>$\frac{1}{5}$</td><td>$\frac{15}{4}$</td><td>$\frac{8}{5}$</td><td>$\frac{17}{2}$</td></tr></table>	x	1	3	3.2	4	6.1	7	8	y	$\frac{3}{2}$	$\frac{5}{2}$	$\frac{18}{4}$	$\frac{1}{5}$	$\frac{15}{4}$	$\frac{8}{5}$	$\frac{17}{2}$		6	LJU-2025						
x	1	3	3.2	4	6.1	7	8																				
y	$\frac{3}{2}$	$\frac{5}{2}$	$\frac{18}{4}$	$\frac{1}{5}$	$\frac{15}{4}$	$\frac{8}{5}$	$\frac{17}{2}$																				
700	9	Fit a second-degree polynomial using the least square method to the following data: <table><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td></tr><tr><td>y</td><td>8</td><td>13</td><td>20</td><td>29</td><td>40</td><td>53</td><td>68</td><td>10</td></tr></table>	x	1	2	3	4	5	6	7	8	y	8	13	20	29	40	53	68	10		6	LJU-2025				
x	1	2	3	4	5	6	7	8																			
y	8	13	20	29	40	53	68	10																			
701	9	Fit a straight line of the form $y = a + bx$ to the following data, and then find the value of y when $x = 12$. <table><tr><td>x</td><td>5</td><td>7</td><td>9</td><td>10</td><td>12</td><td>15</td><td>18</td><td>20</td></tr><tr><td>y</td><td>7</td><td>10</td><td>15</td><td>17</td><td>19</td><td>25</td><td>27</td><td>30</td></tr></table>	x	5	7	9	10	12	15	18	20	y	7	10	15	17	19	25	27	30		5	LJU-2025				
x	5	7	9	10	12	15	18	20																			
y	7	10	15	17	19	25	27	30																			
702	10	If $\pi p = \pi$, where $P = \begin{bmatrix} 0 & 1 \\ 1/2 & 1/2 \end{bmatrix}$ then values of (π_1, π_2) is	$\left(\frac{1}{3}, \frac{2}{3}\right)$	1	LJU-2022 LJU-2023 LJU-2025	$\left(\frac{1}{3}, \frac{2}{3}\right)$	$\left(\frac{1}{2}, \frac{1}{2}\right)$	$\left(\frac{2}{3}, \frac{1}{3}\right)$	(0, 1)																		
703	10	Let $\pi = [\pi_1, \pi_2]$ and $\pi p = \pi$, where $P = \begin{bmatrix} 0.9 & 0.1 \\ 0.2 & 0.8 \end{bmatrix}$ then values of (π_1, π_2) is	$\left(\frac{2}{3}, \frac{1}{3}\right)$	1	LJU-2024 LJU-2025	$\left(\frac{1}{3}, \frac{2}{3}\right)$	$\left(\frac{1}{2}, \frac{1}{2}\right)$	(0,1)	$\left(\frac{2}{3}, \frac{1}{3}\right)$																		
704	10	Consider the following transition probability matrix $P = \begin{bmatrix} 0.1 & 0.5 & 0.4 \\ 0.6 & 0.2 & 0.2 \\ 0.3 & 0.4 & 0.3 \end{bmatrix}$ and the initial probability is (0.7, 0.2, 0.1) then the value of $q_2(3)$ is	0.279	1	LJU-2025	0.279	0.336	0.34	0.385																		
705	10	If the TPM of a Markov chain is $= \begin{bmatrix} 0.1 & 0.5 & 0.4 \\ 0.6 & 0.2 & 0.2 \\ 0.3 & 0.4 & 0.3 \end{bmatrix}$ find $P[X_1 = 3 / X_0 = 2]$		1																							
706	10	Consider the Markov chain with three states $S = \{1,2,3\}$, that has the following transition matrix $P = \begin{pmatrix} 1/2 & 1/4 & 1/4 \\ 1/3 & 0 & 2/3 \\ 1/2 & 1/2 & 0 \end{pmatrix}$		3	LJU-2025																						

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Sr. No.	unit_number	Question text	Answer	Marks	Previous Year	Option 1	Option 2	Option 3	Option 4
		If we know $P(X_1 = 1) = P(X_1 = 2) = \frac{1}{4}$ find $P(X_1 = 3, X_2 = 2, X_3 = 1)$.							
707	10	Three boys A, B and C are throwing a ball to each other. A always throws the ball to B and B always throws the ball to C, but C is just as likely to throw the ball to B as to A. If the initial probability distribution of three states A, B and C is 0.3, 0.4 and 0.3 respectively. Find (1) the transition matrix (2) $P(X_2 = B)$ (3) The distribution of the balls after two rounds.		3					
708	10	In the Dark Ages, Harvard, Dartmouth, and Yale admitted only male students. Assume that, at that time, 80 percent of the sons of Harvard men went to Harvard and the rest went to Yale, 40 percent of the sons of Yale men went to Yale, and the rest split evenly between Harvard and Dartmouth; and of the sons of Dartmouth men, 70 percent went to Dartmouth, 20 percent to Harvard, and 10 percent to Yale. (a) Construct the TPM. (b) Find the probability that the grandson of a man from Harvard went to Harvard (c) Find the probability that the great - grandson of a man from Harvard went to Harvard.		3	LJU-2022				
709	10	In a certain market, there are three brands of cosmetic A, B and C. Given that a lady last purchased cosmetic of brand A, there is 70% chance that she would continue with brand A, 20% and 10% chances that she would shift to brands B and C, respectively. Given that a lady last purchased cosmetic of brand B, there is 50% chances that she would shift to brand A and 10% chance to brand C. Given that a lady last purchased cosmetic of brand C, there is 60%, 20% chance that she would shift to brands A and B respectively. The present market shares of the three brands A, B and C is 60%, 30% and 10% respectively. Using this information, find a. TPM b. The probability that a customer who is currently a purchaser of brand A will purchase brand B after two time periods. c. The probability that a customer who is currently a purchaser of brand A will purchase brand C after two time periods. d. The probability that brand C will be able to retain its customer after two time periods? e. The probability that a purchaser of brand B will purchase brand A three time periods from now.		4	LJU-2025				
710	10	 A professor of Statistics not wanting to be predictable decides on an innovative way of assigning homework based on probabilities. On the first day of the week, he draws a		3					

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Sr. No.	unit_number	Question text	Answer	Marks	Previous Year	Option 1	Option 2	Option 3	Option 4
		transition diagram as shown in Figure. The nodes of the diagram represent full credit (F), half credit (H) and no credit (N) assignments. The transition probabilities for day 1 are as shown in the figure. Construct TPM and compute: $P(X_3 = F/X_2 = N)$ $P(X_2 = N/X_1 = H)$ $P(X_4 = H/X_2 = F)$							
711	10	The TPM of the Markov chain with three states 1, 2, 3 is $P = \begin{bmatrix} 0.2 & 0.3 & - \\ - & 0.6 & 0.3 \\ 0.4 & - & 0.3 \end{bmatrix}$ And the initial probability is (0.5, 0.3, 0.2). Complete the TPM and calculate (1) $P(X_3 = 3, X_2 = 2, X_1 = 1, X_0 = 3)$ (2) $P(X_3 = 3, X_1 = 1, X_0 = 3)$		3	LJU-2023 LJU-2025				
712	10	The TPM of the Markov chain with three states 1, 2, 3 is $P = \begin{bmatrix} 0.3 & 0.2 & 0.5 \\ 0.6 & 0.1 & 0.3 \\ 0.3 & 0.3 & 0.4 \end{bmatrix}$ and the initial probability is (0.1, 0.4, 0.5). Calculate (i) $P(X_4 = 2, X_2 = 3, X_1 = 1, X_0 = 2)$ (ii) $P(X_0 = 1, X_2 = 3, X_4 = 2, X_7 = 1)$		3	LJU-2024				
713	10	Consider a continuous-time Markov chain $X(t)$ with the jump chain shown in Figure. Assume $\lambda_1 = 2$, $\lambda_2 = 3$, and $\lambda_3 = 4$.  (a) Find the stationary distribution of the jump chain $\bar{\pi} = \bar{\pi}_1, \bar{\pi}_2, \bar{\pi}_3$ (b) Using $\bar{\pi}$, find the stationary distribution for $X(t)$.		4	LJU-2022				
714	10	Given that a person last cola purchase was COKE, there is a 90% chance that his next cola purchase will also be COKE. If a person last cola purchase was PEPSI, there is a 80% chance that his next cola purchase will also be PEPSI. The present market share of the COKE and PEPSI is 55% and 45% respectively. Construct the TPM. In the long run, what is the market share of such cola?		3					
715	10	In a certain market, there are three brands of cosmetics A, B and C. Given that a lady last purchased cosmetics of brand A, there is 70% chance that she would continue with brand A, 20% and 10% chances that she would shift to brands B and C, respectively. Given that a lady last purchased cosmetics of brand B, there is 50% chances that she would shift to brand A and 10% chance to brand C. Given that a lady last purchased cosmetics of brand C, there is 60%, 20% chance		4	LJU-2023 LJU-2024				

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Sr. No.	unit_number	Question text	Answer	Marks	Previous Year	Option 1	Option 2	Option 3	Option 4
		that she would shift to brands A and B respectively. The present market share of the three brands A, B and C is 60%, 30% and 10% respectively. Using this information, find the market share of the brands A, B and C in the steady-state.							
716	10	At Komo Cafe, there are three types of daily customers: Regulars, Occasional and First-timers. The cafe owner has been tracking customer visits and notices patterns in how customers return from one day to the next: If a customer is a Regular today, there is a 50% chance they will be a Regular the next day and a 30% chance they will become an Occasional. If a customer is an Occasional today, there is a 40% chance they will become a Regular next time and a 20% chance they will be a First-timer and if a customer is a First-timer today, there is a 30% chance they will come back as an Occasional and an equal chance for them to be either a Regular or a First-timer the next time. Currently, the initial distribution of customers visiting the cafe is 25% Regulars, 40% Occasional and 35% First-timers. (a) Construct a transition probability matrix that represents the daily movement of customers between these three categories. (b) If a customer is a First-timer today, what is the probability that they will be a Regular two days later? (c) In the long run, as the cafe continues to observe customer patterns, what will be the market share of each customer type, and which type is expected to dominate the cafe? (d) Calculate $P(X_0 = 2, X_1 = 2, X_2 = 1, X_4 = 3, X_5 = 1)$		6	LJU-2025				
717	10	In an earlier period, three universities — Oxford, Cambridge, and London University — admitted only male students. It was observed that admission patterns tended to follow family tradition. The following information was collected: Of the sons of Oxford graduates, 70% enrolled at Oxford, 20% enrolled at Cambridge, and the remaining enrolled at London University. Of the sons of Cambridge graduates, 50% enrolled at Cambridge, while the remaining sons were divided between Oxford and London University in the ratio 3:2. Determine whether the system tends toward a steady-state distribution, and if so, state its significance. Of the sons of London University graduates, 60% enrolled at London University, 25% enrolled at Oxford, and the rest enrolled at Cambridge. Assume that these transition probabilities remain unchanged from one generation to the next. Determine whether the system tends toward a steady-state distribution.		6	LJU-2025				
718	10	Three friends A, B, C are throwing a ball among themselves. The rules of throwing are as follows: If A has the ball, he throws it to B with probability $\frac{1}{2}$ and to C with probability $\frac{1}{2}$. If B has the ball, he throws it to A with probability $\frac{1}{3}$ and to C with probability $\frac{2}{3}$. If C has the ball, he throws it to A with probability $\frac{3}{4}$ and to B with probability $\frac{1}{4}$.		5	LJU-2025				

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Sr. No.	unit_number	Question text	Answer	Marks	Previous Year	Option 1	Option 2	Option 3	Option 4
		In the long run, how often does each friend receive the ball?							
719	10	Minimize $f(x_1, x_2) = x_1 - x_2 + 2x_1^2 + 2x_1x_2 + x_2^2$ starting from the point $X_1 = (0,0)$ using Gradient Decent method.		4	LJU-2025				